Medicinal chemistry of diuretics and cardiovascular Agents

By

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Medicinal chemistry of diuretics

Outline

- Diuretics
  - ✓Carbonic anhydrase inhibitors
  - ✓High-ceiling or loop diuretics
  - ✓The thiazide and thiazide-like diuretics
  - ✓Potassium-sparing and other diuretics

Introduction [1]

- Diuretics are agents that increase the rate of urine formation.
  - excretion of electrolytes and water from the body.
- Diuretics used in the treatment of edematous conditions and in the management of hypertension.

Introduction [2]

- The primary target organ for diuretics is the kidney,
  - where these drugs interfere with the reabsorption of sodium and other ions from the lumina of the nephrons.
- The amount of ions and accompanying water that are
  - excreted as urine following administration of a diuretic is determined by many factors.

Introduction [3]

- Thus, it is important to be aware of the normal mechanisms of urine formation and renal control mechanisms
  - to understand clearly the ability of chemicals to induce diuresis.

Normal Physiology of Urine Formation

- Two important functions of the kidney are
  - to maintain a homeostatic balance of electrolytes and water
  - to excrete water-soluble end products of metabolism.
- The kidney accomplishes these functions through the formation of urine by the nephrons

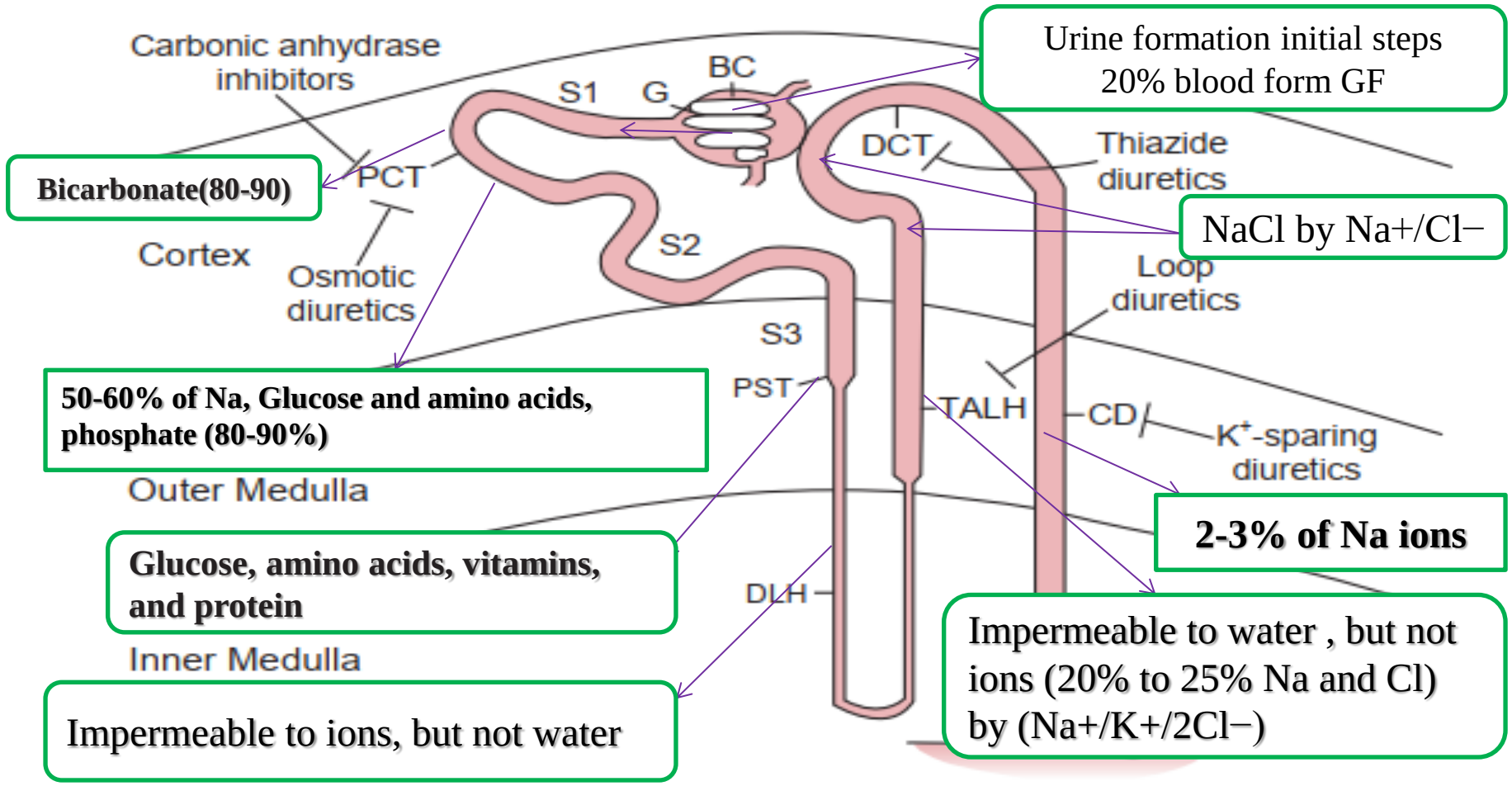


FIGURE 22.1 The nephron. BC, Bowman's capsule; CD, collecting duct; DCT, distal convoluted tubule; DLH, descending limb of the Loop of Henle; G, glomerulus; PCT, proximal convoluted tubule; PST, proximal straight tubule; TALH, thick ascending limb of the loop of Henle.

Structure Classification

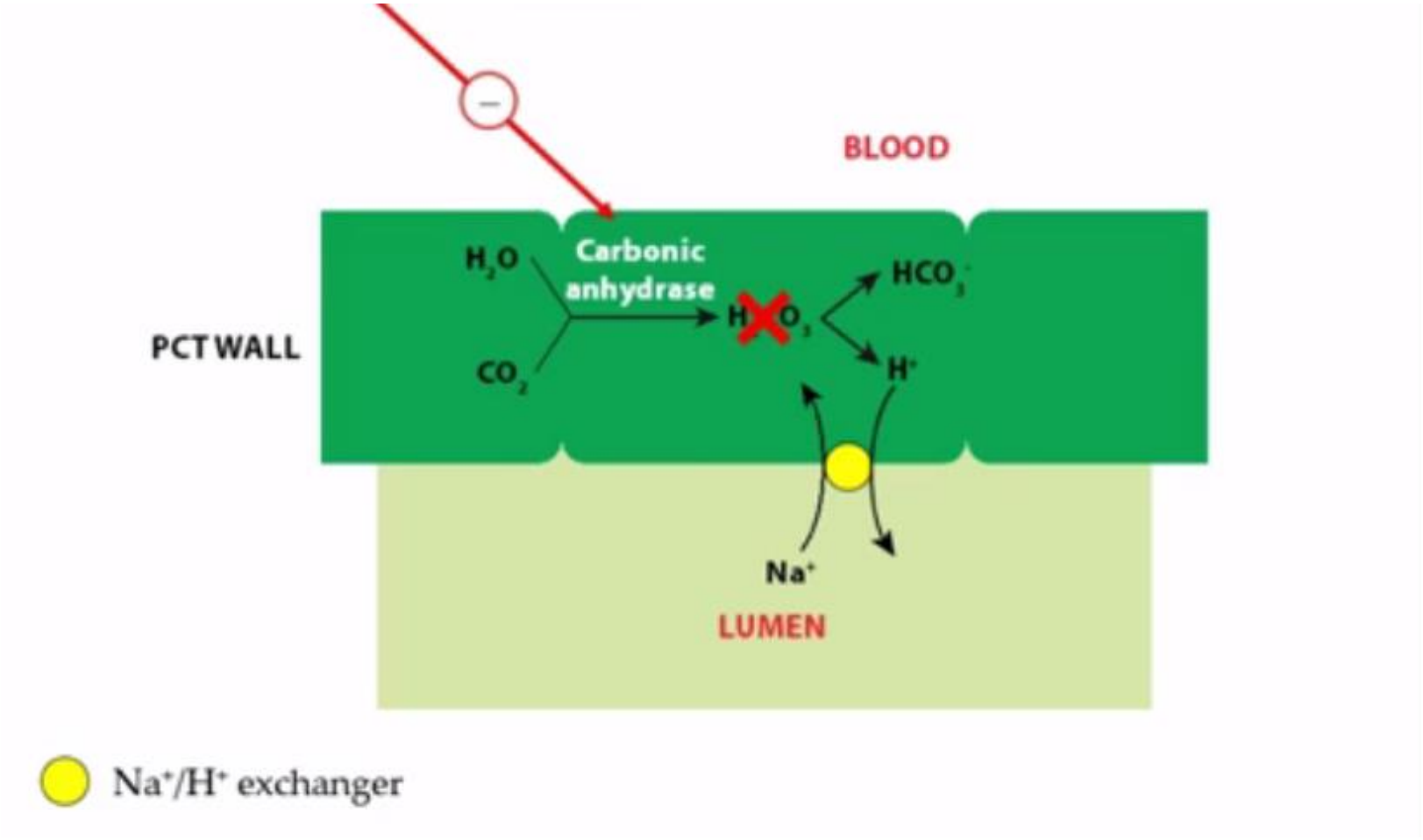
- Current diuretics are classified by their
  - chemical class (thiazides),
  - mechanism of action (carbonic anhydrase inhibitors and osmotics),
  - site of action (loop diuretics), or
  - effects on urine contents (potassium-sparing diuretics)

Carbonic Anhydrase Inhibitors

Mechanism of action

- Inhibit the carbonic anhydrase enzyme in proximal convoluted tubule wall
  - → H_2CO_3 is not formed
    - → H^+ ions are not available for exchange with Na^+ ions → no reabsorption of Na^+.

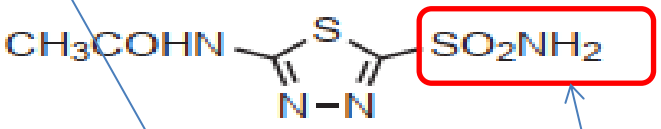
Carbonic Anhydrase Inhibitors



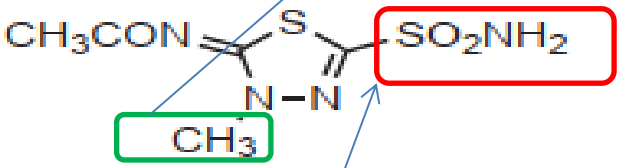
Specific Drugs

The first CAIs, thiadiazole derivative

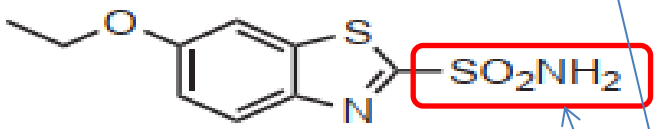
Substitution of the methyl group



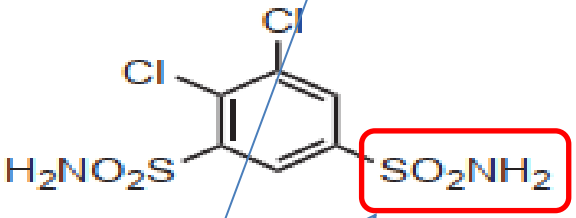
Acetazolamide



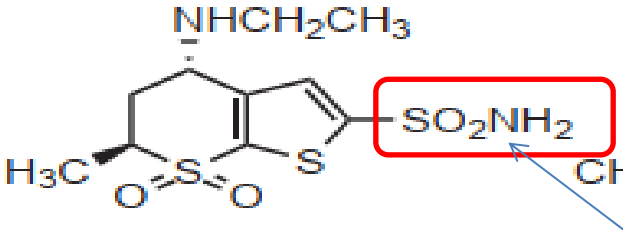
Methazolamide



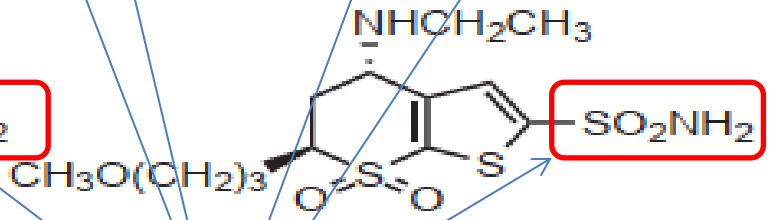
Ethoxzolamide



Dichlorphenamide



Dorzolamide

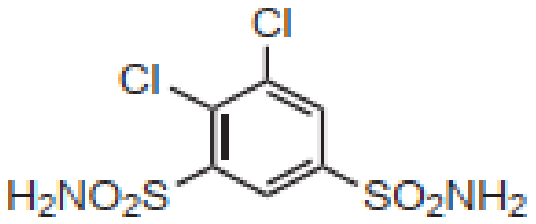


Brinzolamide

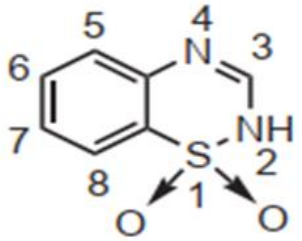
The free sulfamoyl nitrogen.
The mono- and di- substituents at SO2NH2

Thiazide Diuretics

- Benzene disulfonamide derivatives
  - to find more efficacious carbonic anhydrase inhibitors.
- When the amino group was acylated, an unexpected ring closure took place.



Dichlorphenamide

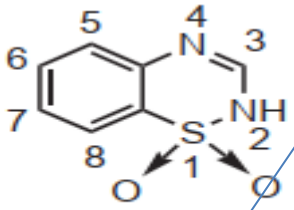


- Diuretic activity independent of the CA activity,
  - a new series of diuretics called the benzothiadiazines.

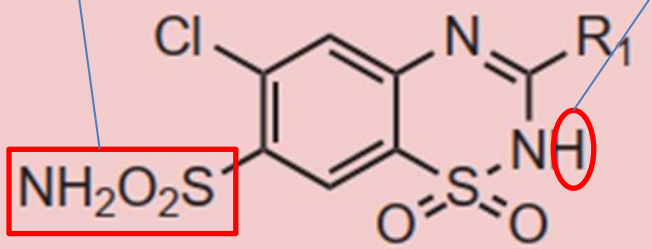
Structure–Activity Relationship[1]

- The thiazide diuretics are weakly acidic, with a benzothiadiazine1,1-dioxide nucleus.

Additional point of acidity in the molecule



- **The most acidic b/c of sulfone group.**

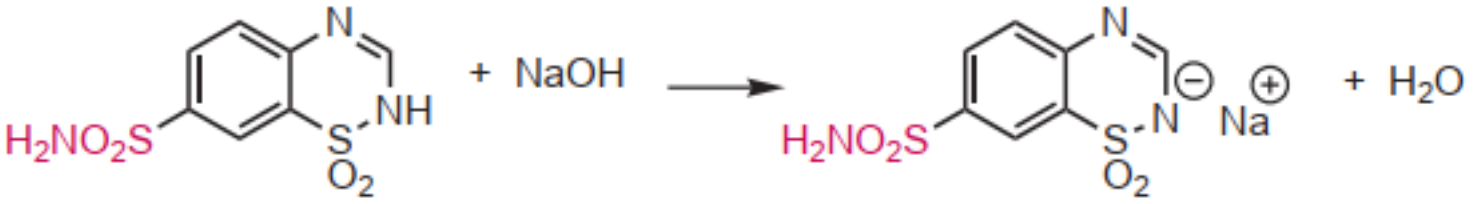


Structure I

- **Alkyl substitution decreases the polarity and increases the duration of diuretic action.**

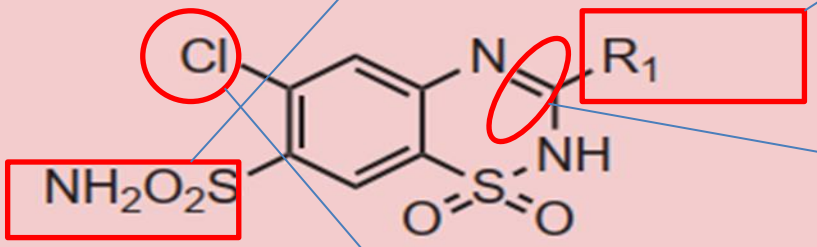
Structure–Activity Relationship[2]

- The acidic protons make possible the formation of a water-soluble sodium salt that can be used for intravenous administration of the diuretics.



- ***The sulfamoyl group in the 7 position is essential for diuretic effect.***
- ***Removal or replacement of this group result in significant reduction or complete loss of the diuretic activity***

- ***Dominant role in determining the potency and duration of action of the thiazides.***
- ***Haloalkyl, aralkyl, or thioether substitution increases the lipid solubility of the molecule.***

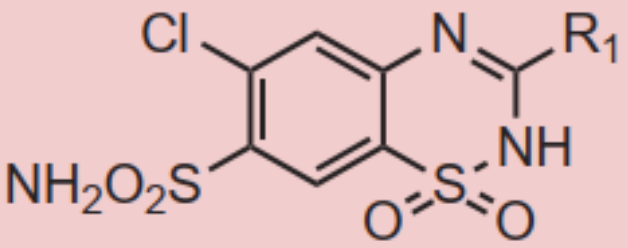


Structure I

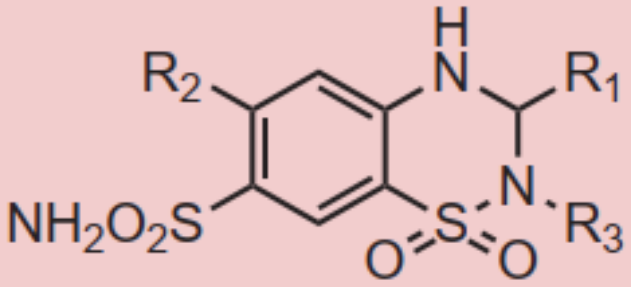
Saturation of the double bond to give a 3,4-dihydro derivative produces a diuretic that is more active than the unsaturated derivative.

- **An electron-withdrawing group is necessary for diuretic activity.**
  - **H atom Vs Chloro or trifluoromethyl .**
- **The trifluoromethyl-substituted diuretics Vs their chloro-substituted analogs**
- **Electron-releasing groups, such as methyl or methoxyl**

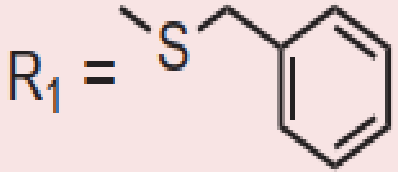
Specific examples [1]



Structure I



Structure II

Chlorothiazide	Diuril	Structure I: $R_1 = H$
Benzthiazide	Exna	Structure I: $R_1 =$ 

Specific examples [2]

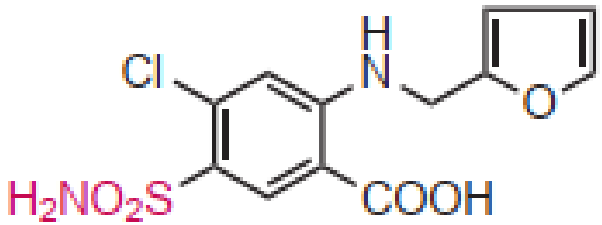
Hydrochlorothiazide	HydroDiuril Esidrix	Structure II: R_1 = H, R_2 = Cl, R_3 = H
Trichlormethiazide	Diurese Metahydrin Naqua	Structure II: R_1 = $CHCl_2$, R_2 = Cl; R_3 = H
Methyclothiazide	Enduron Aquatensen	Structure II: R_1 = CH_2Cl R_2 = Cl; R_3 = CH_3
Polythiazide	Renese	Structure II: R_1 = $-CH_2-S-CH_2-CF_3$, R_2 = Cl, R_3 = CH_3
Hydroflumethiazide	Saluron Diucardin	Structure II: R_1 = H, R_2 = CF_3, R_3 = H
Bendroflumethiazide	Naturetin	Structure II: R_1 = benzyl, R_2 = CF_3, R_3 = H

High-Ceiling or Loop Diuretics

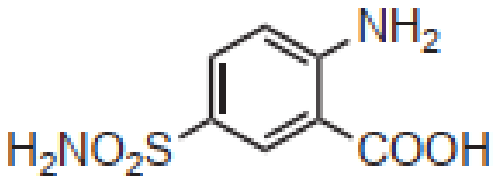
- Characterized more by its *pharmacologic similarities* than by its chemical similarities.
  - Examples include furosemide, bumetanide, torsemide, and ethacrynic acid.
- These drugs produce a peak diuresis.
  - hence the name high-ceiling diuretics.

Specific Drugs-Furosemide

- ***Structure–Activity Relationships***
  - Furosemide is a derivative of anthranilic acid.



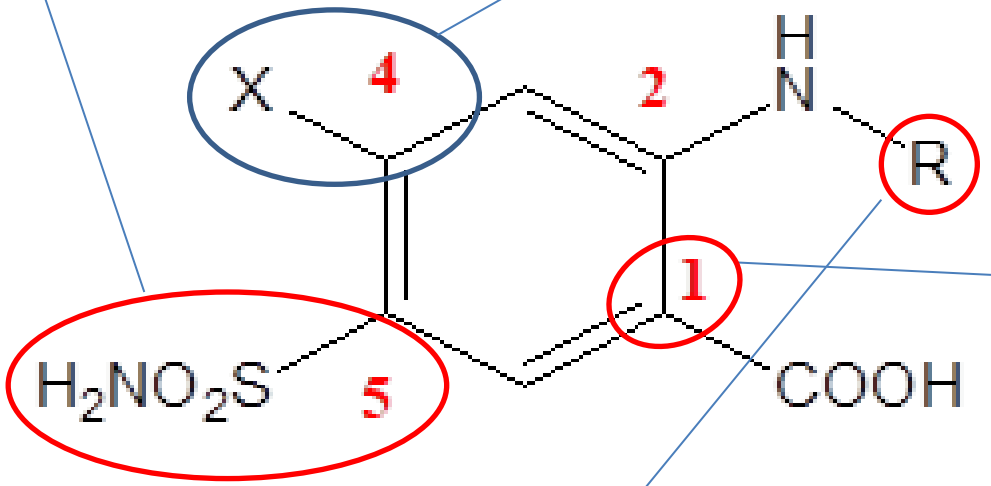
Furosemide



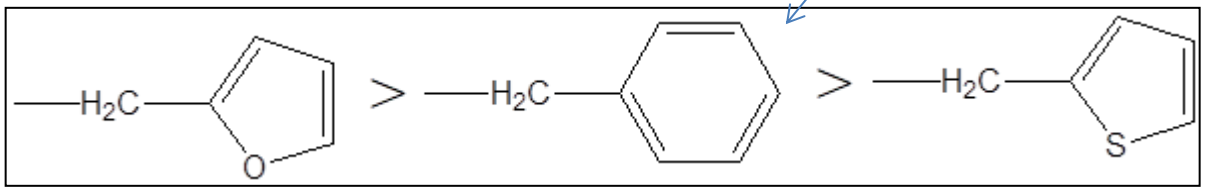
5-Sulfamoyl-anthranilic acid

- ***The activating group (-X) in the 4 position can be Cl^- or CF_3^-***
- ***Phenoxy, alkoxy, anilino, benzyl or benzoyl groups substituted at 4th position decreases diuretic activity.***

A sulfamoyl group in the 5 position is a prerequisite for optimal high-ceiling diuretic activity.

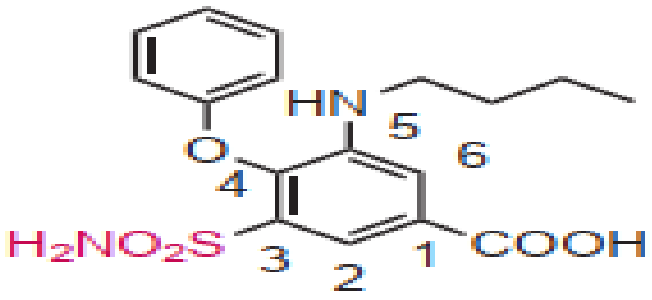


Substituent at the this position must be acidic



Bumetanide

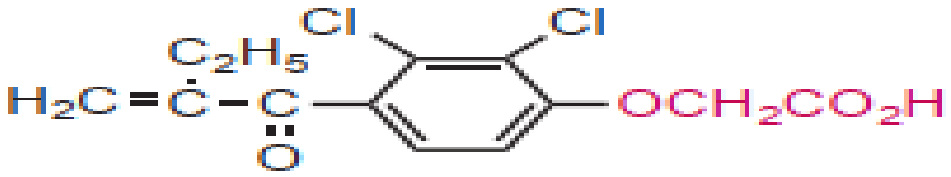
- A phenoxy group.
- The amine group.
- Butyl group on the C-5 amine is replaced with a furanylmethyl group.



Bumetanide

Ethacrynic acid

- The phenoxyacetic acid derivatives, of which ethacrynic acid is the prototypical agent.
- Position of oxyacetic acid group with unsaturated carbonyl and chloro
  - important for optimal diuretic activity .
- Hydrogen atoms on the terminal alkene carbon.



Ethacrynic acid

Potassium Sparing Diuretics

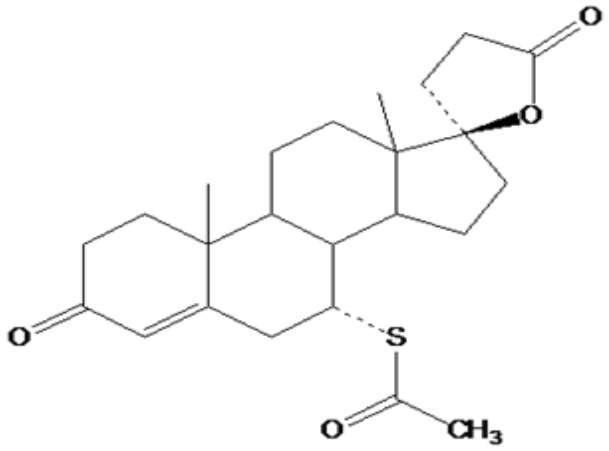
This class of diuretics are characterized by their ability to increase **Na^+ and Cl^-** excretion without a concomitant increase in the urinary excretion rate of K^+ in collecting duct.

They are also known as antikaliuretic agents.

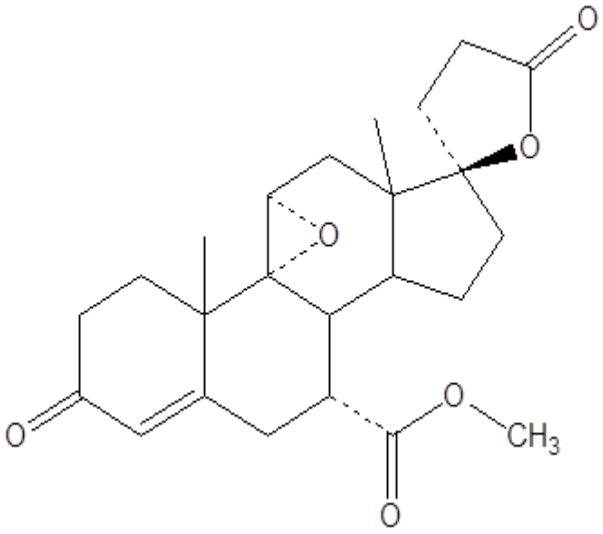
They can be classified into 2 classes with respect to mechanism of action:

1. **Aldosterone antagonists**
2. **Na^+-channel blockers**

1. Aldosterone antagonists

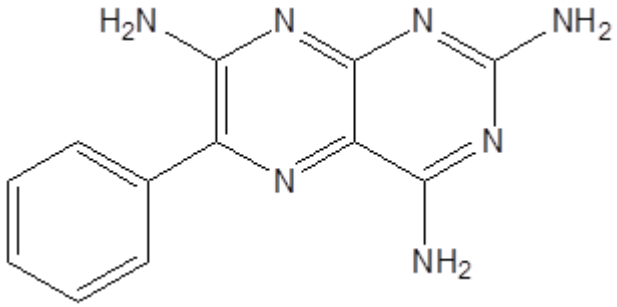


Spironolactone

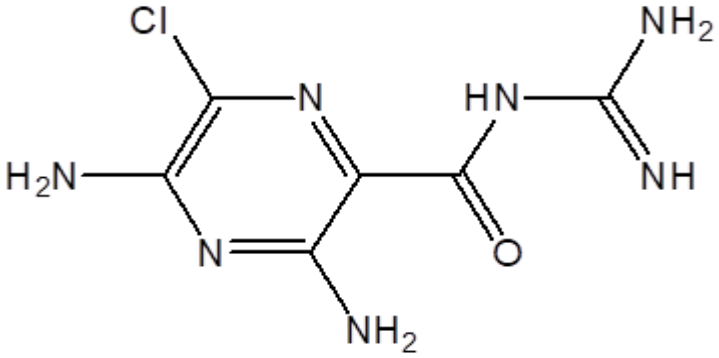


Eplerenone

2. Na+-channel blockers



Triamterene



Amiloride

Cardiovascular Agents

Outline

- Anti-hypertensive agents
- **Cardiac Agents**: Cardiac Glycosides, Antianginal agents, Antiarrhythmic drugs
- Antilipemic drugs
- Coagulants & anticoagulants

Cardiac Agents:

Cardiac Glycosides, Antianginal, and Antiarrhythmic Drugs

Introduction

- Heart diseases can be grouped into three major disorders:
  - Cardiac failure, contractile dysfunction (congestive heart failure)
  - Ischemia (with angina as its primary symptom) and
  - Cardiac arrhythmia

Congestive heart failure

❑ What is cardiac failure?

- ✓ Cardiac failure can be described as inability of the heart to pump blood effectively at a rate that meets the needs of metabolizing tissues



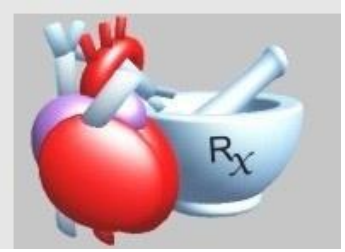
Congestive heart cont'd…

- is the direct result of a reduced contractility of the cardiac muscles,
  - especially those of the ventricles, which causes a decrease in cardiac output, increasing the blood volume of the heart
  - hence the term "congested"
- the systemic blood pressure and the renal blood flow are both reduced,
  - which often lead to the development of edema in the
    - lower extremities and
    - the lung (pulmonary edema) as well as renal failure
- A group of drugs known as the cardiac glycosides were found to reverse most of these symptoms and complications

Congestive heart cont'd…

- Group of drugs known as cardiac glycosides
  - ✓ Are important class of naturally occurring drugs whose actions include both beneficial and toxic effects on the heart
  - ✓ Their desirable cardiotonic action is of particular benefit in the treatment of congestive heart failure and associated edema and their preparations have been used as a medicinal agents as well as poisons since 1500 B.C.
  - ✓ One of the most important drug classes available to the physician to treat congestive heart failure

Cardiac Glycosides

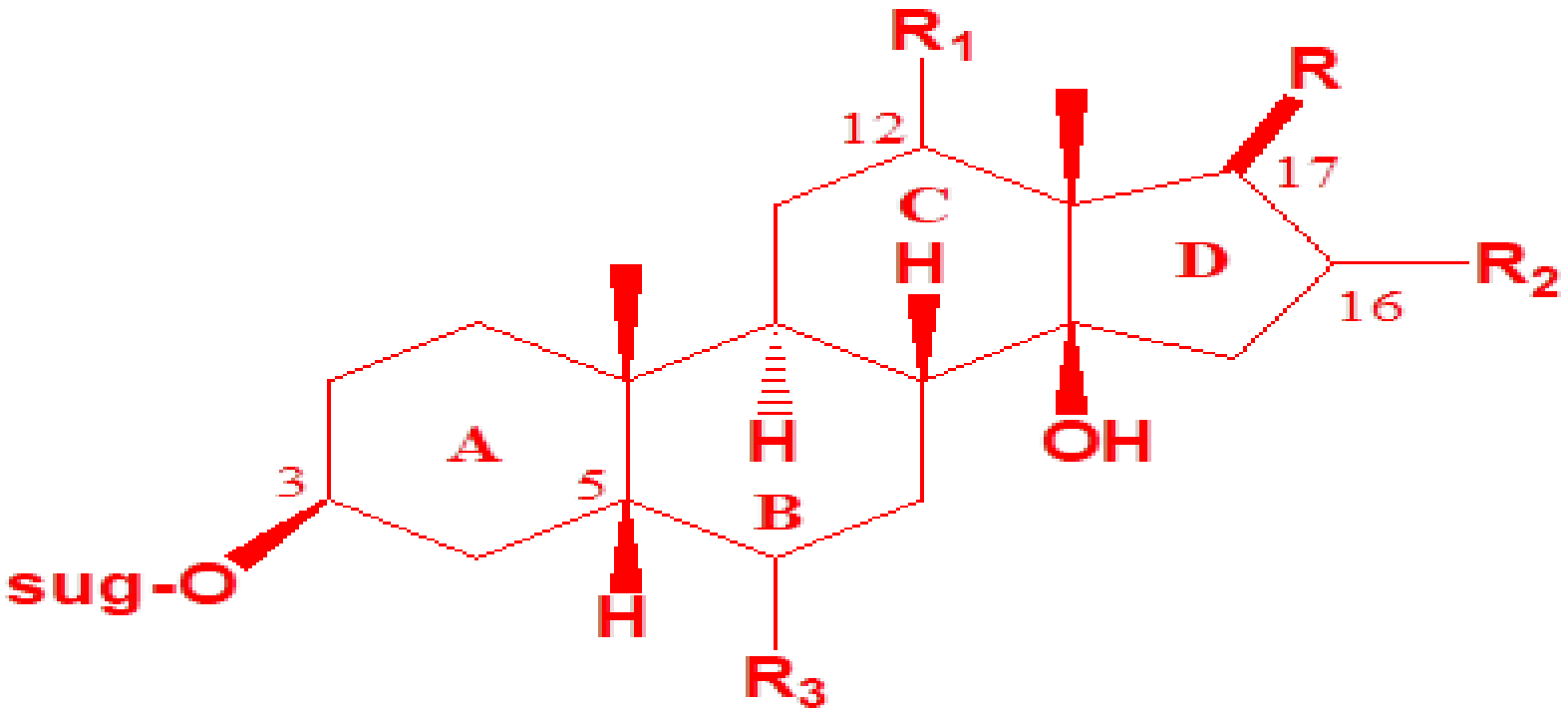


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Chemistry of the cardiac glycosides

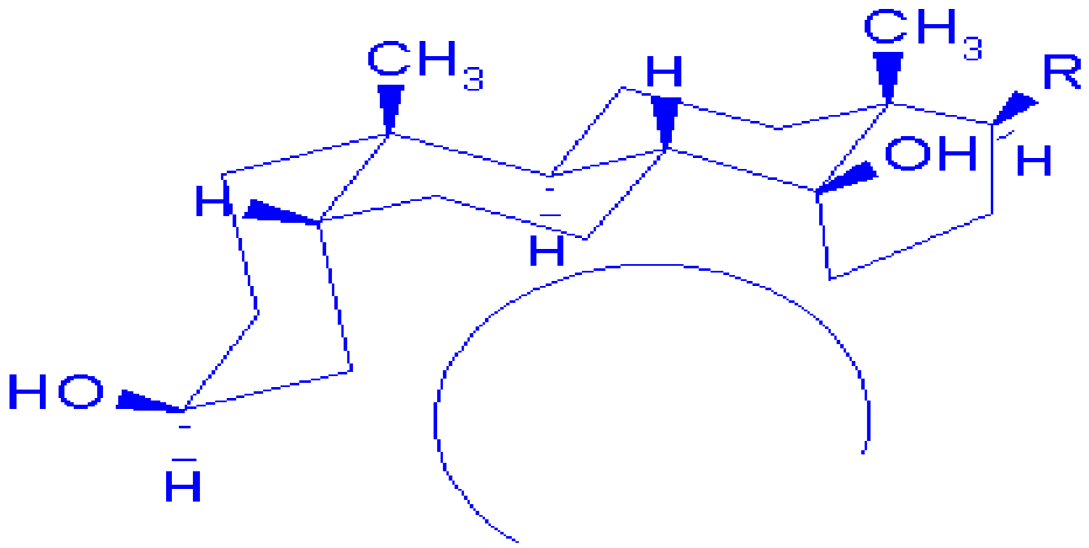
- Composed of two portions:
  - ✓the sugar moiety, and
  - ✓the non-sugar moiety
- Aglycones
  - ✓is a steroid nucleus with a unique set of fused rings, which makes these agents easily distinguished from the other steroids

Chemistry of glycosides cont'd…



Chemistry of glycosides cont'd…

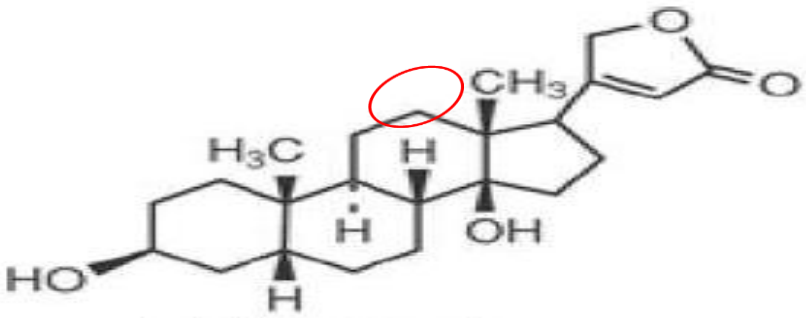
- ✓ **Rings A-B and C-D are *cis* fused, whereas rings B-C have a *trans* configuration**
- ✓ **Such ring fusion gives the aglycone nucleus of cardiac glycosides the characteristic "U-shape"**



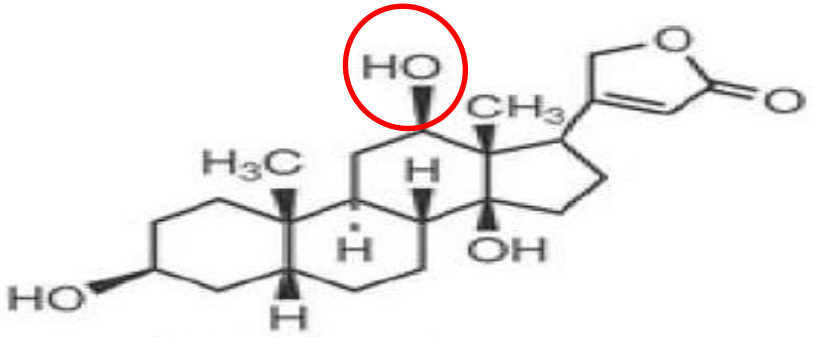
Chemistry of glycosides cont'd…

- The steroid nucleus also carries, in most cases, two angular methyl groups at C-10 and C-13.
- Hydroxyl groups are located at C-3, the site of the sugar attachment, and at C-14.
- The C-14 hydroxyl is normally unsubstituted;
- Additional hydroxyl groups may be found at C-12 and C-16, the presence or absence of which distinguishes the important genins:
  - ✓ digitoxigenin,
  - ✓ digoxigenin, and gitoxigenin

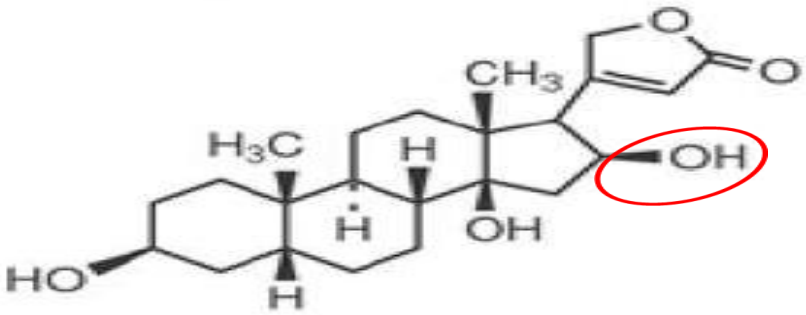
Chemistry of glycosides cont'd…



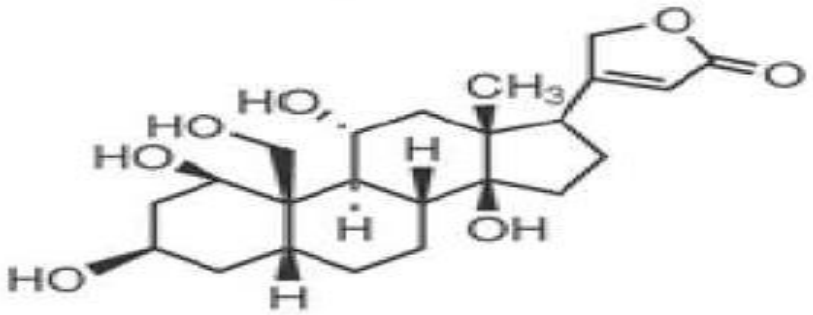
Digitoxigenin



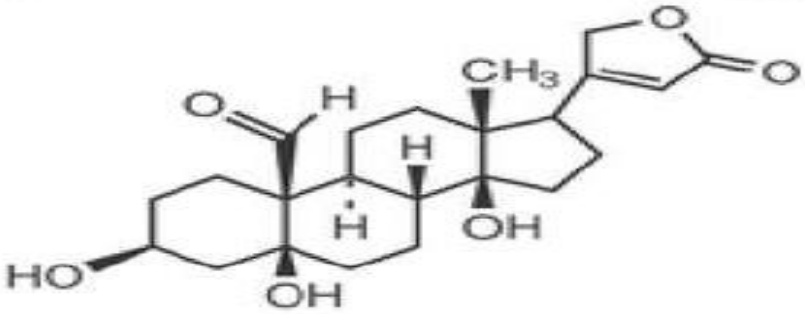
Digoxigenin



Gitoxigenin



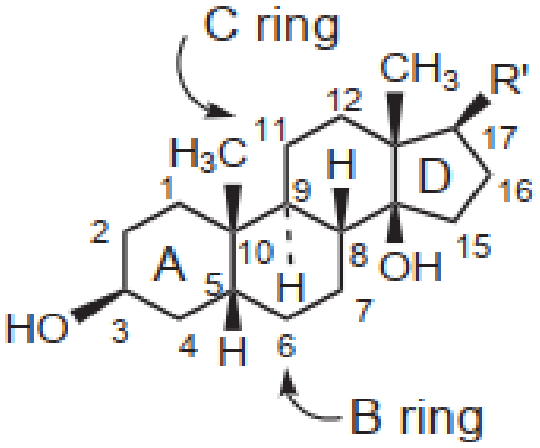
Ouabagenin



Strophanthidin

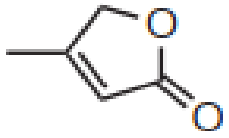
Chemistry of glycosides cont'd…

- The lactone ring at C-17 is another major structural feature of the cardiac aglycones
  - ✓ The size and degree of unsaturation of the lactone ring varies with the source of the glycoside
  - Plant source: the cardenolides, possess a 5-membered, α,β-unsaturated lactone ring,
  - Animal origin: the bufadienolides, possess a 6-membered lactone ring with two conjugated double bonds (generally referred to as α-pyrone)

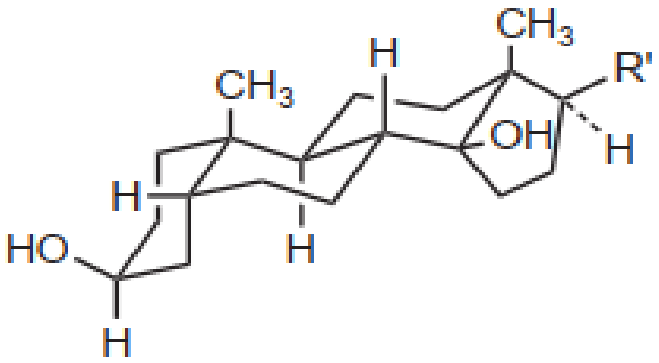


R'

Class



Cardenolide
(digitoxigenin shown)

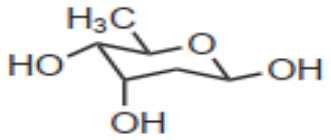




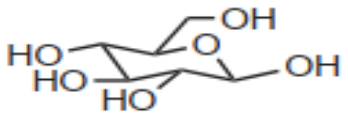
Bufadienolide
(bufalin shown)

Chemistry of glycosides cont'd…

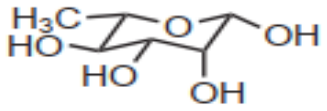
- Sugars
  - ✓ The hydroxyl group at C-3 of the aglycone portion usually is conjugated to a monosaccharide or a polysaccharide with β-1,4-glucosidic linkages
  - ✓ The number and identity of sugars vary from one glycoside to another



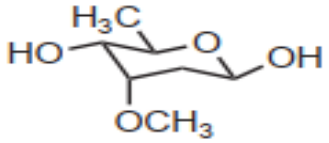
β-D-Digitoxose



β-D-Glucose



β-L-Rhamnose



β-D-Cymarose

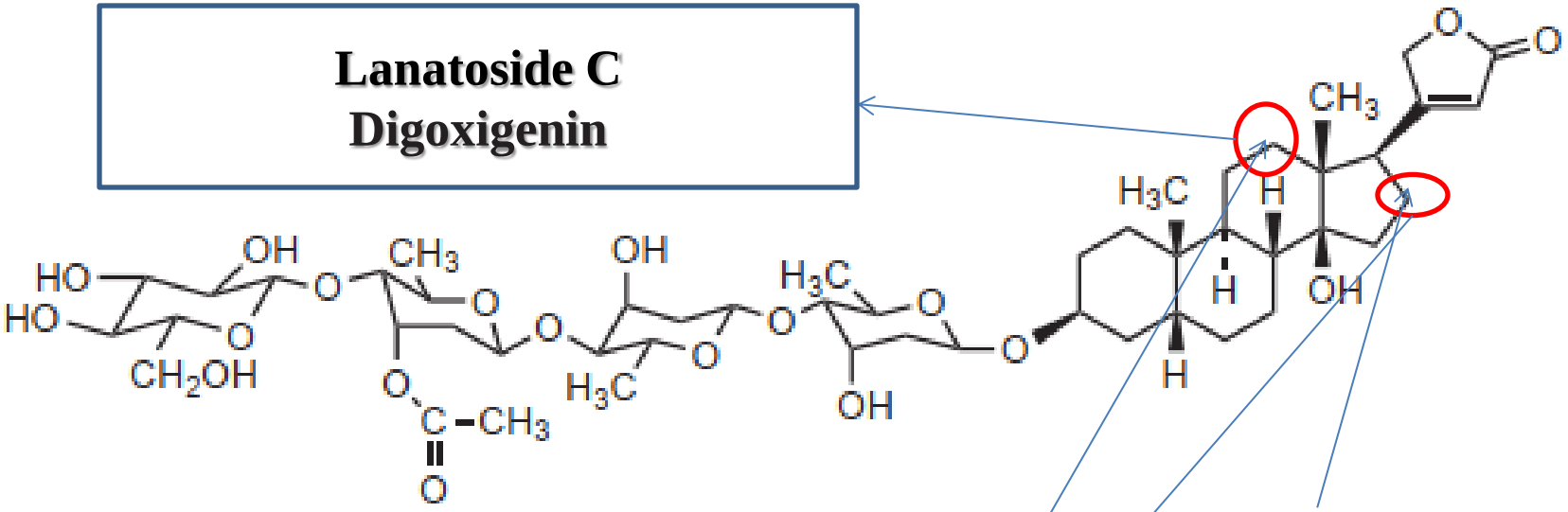
Sources and common names of cardiac glycosides

- Occur mainly in plants and in rare cases in animals,
  - *Digitalis purpurea* or the foxglove plant,
  - *Digitalis Lanata*,
  - *Strophanthus Gratus*, and
  - *Strophantus Kombe'* are the major plant sources of the cardiac glycosides.
- Based on the nature and number of sugar molecules and the number of hydroxyl groups on the aglycone moiety, each combination of sugars and aglycones assumes different generic names.

Source	Glycoside	Aglycone	Sugar[a]
Digitalis lanata (leaf)	Lanatoside A (digilanide A)	Digitoxigenin	Glucose-3-acetyldigitoxose-digitoxose-digitoxose
	Lanatoside B (digilanide B)	Gitoxigenin	
	Lanatoside C (digilanide C)	Digoxigenin	
Digitalis purpurea (leaf)	Purpurea glycoside A (desacetyl digilanide A)	Digitoxigenin	Glucose-digitoxose-digitoxose-digitoxose
	Purpurea glycoside B (desacetyl digilanide B)	Gitoxigenin	
Strophanthus gratus (seed)	G-Strophanthin	Oubagenin	Rhamnose
Strophanthus kombe (seed)	k-Strophanthoside	Strophanthidin	Glucose-glucose-cymarose

Digitalis Lanata

Lanatoside C
Digoxigenin



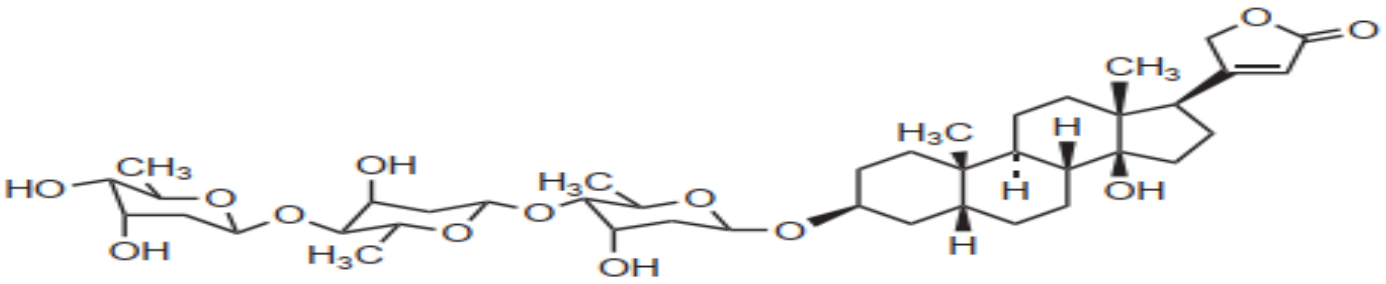
Lanatoside A

Digitoxigenin

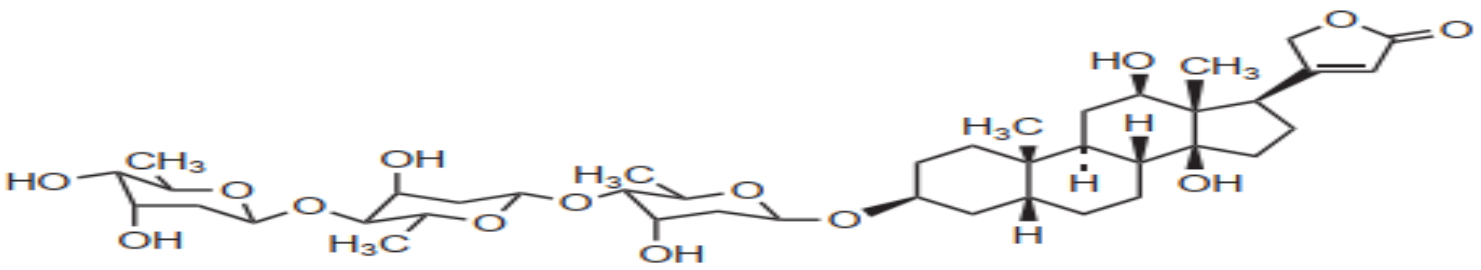
Lanatoside B
Gitoxigenin

Cont'…d

- Partial hydrolysis of the glucose molecule and the acetate group from lanatoside A and C produces, respectively, digitoxin and digoxin



Digitoxin



Digoxin

Digitalis Purpurea

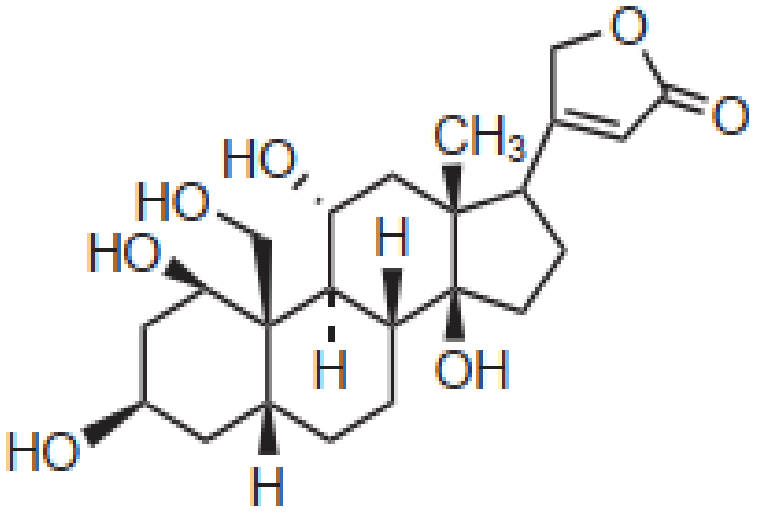
- Purpurea glycosides A and B have structures identical to those of lanatosides A and B,
  - but with no acetyl group on the third digitoxose.
- Therefore, the purpurea glycosides A and B are sometimes called desacetyl digilanides A and B.
- There is no purpurea glycoside C.

Strophanthus Gratus and Strophanthus Kombe

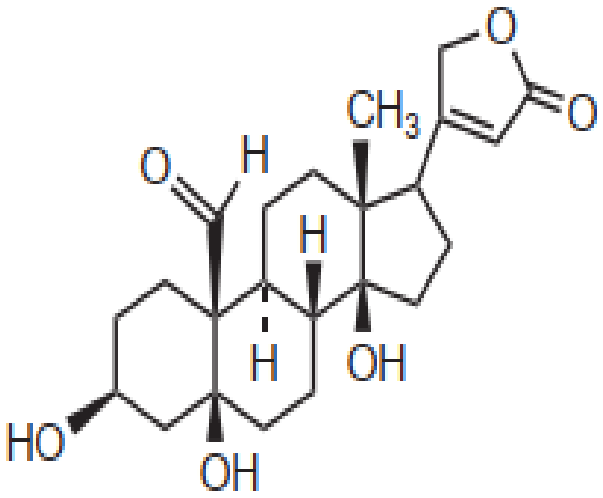
- The glycosides extracted from the plants
  - *Strophanthus gratus*… (ouabain) G-strophanthin
  - *Strophanthus kombe*…k-strophanthoside.
- The corresponding aglycone for
  - ouabain is ouabagenin,
  - k-strophanthoside is strophanthidin.

Cont'…d

- Ouabagenin has a polyhydroxylated steroidal nucleus, and
- Strophanthidin has an additional hydroxyl group at C-5 with an angular aldehyde group at C-10, replacing the traditional methyl group at that position.



Ouabagenin



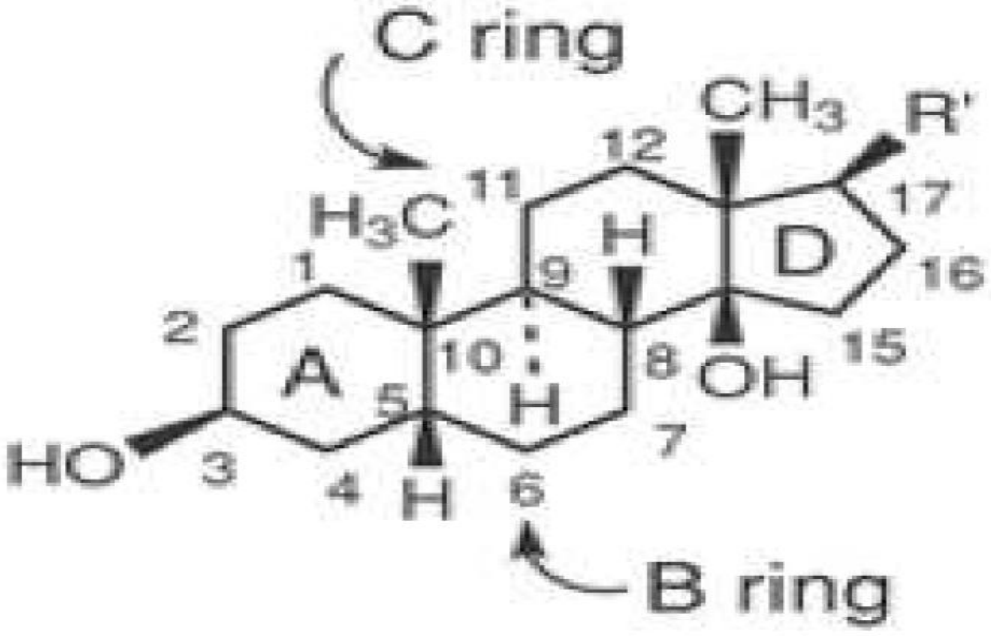
Strophanthidin

Cont'…d

- The medicinally used preparations are mainly obtained from
  - *Digitalis purpurea* and *Digitalis lanata* plants.
- These glycosides are referred to as digitalis glycosides, cardiac glycosides, or simply, cardenolides.
- *Strophanthus* glycosides (e.g., ouabain) are no longer used therapeutically but were previously administered only intravenously because of poor oral absorption.
- Cardiac glycosides from animal sources (generally referred to as bufadienolides) are rare and of far less medicinal importance because of their high toxicity.

- Pharmaceutical preparations of whole plants and partially hydrolyzed glycosides of *Digitalis lanata* and *Digitalis purpuea* have been used clinically.
- Advancements in isolation and purification techniques, however, have made it possible to obtain highly purified digoxin preparations.

SAR of cardiac glycosides



SAR of cardiac glycosides

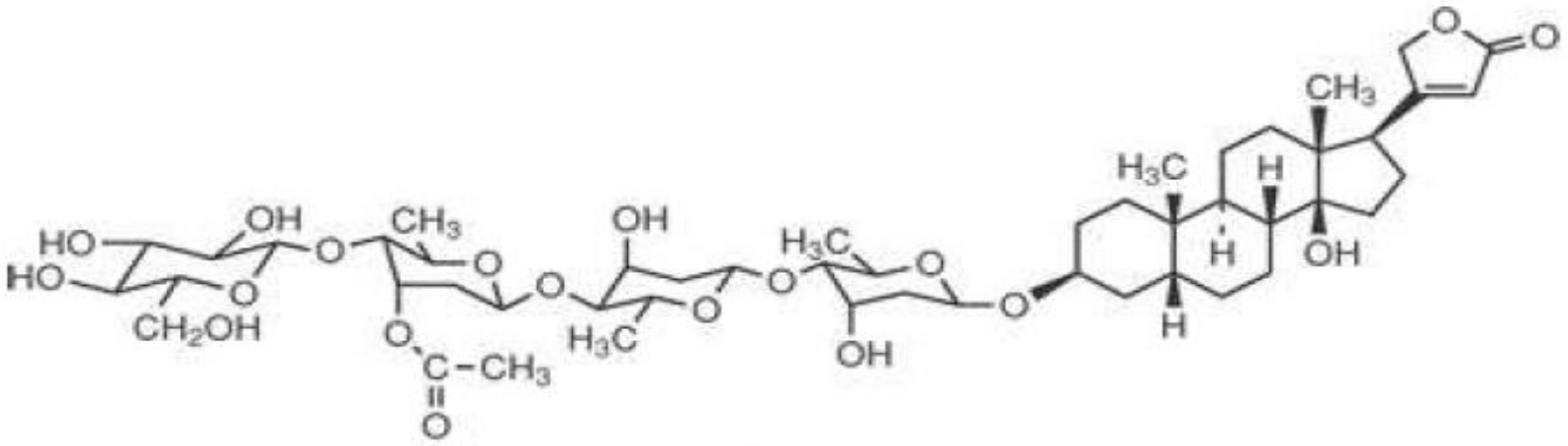
- The 17-lactone plays an important role in receptor binding.
- Replacement of the lactone with open chain structures of varying electronic and steric resemblance to the lactone showed the σ, β- unsaturated lactone ring at C-17 is an absolute requirement.

SAR cont'd…

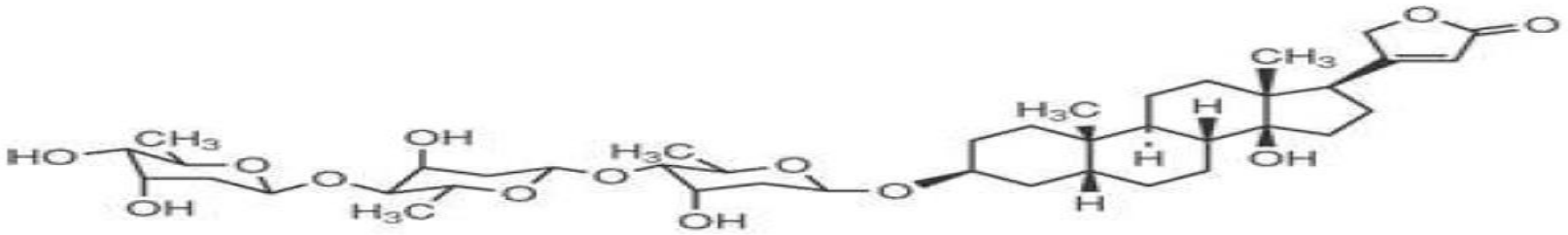
- The ring structure A-B-C-D is important
  - Lactones alone without the steroid ring are inactive
- The C-D *cis* ring juncture appears to be critical for activity
- But, the A-B *cis* ring juncture appears not to be mandatory for cardiac glycosides activity.

Absorption, metabolism, and excretion

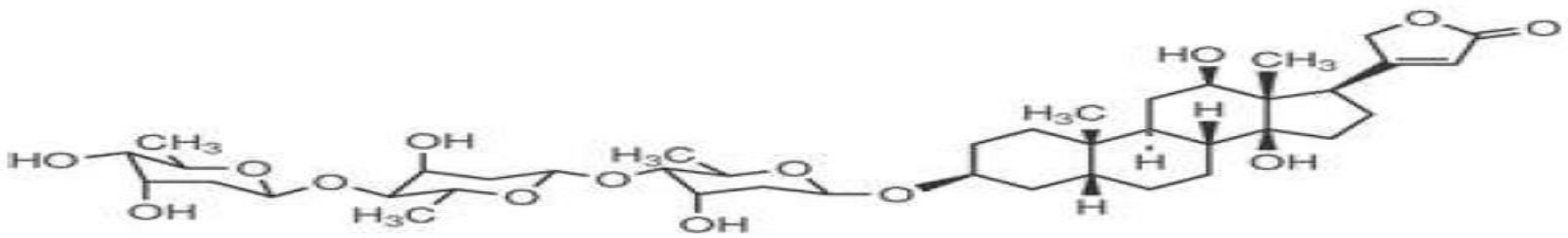
- Cardiac glycosides with more lipophilic character are absorbed faster and exhibit longer duration of action as a result of a slower urinary excretion rate.
- Lipophilicity is markedly influenced by the number of sugar molecules and the number of hydroxyl groups on the aglycone part of a given glycoside.
- Presence of the 3-O-acetyl group on acetyldigoxin enhances its lipophilic character.



Lanatoside A



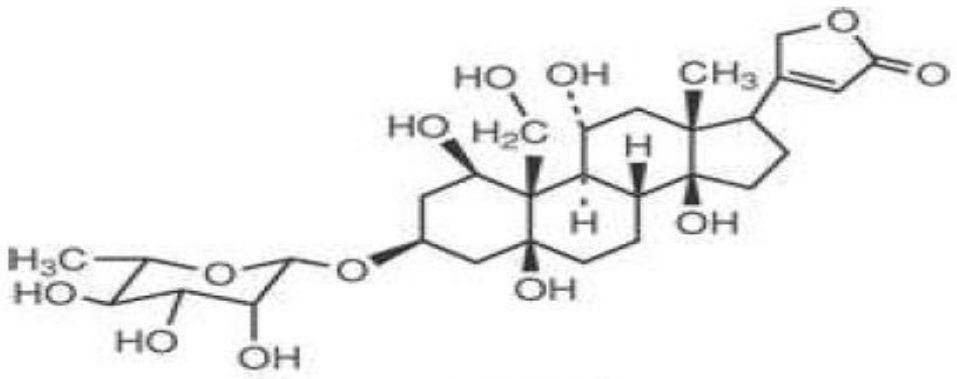
Digitoxin



Digoxin

Absorption, metabolism cont'd…

- The glycoside G-strophanthin (Ouabain) possesses a very low lipophilic character because of
  - the presence of five free hydroxyl groups on the steroid nucleus of the aglycone ouabagenin
- Digoxin is eliminated primarily unchanged by renal tubule excretion.



Ouabain

Medicinal Chemistry of Antihypertensive

Hypertension

- Hypertension is defined as either
  - a sustained systolic or diastolic blood pressure of greater than 140 mm Hg or 90 mm Hg respectively.
- Hypertension is the leading cause of cardiovascular disease and mortality.
- Disease symptoms and organ damage caused by hypertension are not evident
  - until 10–15 years after the disease has started.

Causes of Hypertension

- Most people have essential hypertension where the exact cause is not known.
- Increased sympathetic activity and sodium overload increase blood pressure (BP).
- Renal disease and increased renin-angiotensin-aldosterone activity raise BP and cause sodium and fluid retention.

Cont'd…

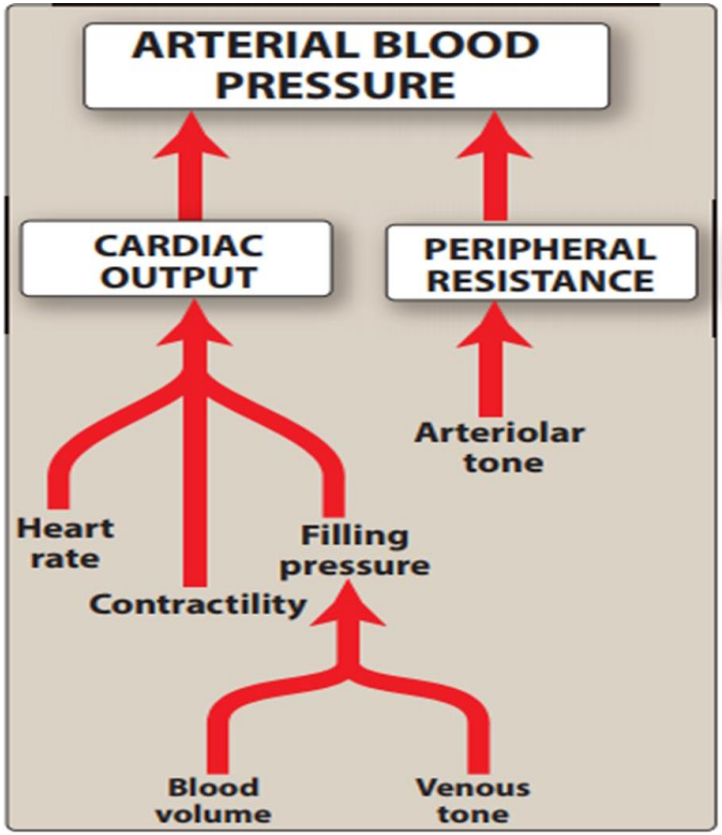
- Persons with diabetes, obesity, or disability status are all more likely to have hypertension than those without.
- In addition, environmental factors, such as a stressful lifestyle, high dietary intake of sodium, and smoking, may further predispose an individual to hypertension.
- Proper medication and patient compliance will control most cases of hypertension.

Cont'd…

- Hypertension is classified into four categories for the purpose of treatment management.

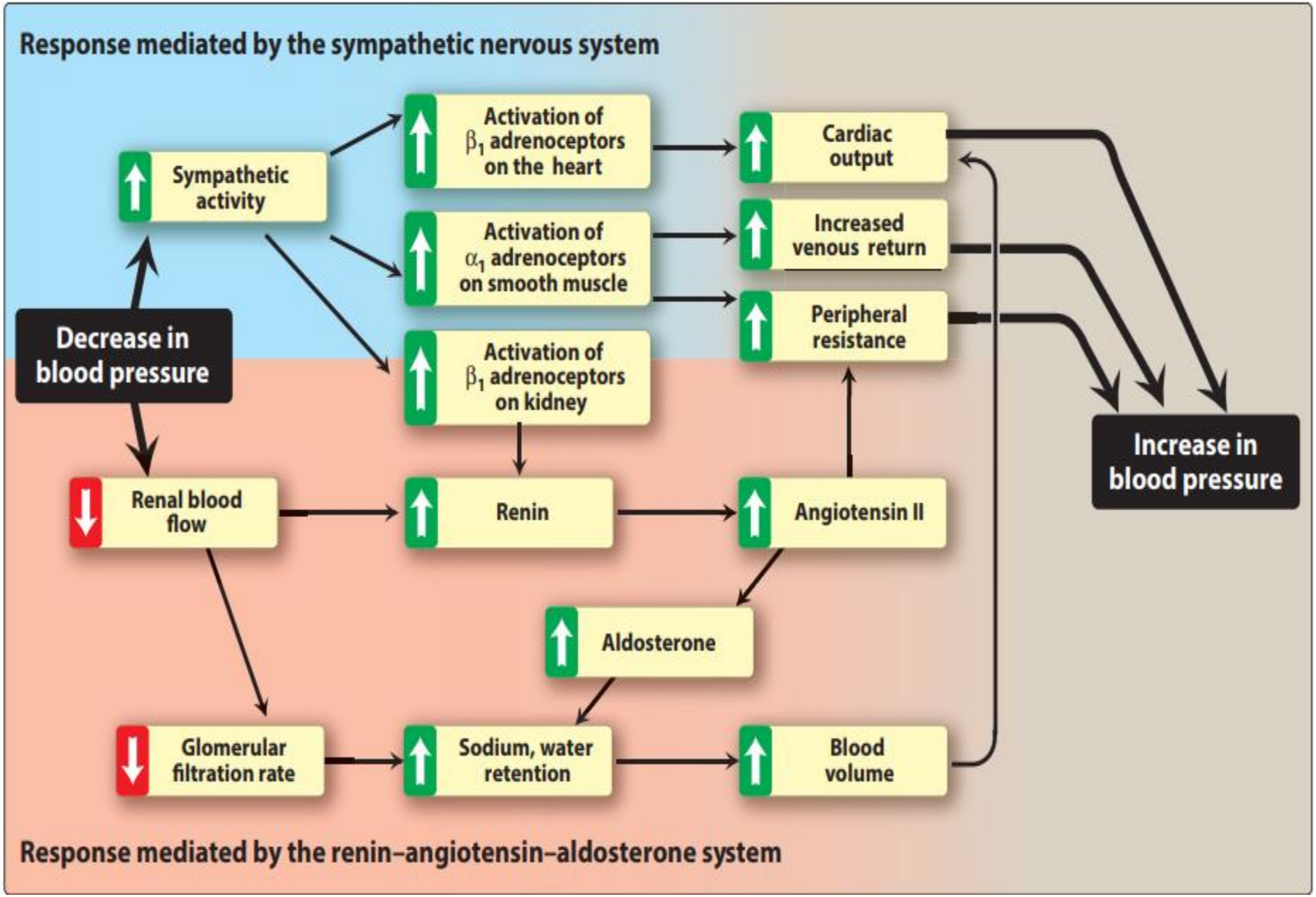
	Systolic mm Hg		Diastolic mm Hg
Normal	<120	and	<80
Prehyper-tension	120–139	or	80–89
Stage I	140–159	or	90–99
Stage II	≥160	or	≥100

Mechanism of controlling blood pressure



Cont’d…

- Cardiac output and peripheral resistance, in turn, are controlled mainly by two overlapping control mechanisms:
  - the baroreflexes and
  - the renin–angiotensin–aldosterone system

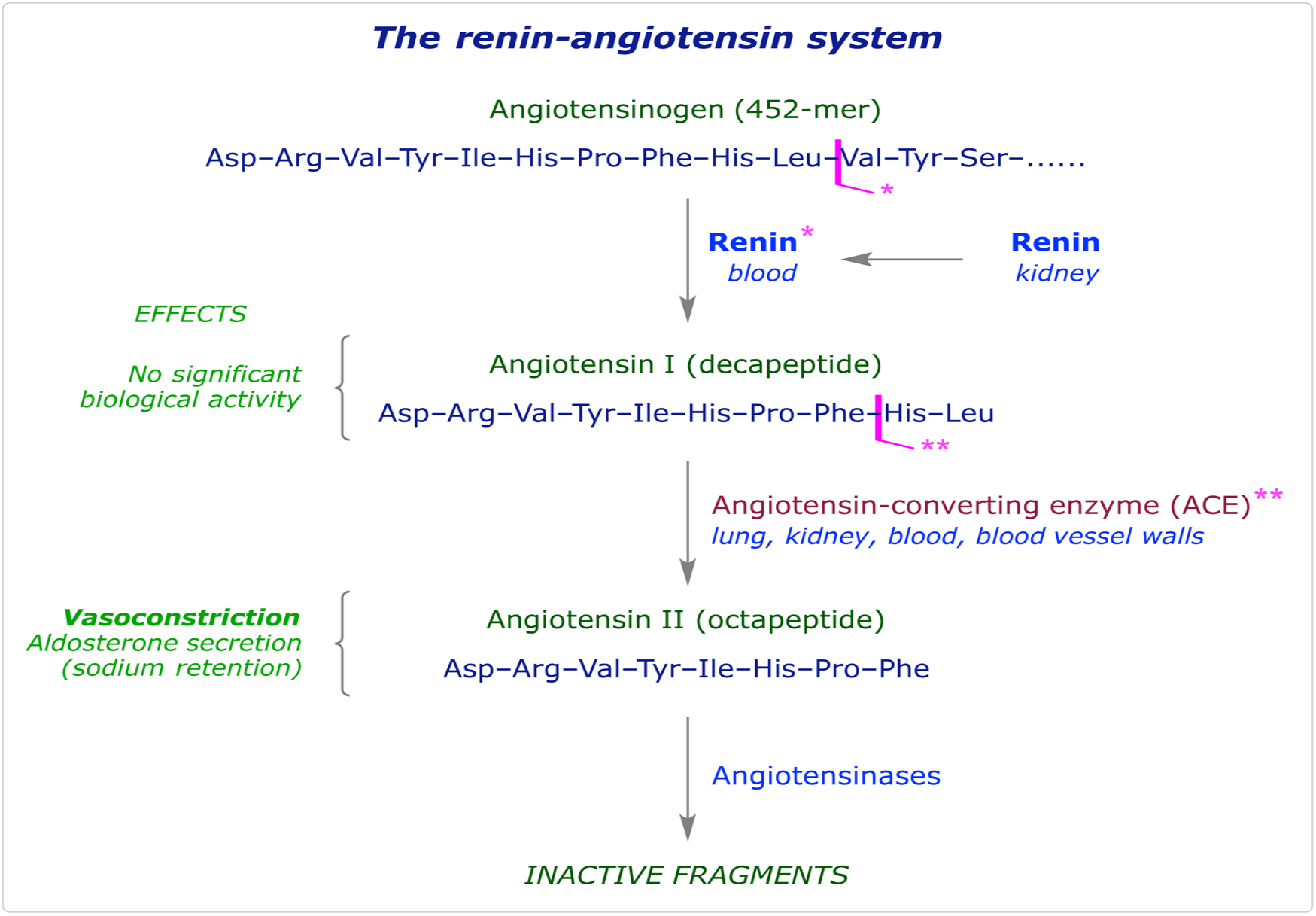


Response of the autonomic nervous system and the renin–angiotensin–aldosterone system to a decrease in blood pressure

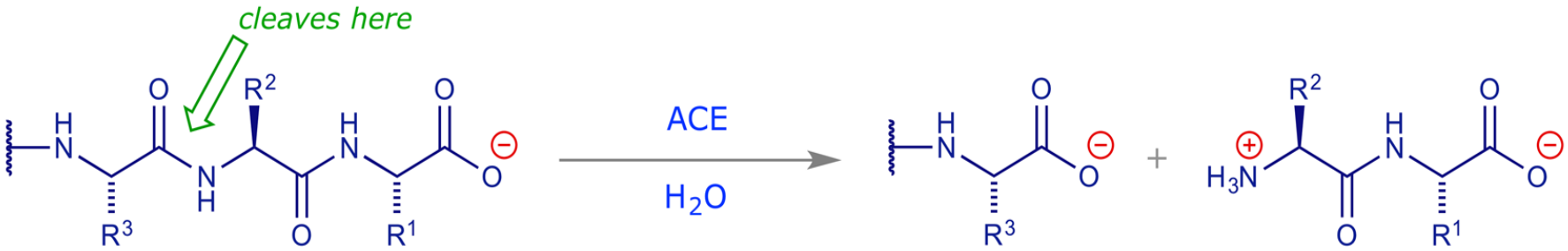
Drug Classes Used to Treat Hypertension

- Angiotensin-converting enzyme inhibitor
- Angiotensin receptor blocking drugs
- Calcium antagonist drugs
- Sympatholytic drugs and Vasodilator drugs (see ANS, antianginal)
- Diuretics (Check previous lecture)

Angiotensin-converting enzyme inhibitor

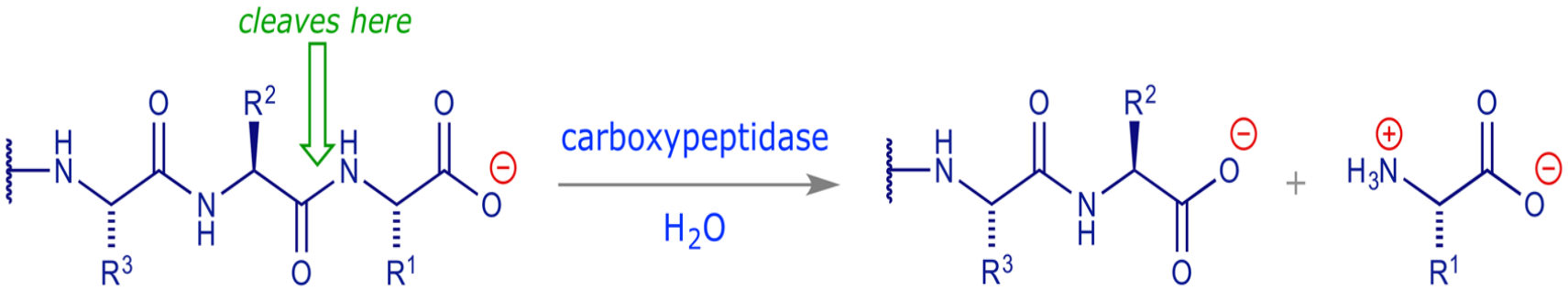


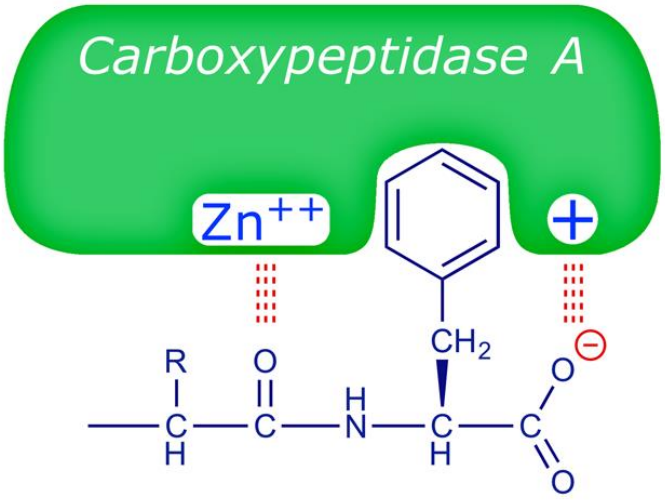
- its main role in the body is to break down *angiotensin I* (decapeptide)
  - into *angiotensin II* (octapeptide).
- It does this by cleaving the second amide bond in from the carboxyl terminus.



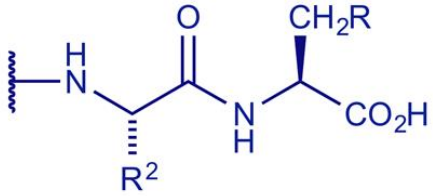
The development of the ACE inhibitors

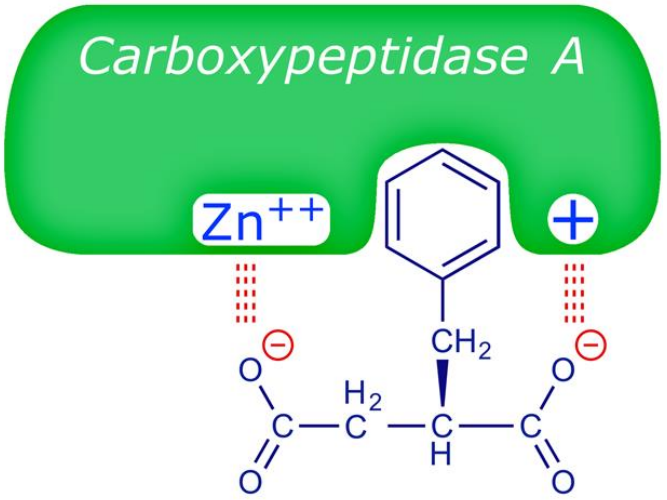
- A hypothetical model of the active site of ACE based on comparisons with zinc metalloproteinase,
  - carboxypeptidase A.



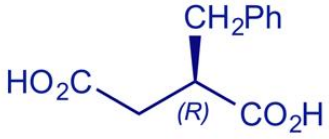


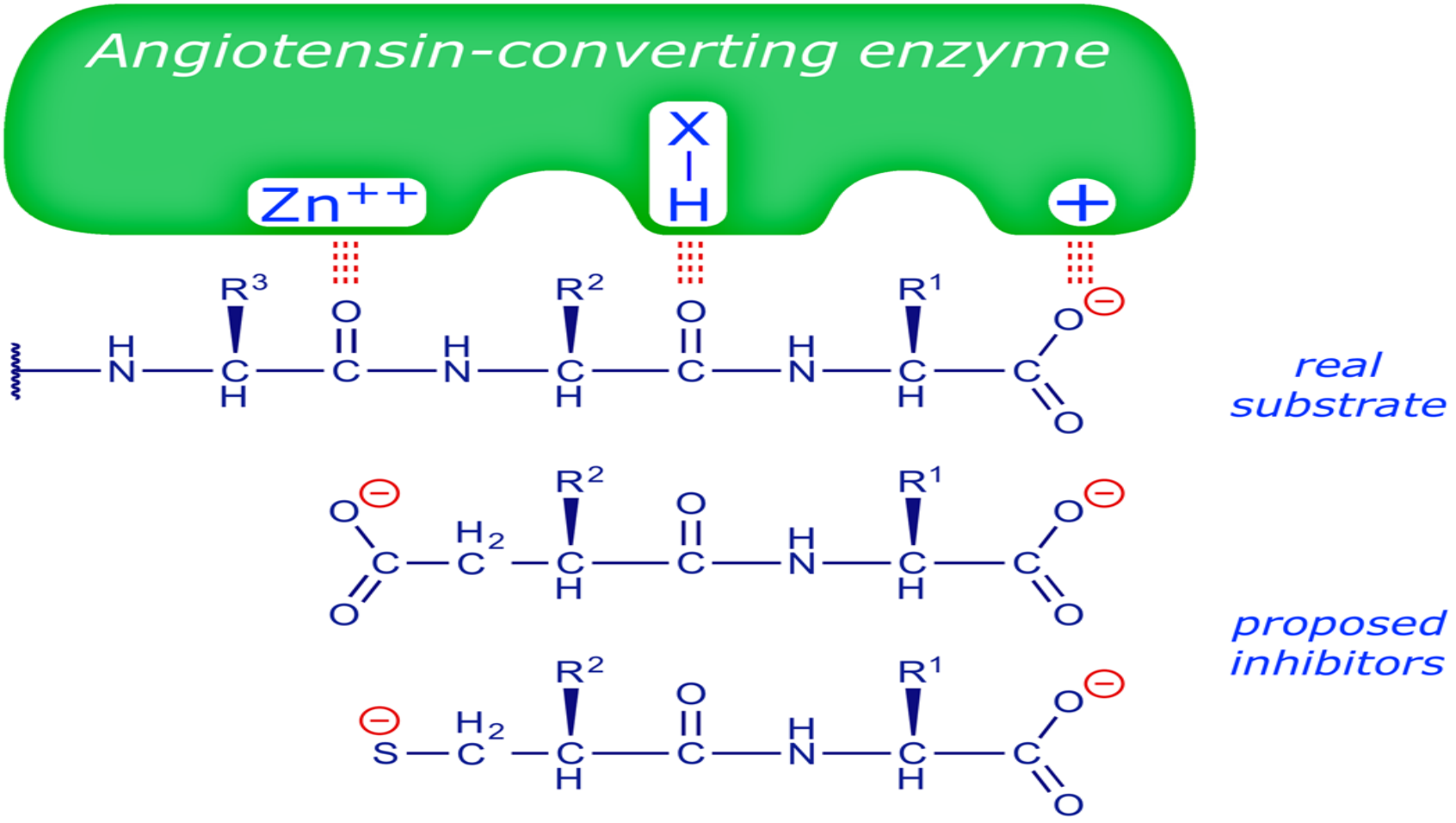
real substrate

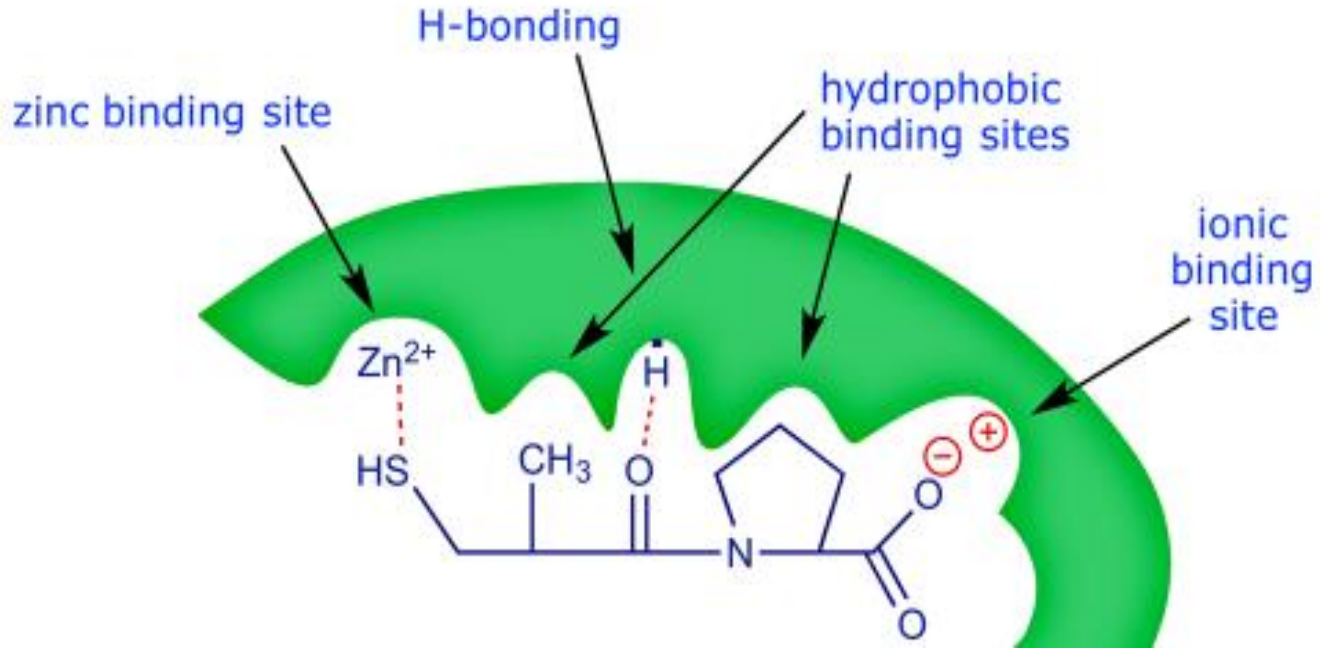




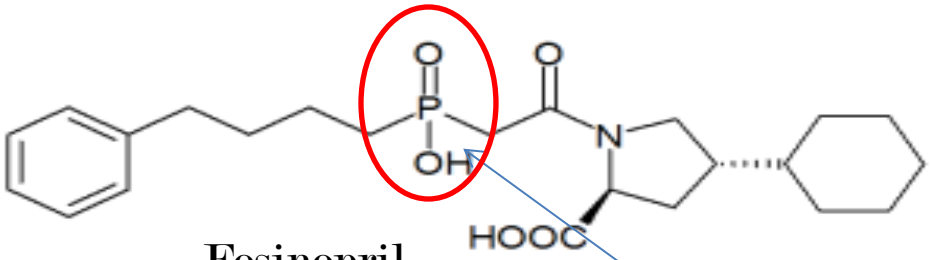
known inhibitor




Angiotensin-converting enzyme
Zn++
X
H
+
real substrate
proposed inhibitors

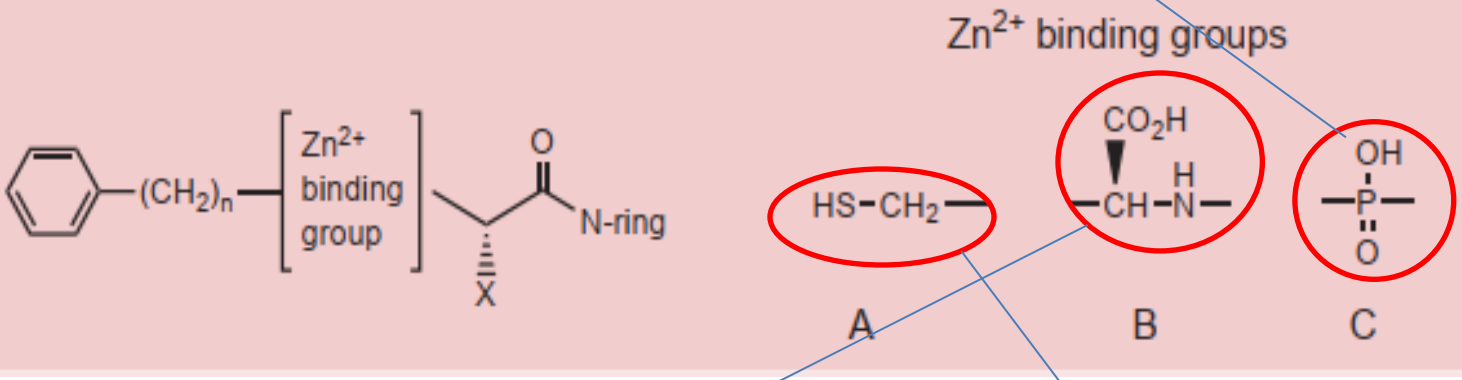

H-bonding
zinc binding site
hydrophobic
binding sites
ionic
binding
site
Zn2+
H
HS
CH3
O
N
O
O

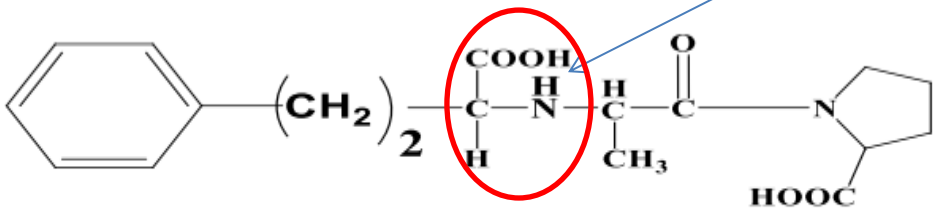
- Sub classified into three groups based on their chemical composition:
  - sulfhydryl-containing inhibitors (captopril),
  - dicarboxylate-containing inhibitors (enalapril),
  - phosphonate-containing inhibitors (fosinopril)



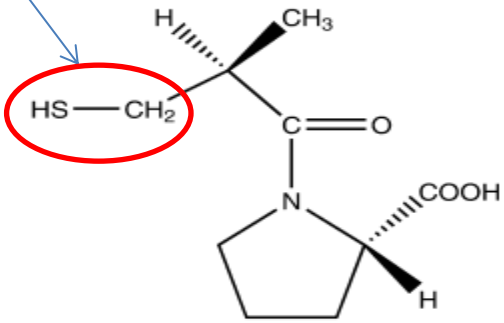
Fosinopril

TABLE 23.2 Structure–Activity Relationship of ACE Inhibitors

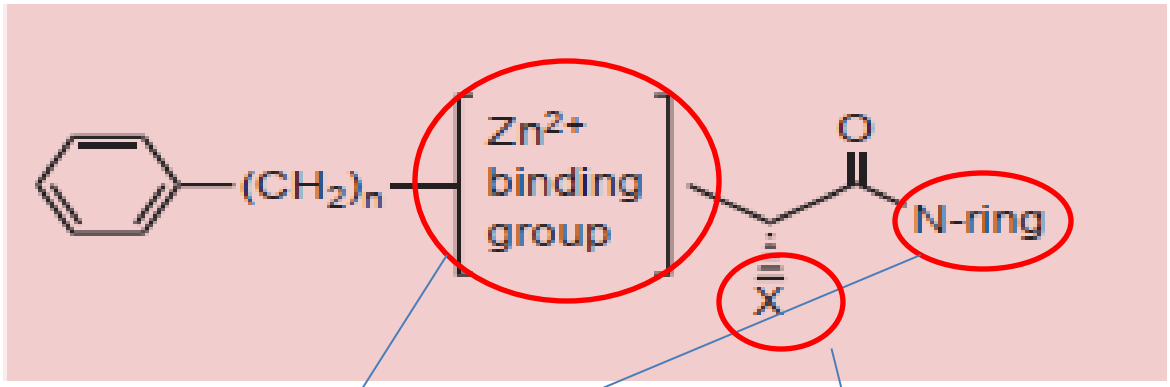




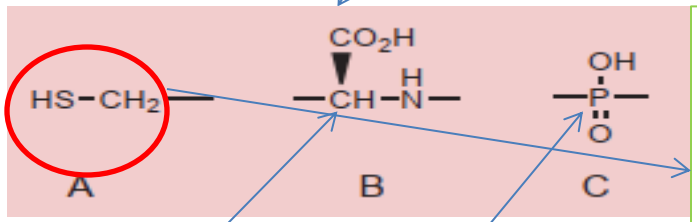
Enalapril



Captopril



The N-ring must contain a carboxylic acid to mimic the C-terminal carboxylate of ACE substrates.
Large hydrophobic heterocyclic rings (i.e., the N-ring) increase potency and alter pharmacokinetic parameters.



The sulfhydryl group shows superior binding to zinc.
Sulfhydryl-containing compounds produce high incidence of skin rash and taste disturbances.
Sulfhydryl-containing compounds can form dimers and disulfides, which may shorten duration of action

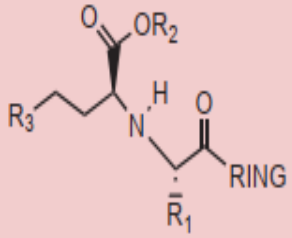
Compounds that bind to zinc through either a carboxylate or phosphinate mimic the peptide hydrolysis transition state and enhance binding.
Esterification of the carboxylate or phosphinate produces an orally bioavailable prodrug.

X is usually methyl

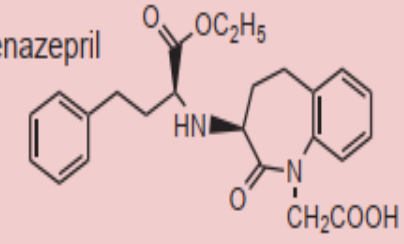
Specific drugs

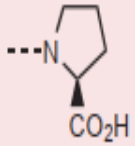
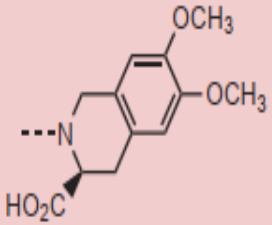
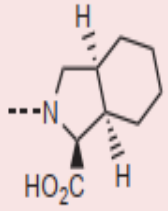
TABLE 23.1 Additional Dicarboxylate-Containing Angiotensin-Converting Enzyme Inhibitors

General structure:

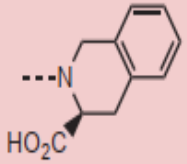
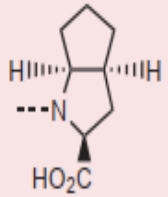
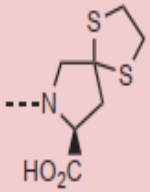
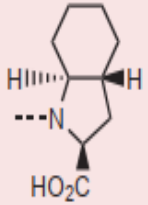


Benazepril



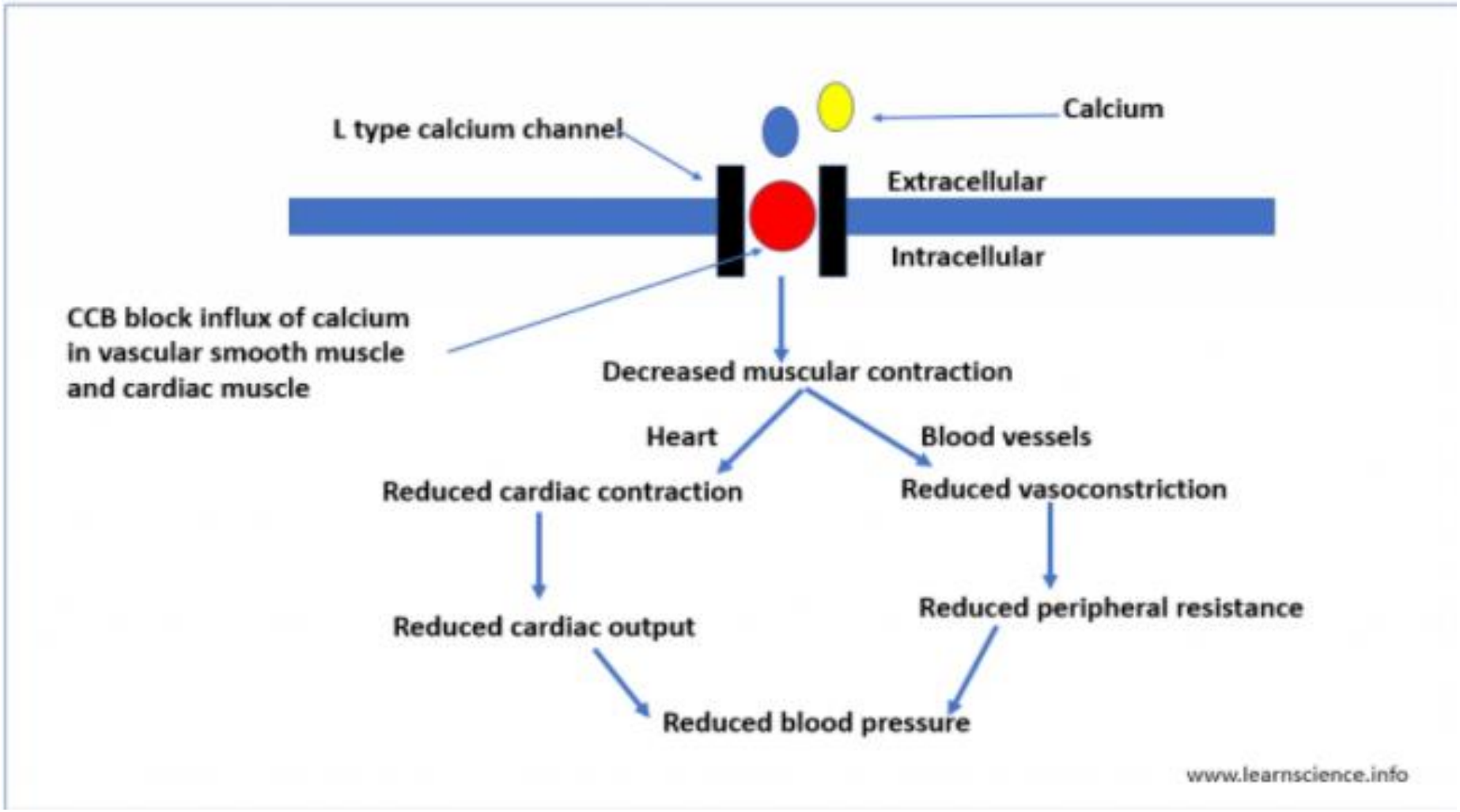
Compounds	Ring	R_1	R_2	R_3
Lisinopril		$(CH_2)_4NH_2$	H	
Moexipril		CH_3	CH_2CH_3	
Perindopril		CH_3	CH_2CH_3	CH_3

Cont'd…

Quinapril		CH_3	CH_2CH_3	
Ramipril		CH_3	CH_2CH_3	
Spirapril		CH_3	CH_2CH_3	
Trandolapril		CH_3	CH_2CH_3	

Kaduse.com

Calcium channel blockers



Cont'd…

Diphenylalkylamines

- *Verapamil* is the least selective of any calcium channel blocker and has significant effects on both cardiac and vascular smooth muscle cells

Benzothiazepines

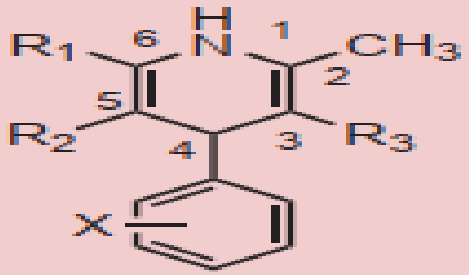
- Like *verapamil*, *diltiazem* affects both cardiac and vascular smooth muscle cells. *Diltiazem* has a favorable side effect profile.

Dihydropyridines

- *nifedipine* (the prototype), *amlodipine*, *felodipine*, *isradipine*, *nicardipine*, and *nisoldipine*
- All dihydropyridines have a much greater affinity for vascular calcium channels than for calcium channels in the heart.
  - They are, therefore, particularly beneficial in treating hypertension

Structure–activity Relationships 1,4-DHP

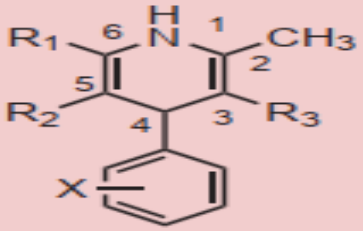
General structure:



- A substituted phenyl ring at the C4 position optimizes activity.
- Compounds with ortho or meta substitutions possess optimal activity.
- The 1,4-DHP ring is essential for activity.
- Substitution at the N1 position
- Ester groups at the C3 and C5 positions optimize activity

Structure of the Dihydropyridine Ca2+ Channel Blockers

General structure:



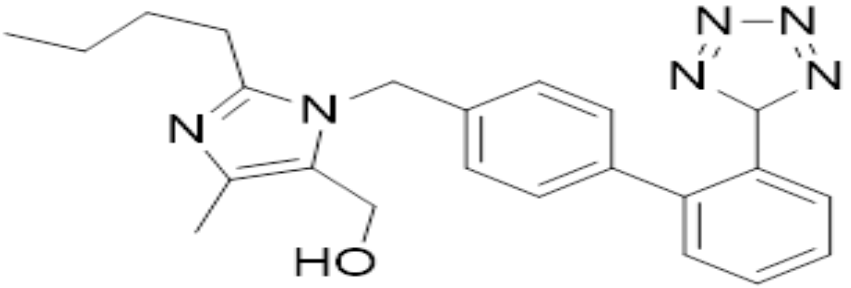
Isradipine:



Compounds	R_1	R_2	R_3	X
Amlodipine	$CH_2O(CH_2)_2NH_2$	$CO_2CH_2CH_3$	CO_2CH_3	2-Cl
Clevidipine	CH_3	$CO_2CH_2O-\overset{O}{\overset{\|}{C}}-nC_3H_7$	CO_2CH_3	2,3-Cl_2
Felodipine	CH_3	$CO_2CH_2CH_3$	CO_2CH_3	2,3-Cl_2
Nicardipine	CH_3	$CO_2(CH_2)_2-N(CH_3)-H_2C-C_6H_5$	CO_2CH_3	3-NO_2
Nifedipine	CH_3	CO_2CH_3	CO_2CH_3	2-NO_2
Nimodipine	CH_3	$CO_2CH_2CH_2OCH_3$	$CO_2CH(CH_3)_2$	3-NO_2
Nisoldipine	CH_3	$CO_2CH_2CH(CH_3)_2$	CO_2CH_3	2-NO_2

Angiotensin receptor blocking drugs

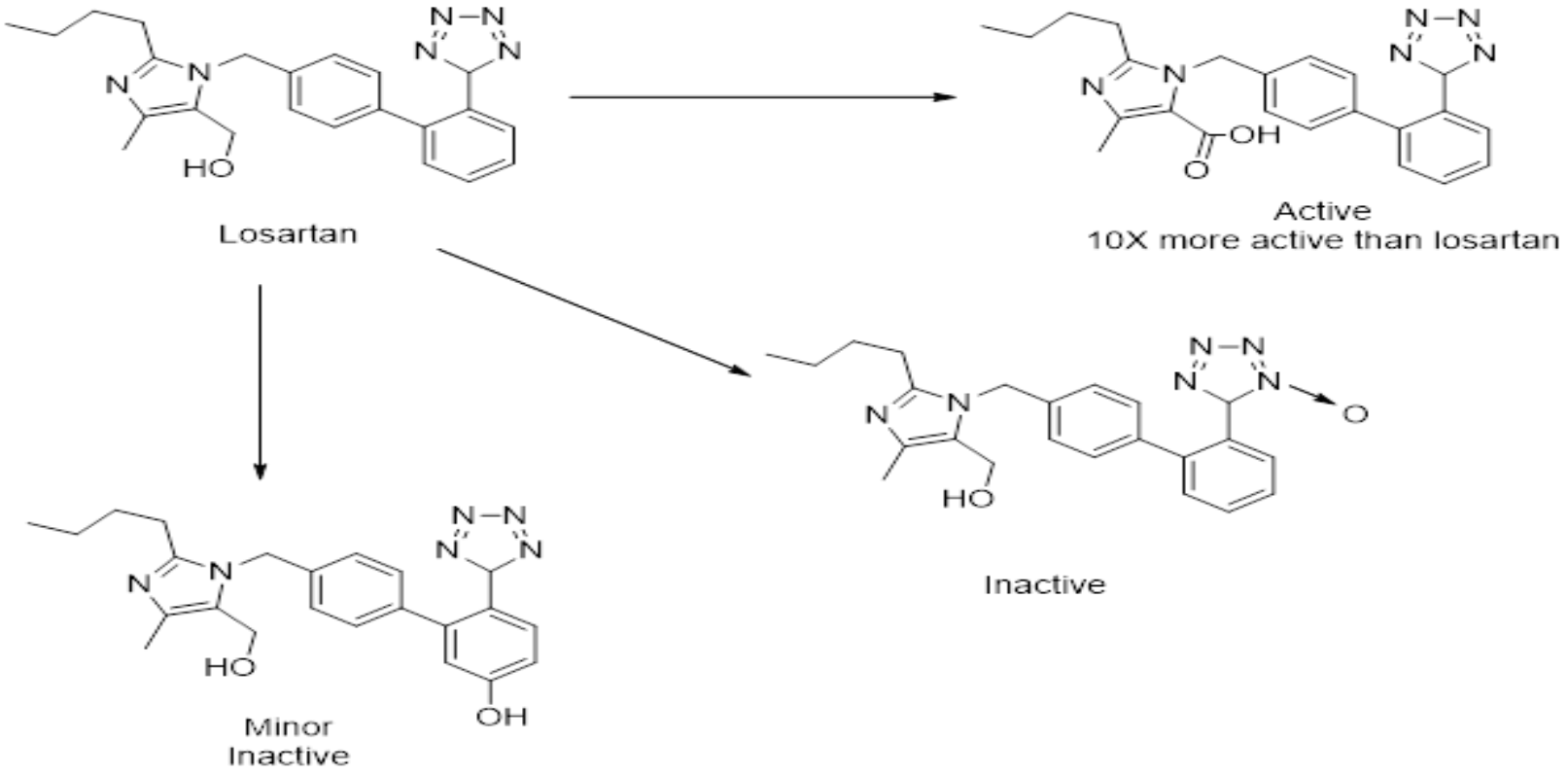
- Novel mechanism of action is targeting angiotensin II receptors…as blockers;
  - they will prevent the binding of angiotensin II to its receptor.
- The prototype is losartan



Losartan

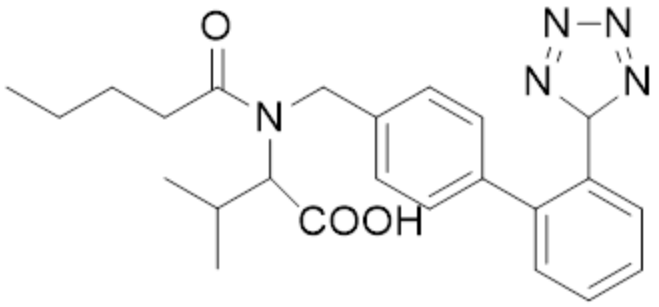
Cont'd…

It is metabolized into an active metabolite; the carboxylic acid metabolite:



Cont'd…

- Valsartan is a new anti-hypertensive agent with the same mechanism as losartan.
- The difference in structure between losartan and valsartan is:



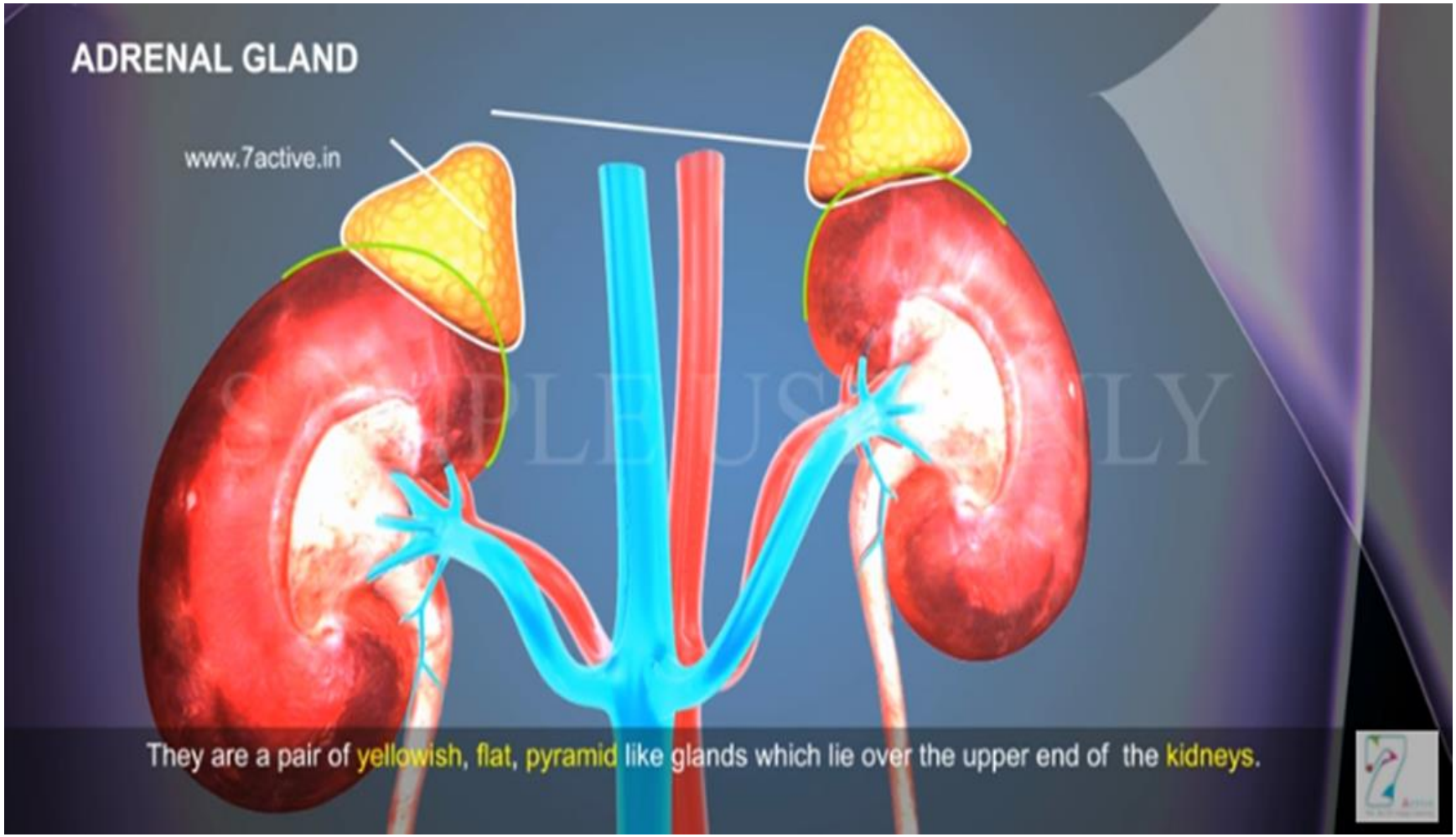
  - Valsartan is more polar…reduces its oral absorption

Group assignment

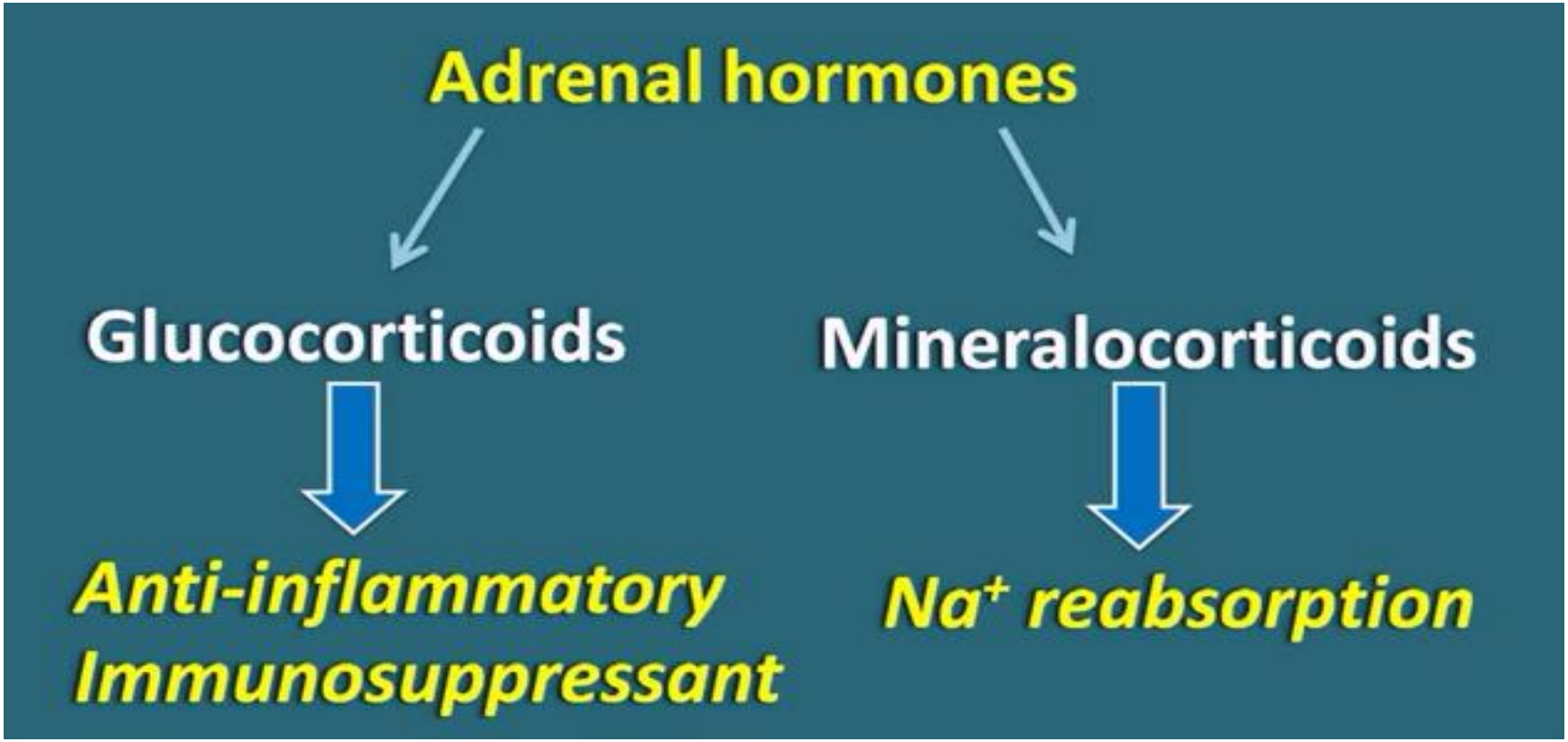
- Antianginal agents…..G1
- Antiarrhythmic drugs…G2
- Antilipemic drugs….G3
- Coagulants & anticoagulants…G4
- Non-steroidal hormones and related drugs…G5
- Local and general anesthetics…..G6

Medicinal Chemistry of Steroidal hormones and related drugs

Introduction

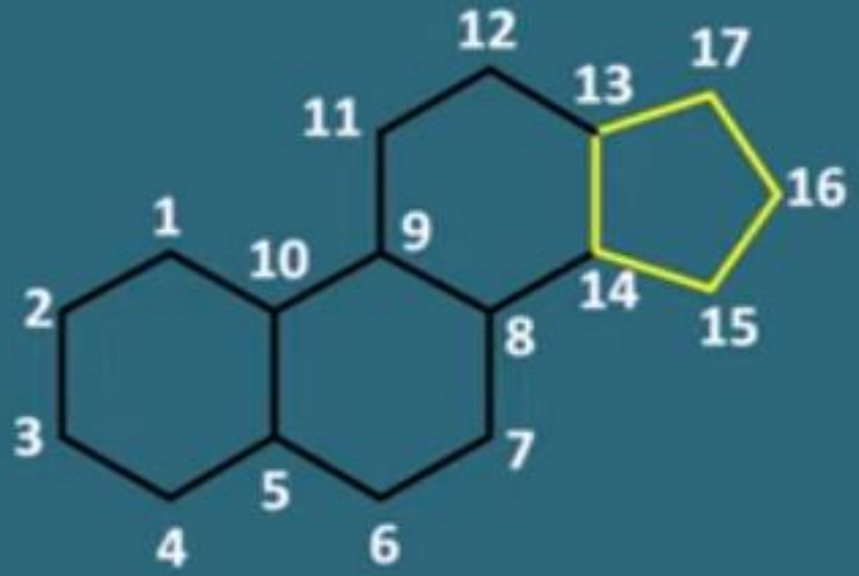


Adrenocorticoids



Cont'd...

Steroidal nucleus

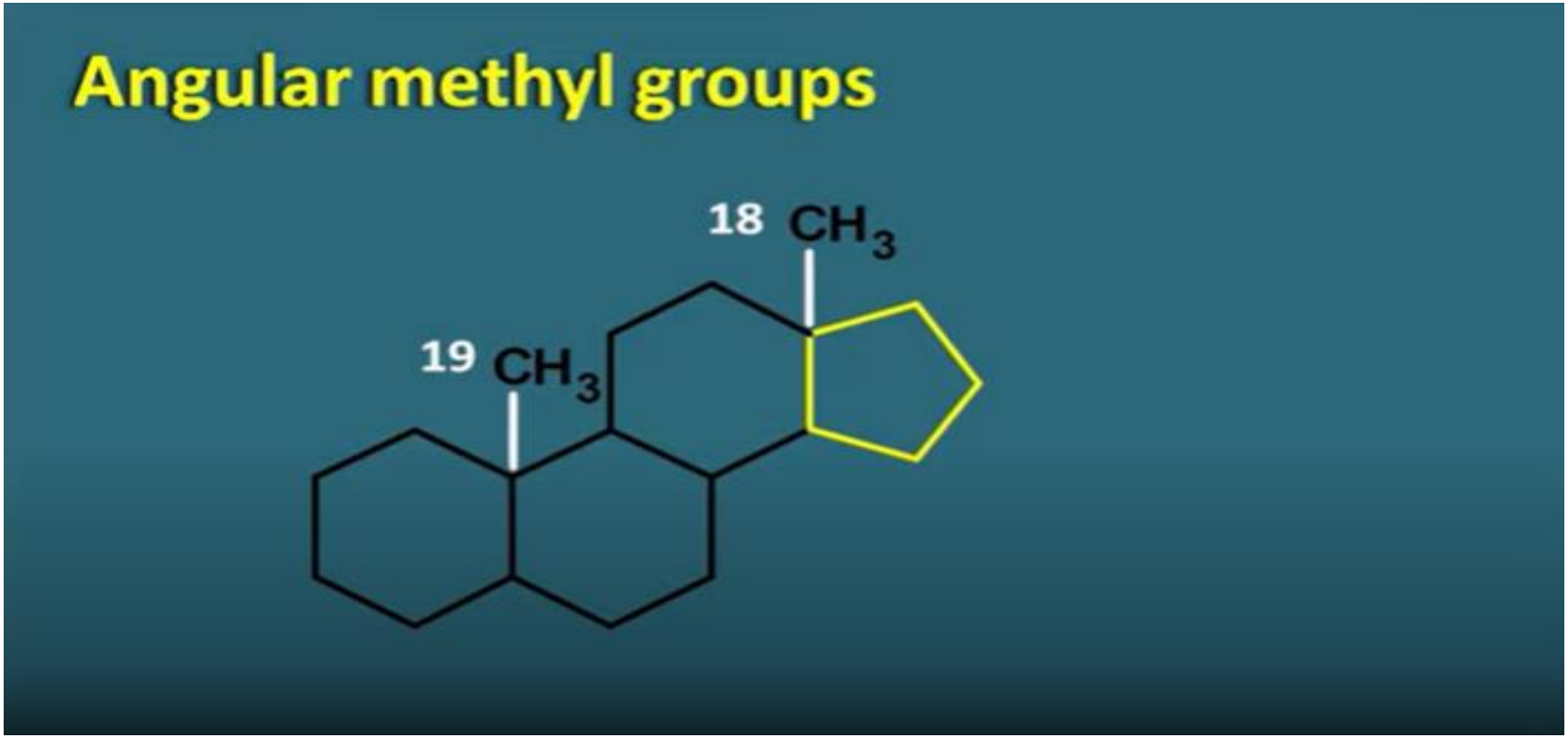


Cyclopenatano-

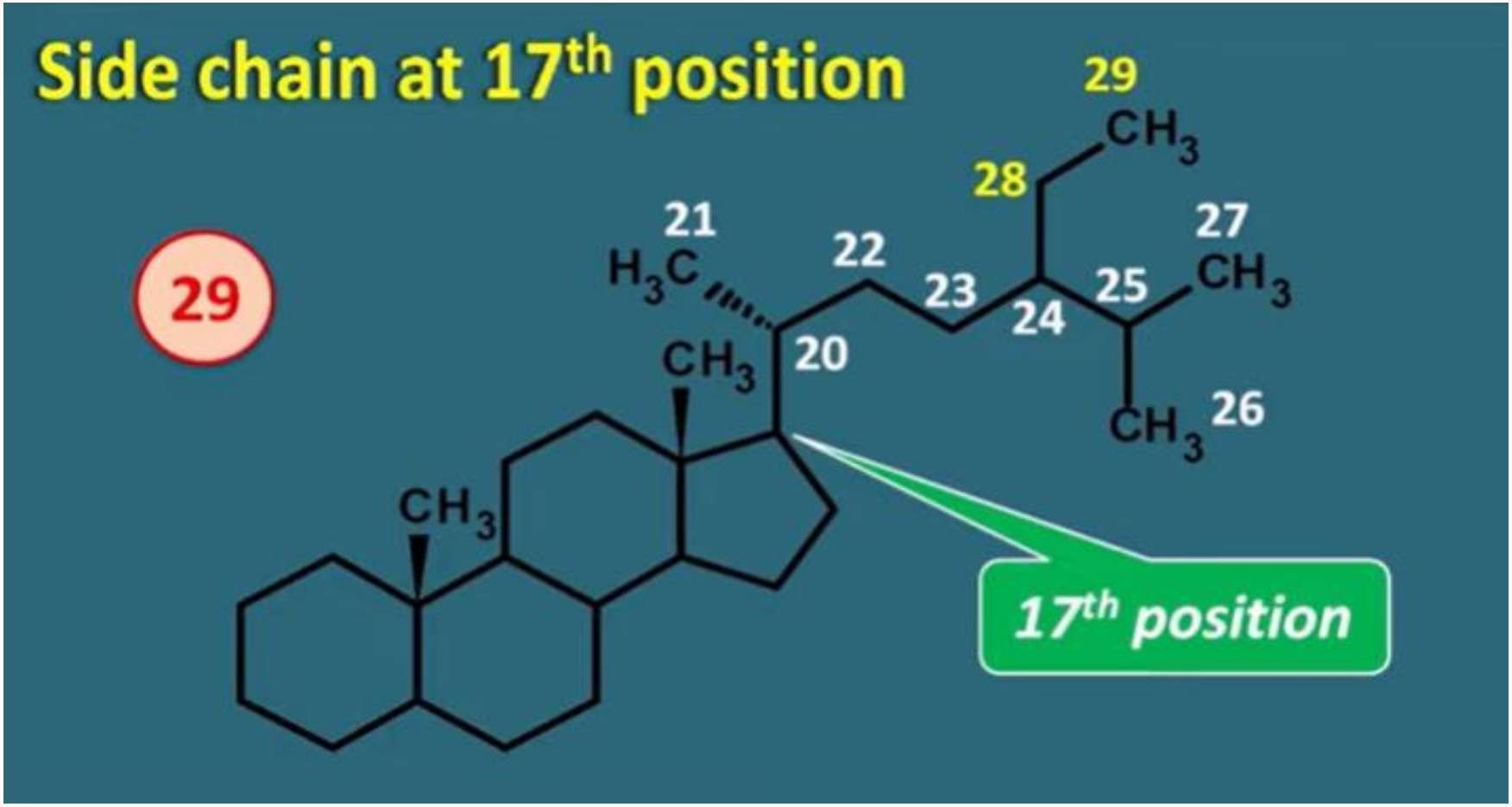
-perhydro-

phenanthrene

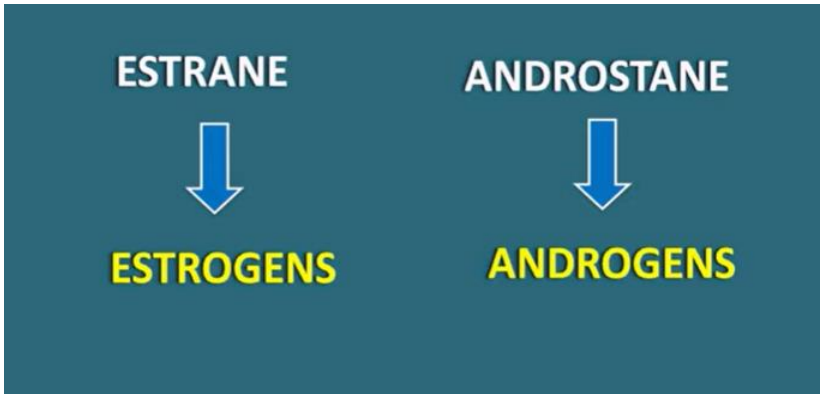
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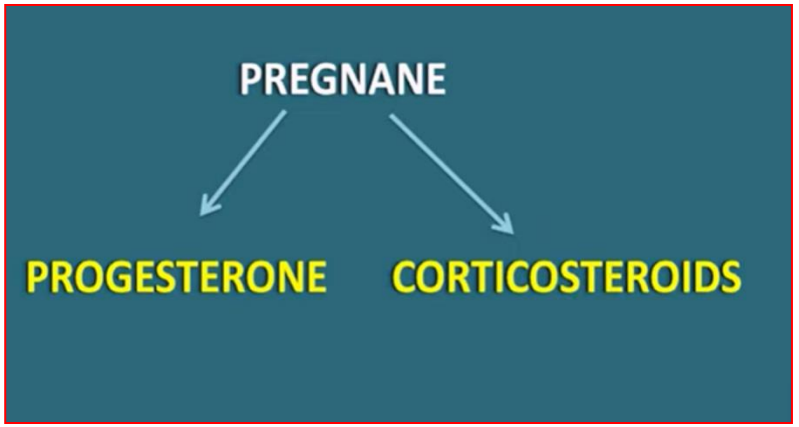


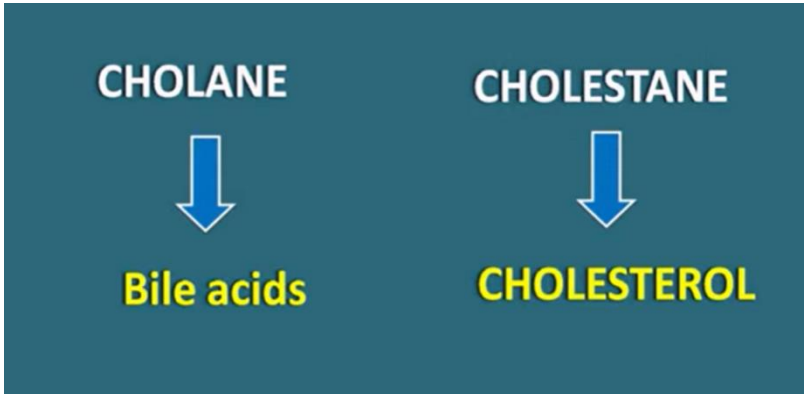
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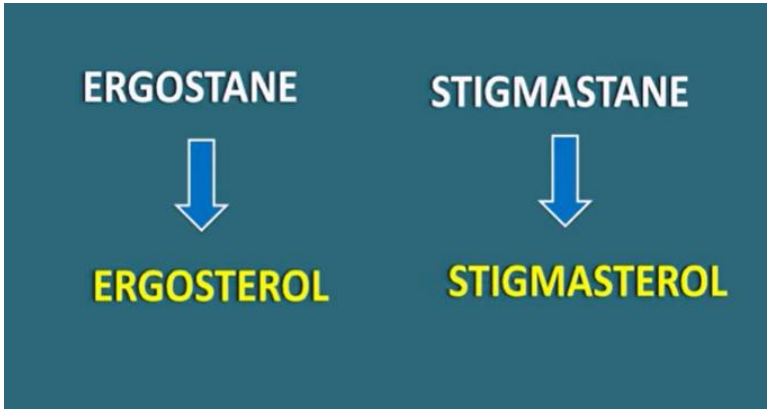

Side chain at 17th position
29
CH3
28
21
H3C
22
27
CH3
29
23
25
24
CH3
20
CH3
26
CH3
CH3
17th position

No. of carbons	Name of ring
17	GONANE
18	ESTRANE
19	ANDROSTANE
21	PREGNANE
24	CHOLANE
27	CHOLESTANE
28	ERGOSTANE
29	STIGMASTANE

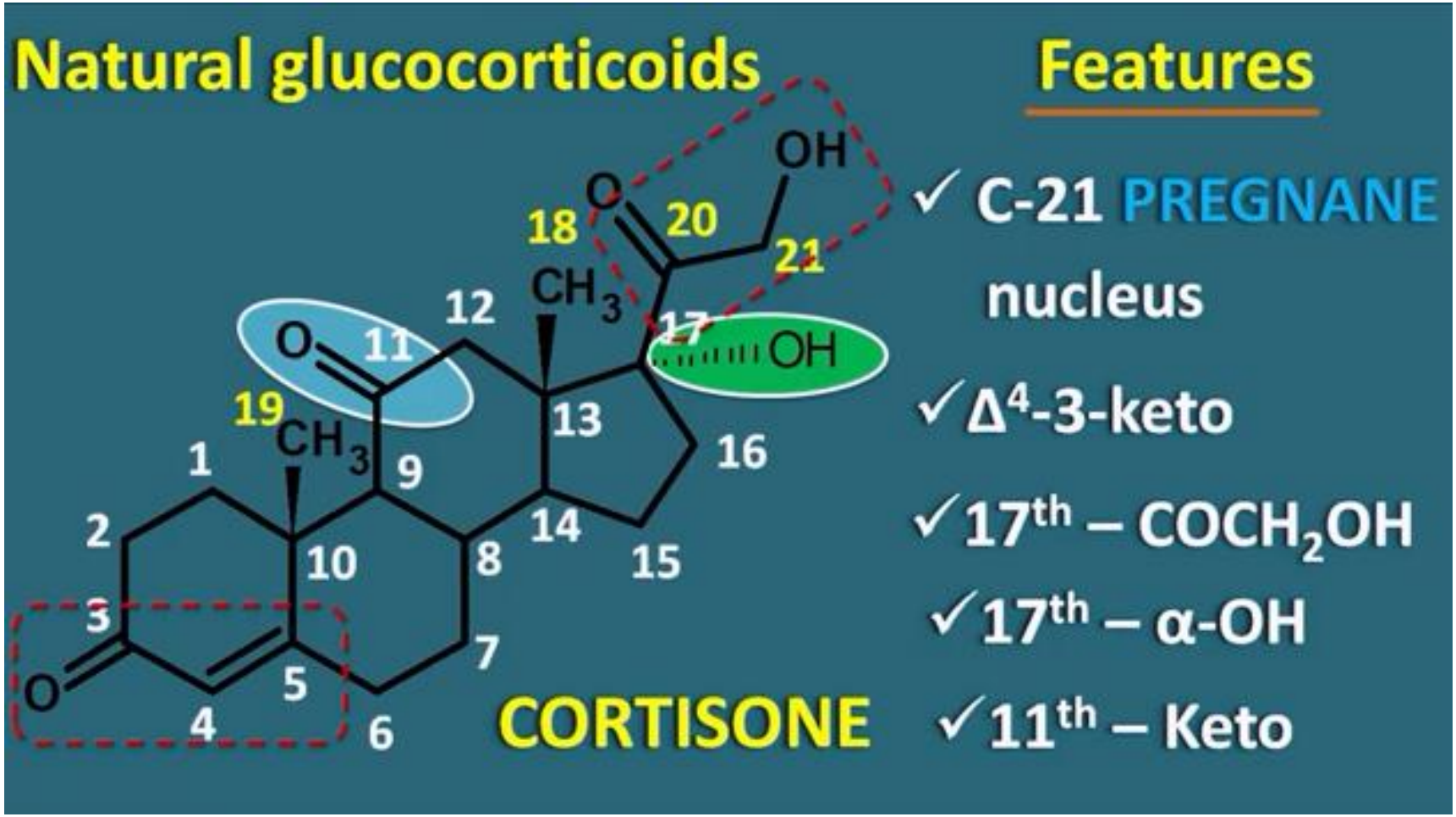

ESTRANE
ESTROGENS
ANDROSTANE
ANDROGENS


PREGNANE
PROGESTERONE
CORTICOSTEROIDS

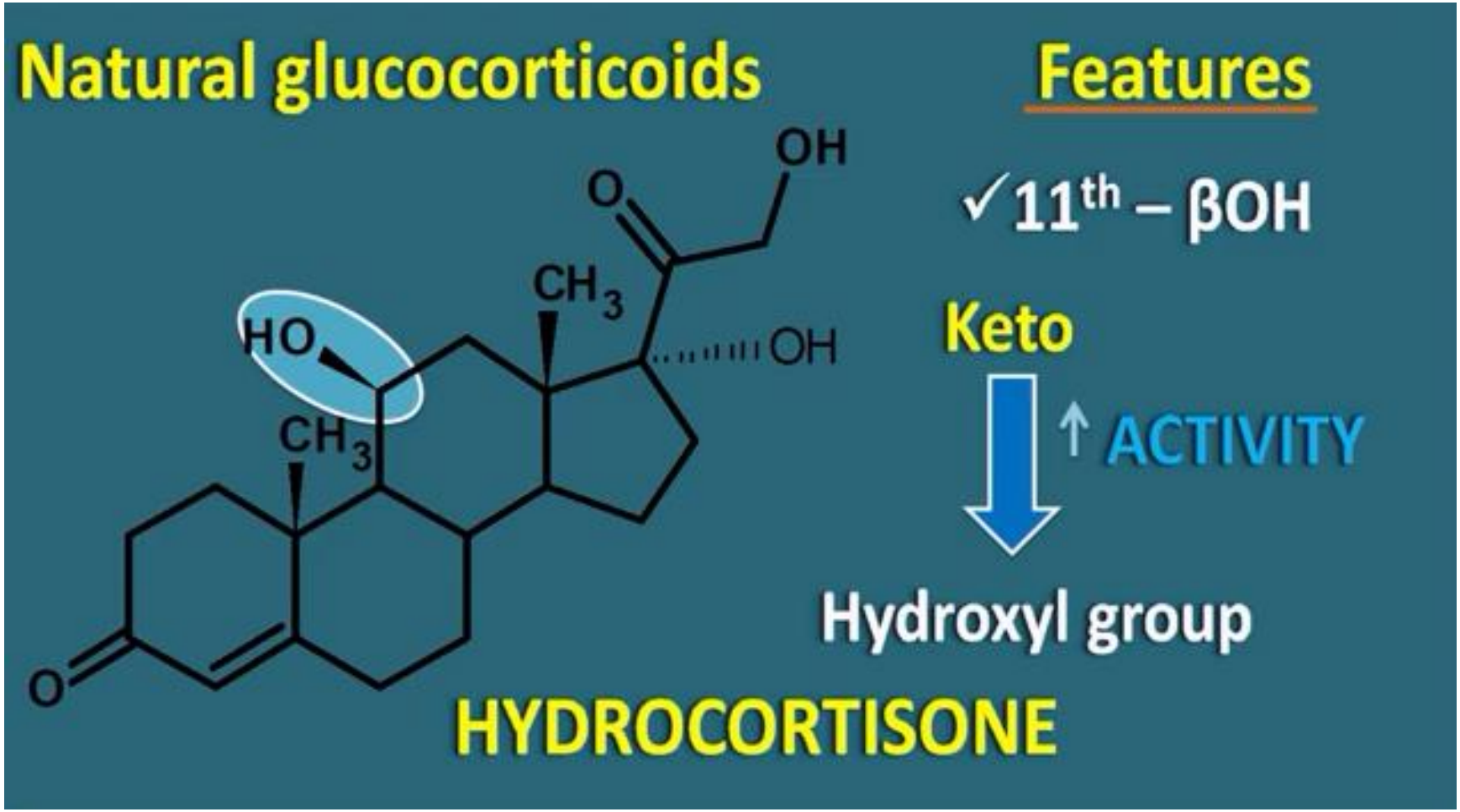

CHOLANE
Bile acids
CHOLESTANE
CHOLESTEROL


ERGOSTANE
ERGOSTEROL
STIGMASTANE
STIGMASTEROL

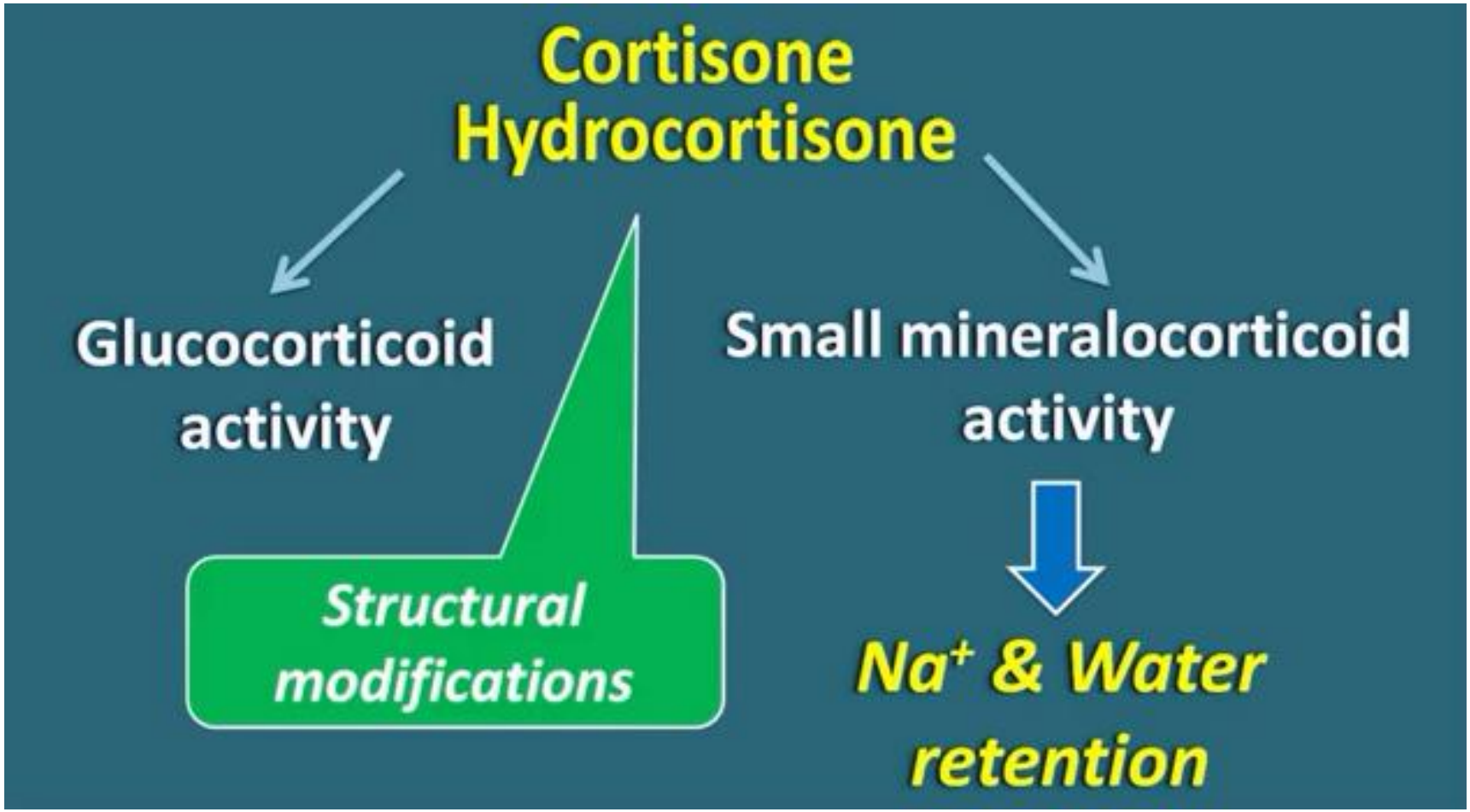
Glucocorticoids



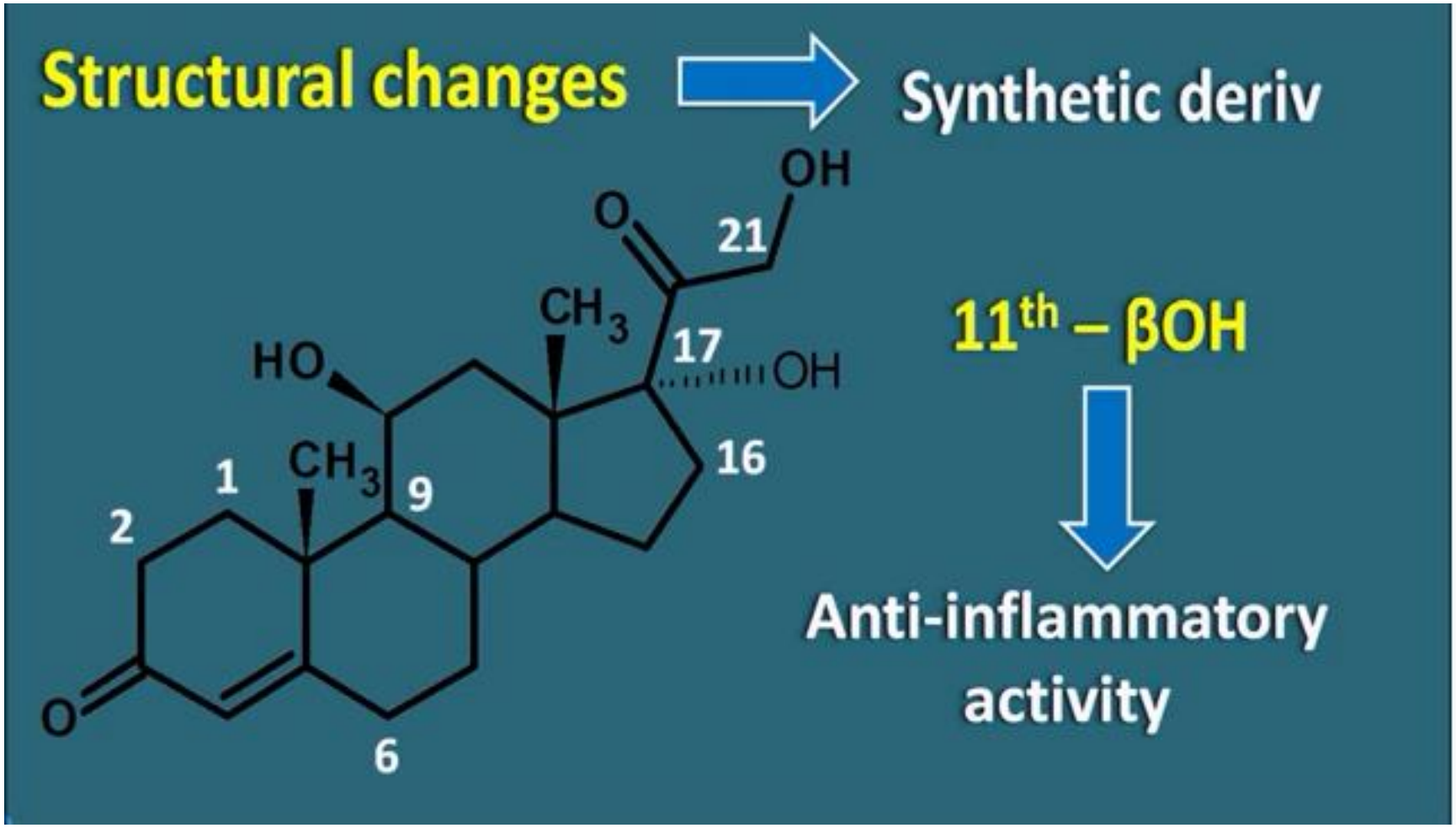
Cont'd...


Natural glucocorticoids
Features
OH
O
CH3
HO
OH
CH3
O
✓11th – βOH
Keto
↑ ACTIVITY
Hydroxyl group
HYDROCORTISONE

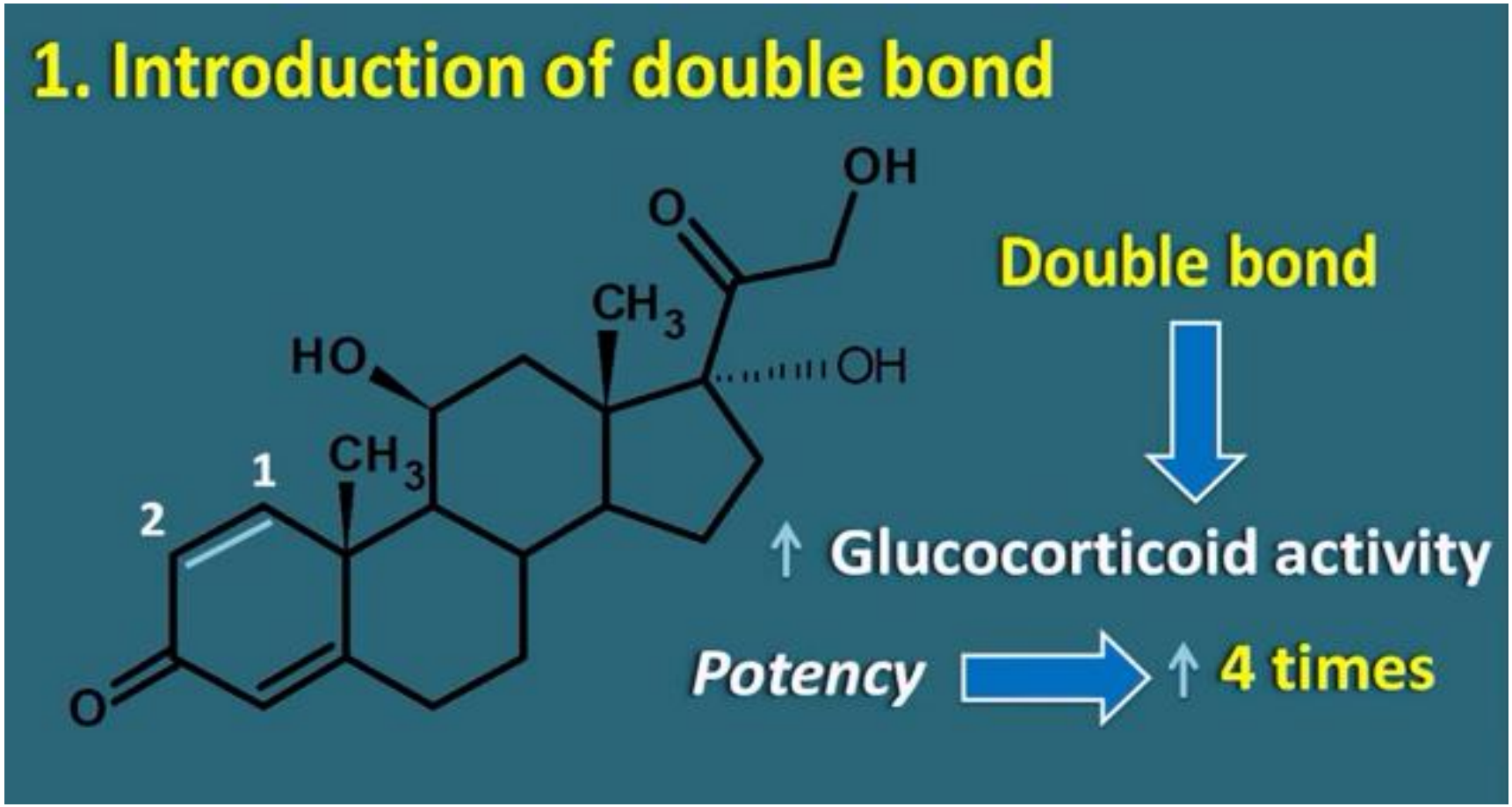
SAR of Adrenocorticoids



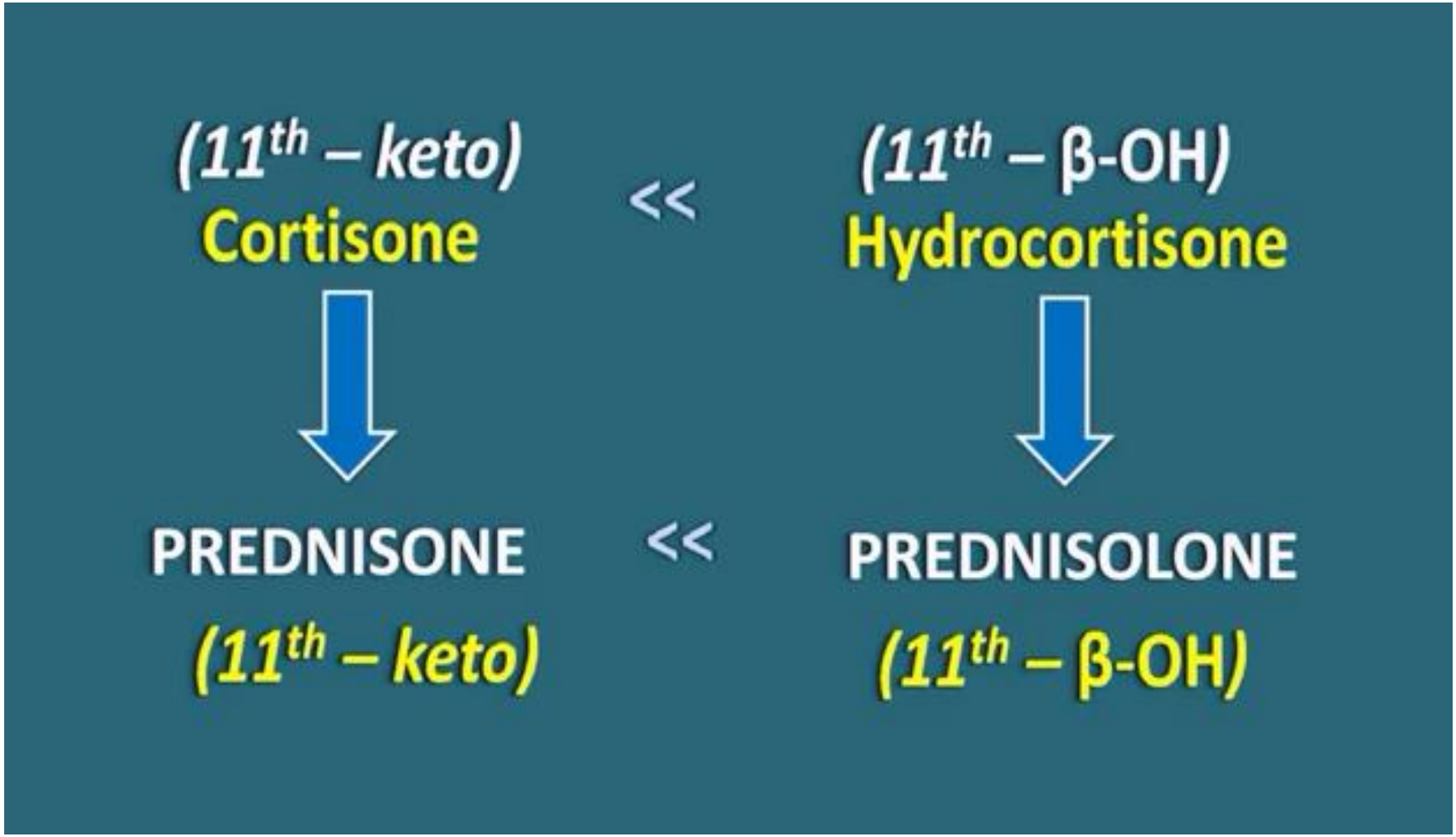
Cont'd...


Structural changes
Synthetic deriv
OH
O
21
CH3
17
HO
OH
16
CH3
1
9
2
O
6
11th – βOH
Anti-inflammatory activity

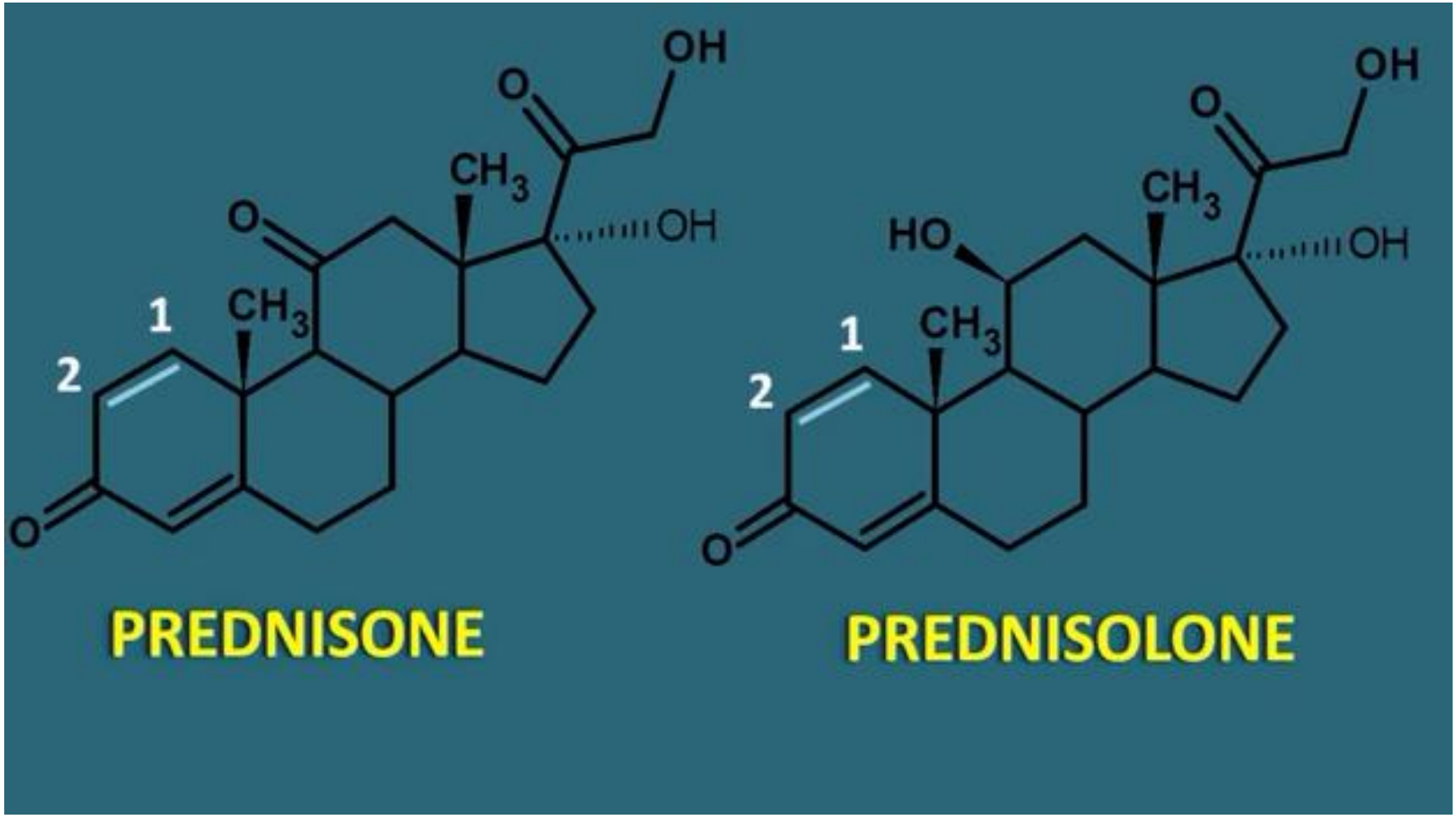
Cont'd...


1. Introduction of double bond
OH
O
CH3
HO
OH
CH3
1
2
O
Double bond
↑ Glucocorticoid activity
Potency
↑ 4 times

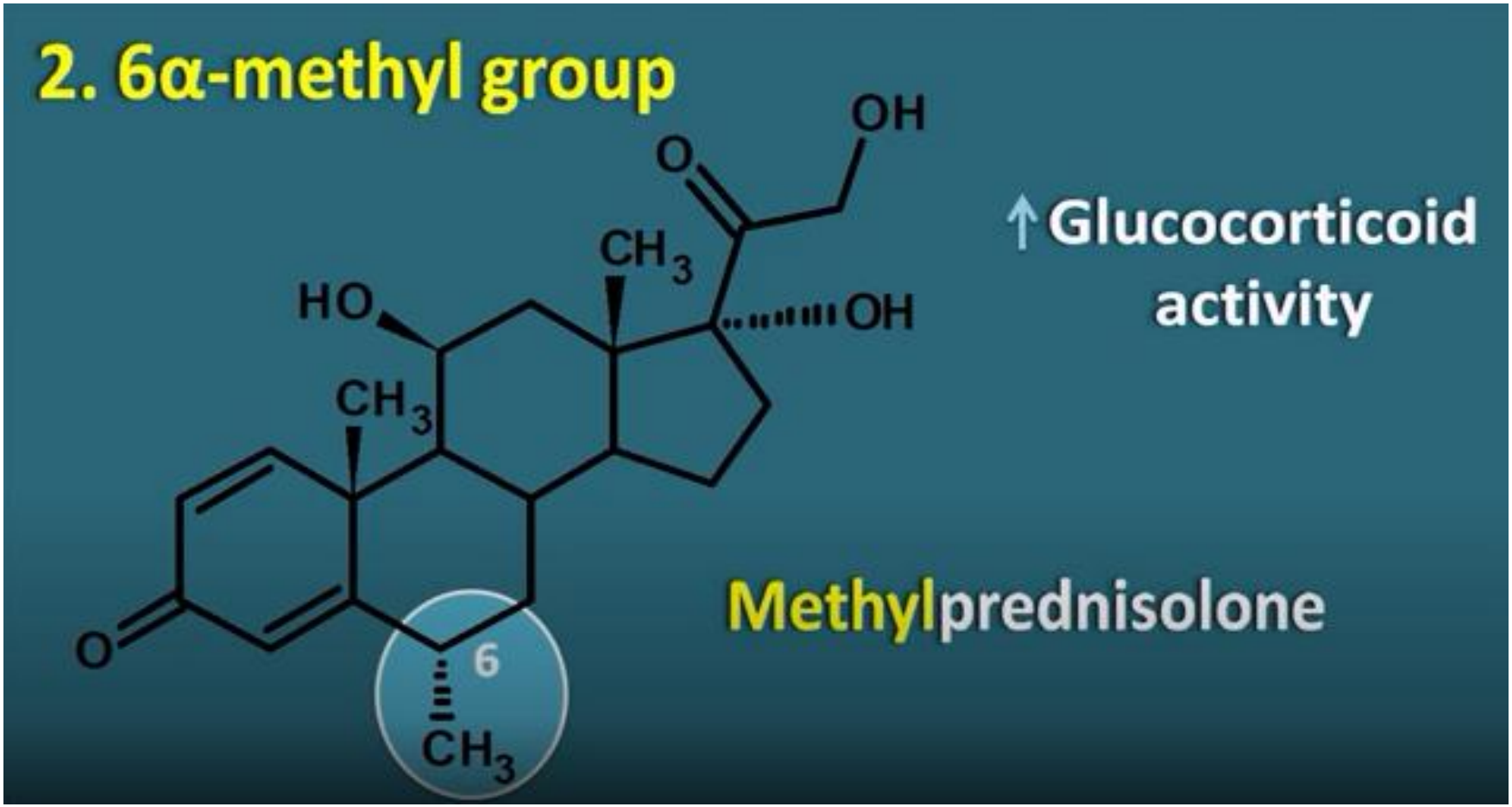
Cont'd...


(11th – keto)
Cortisone
<<
(11th – β-OH)
Hydrocortisone
PREDNISONE
<<
PREDNISOLONE
(11th – keto)
(11th – β-OH)

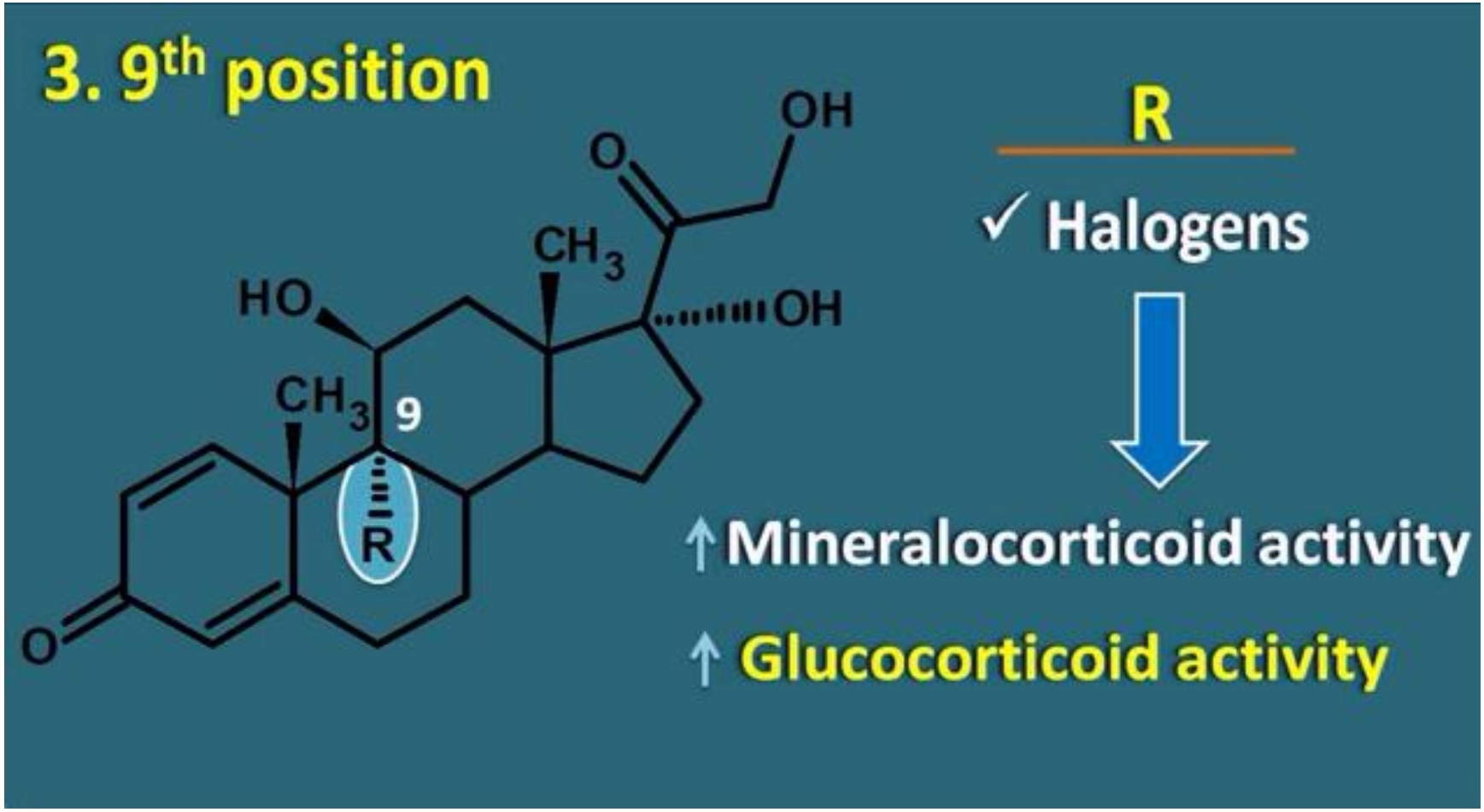
Cont'd...


OH
O
CH3
O
OH
1
CH3
2
O
PREDNISONE
OH
O
CH3
HO
OH
1
CH3
2
O
PREDNISOLONE

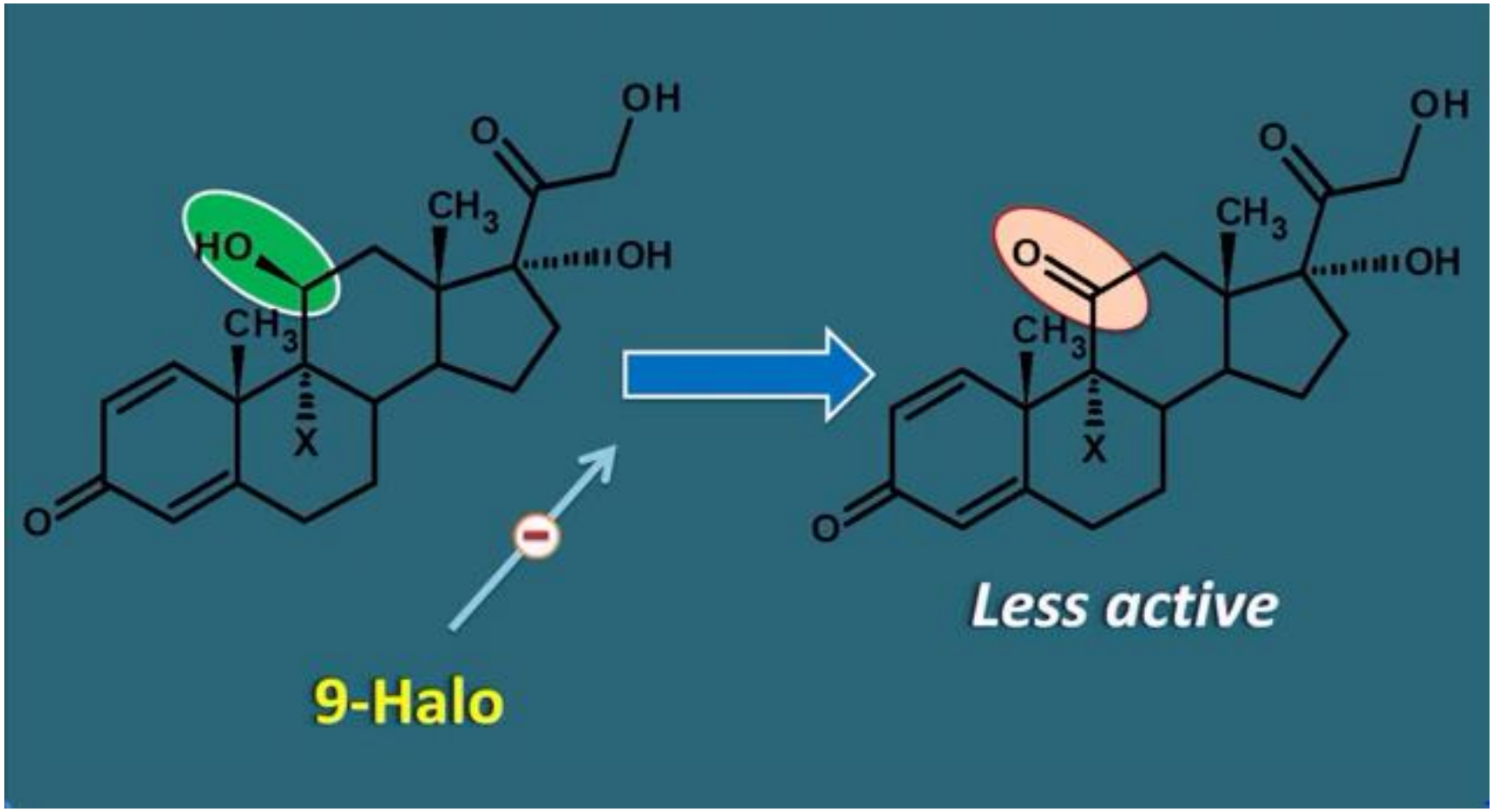
Cont'd...


2. 6α-methyl group
OH
O
CH3
HO
OH
CH3
↑Glucocorticoid activity
O
6
CH3
Methylprednisolone

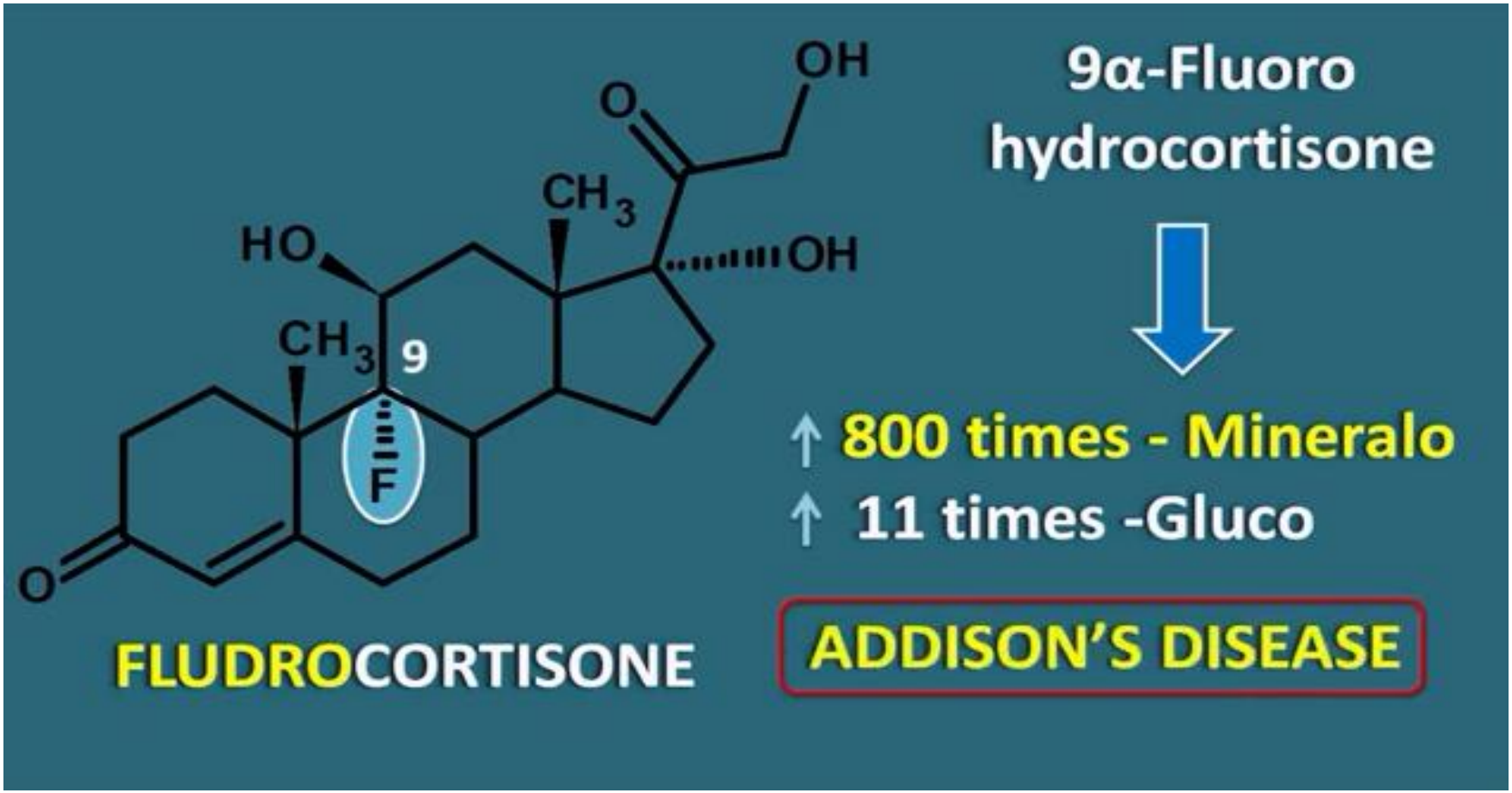
Cont'd...


3. 9th position
OH
O
CH3
HO
OH
CH3
9
R
O
R
✓ Halogens
↑ Mineralocorticoid activity
↑ Glucocorticoid activity

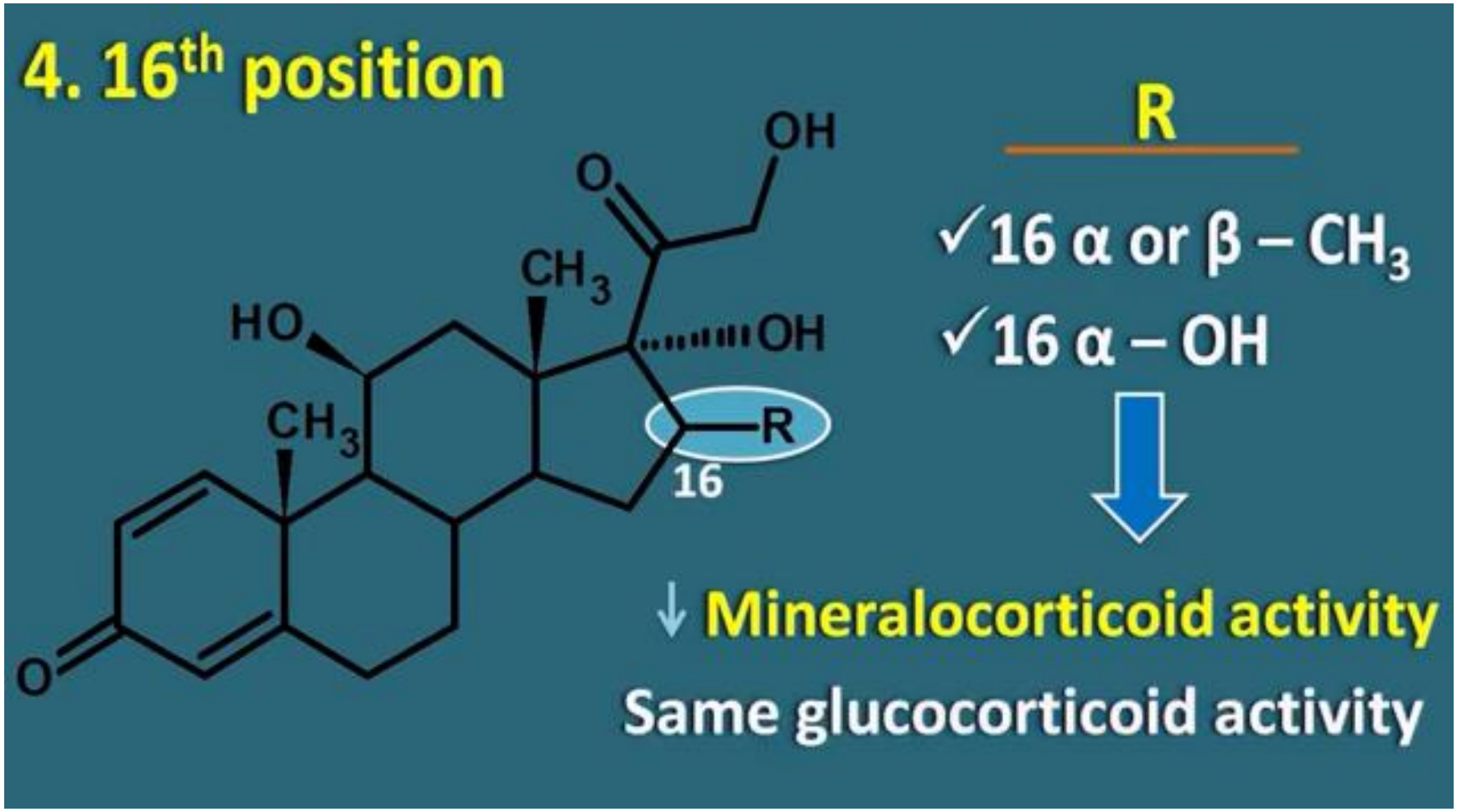
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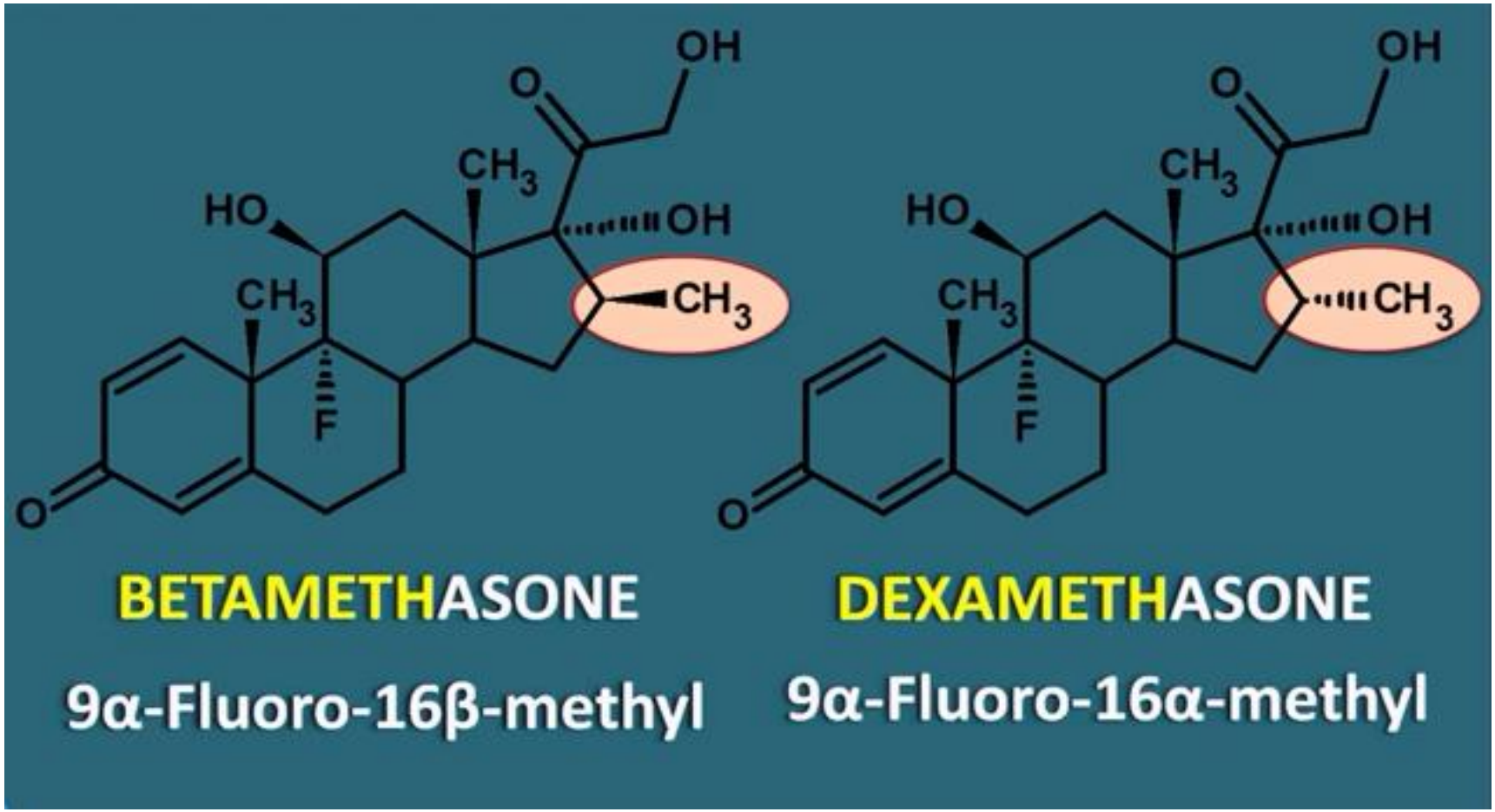
Cont'd...


OH
O
CH3
HO
OH
CH3
9
F
O
9α-Fluoro
hydrocortisone
↑ 800 times - Mineralo
↑ 11 times -Gluco
FLUDROCORTISONE
ADDISON'S DISEASE

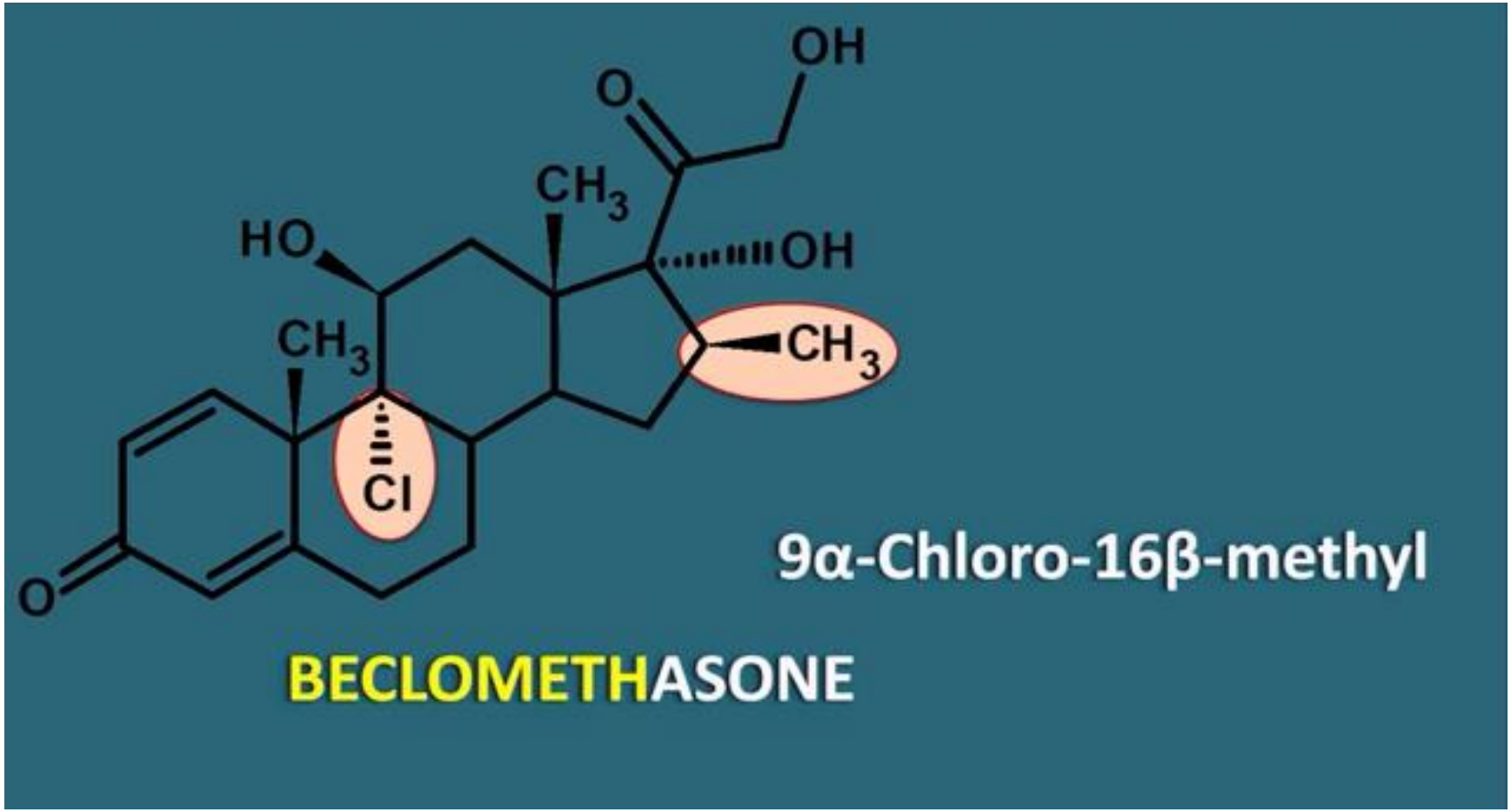
Cont'd...


4. 16th position
OH
O
CH3
HO
OH
CH3
R
16
O
R
✓16 α or β – CH3
✓16 α – OH
↓ Mineralocorticoid activity
Same glucocorticoid activity

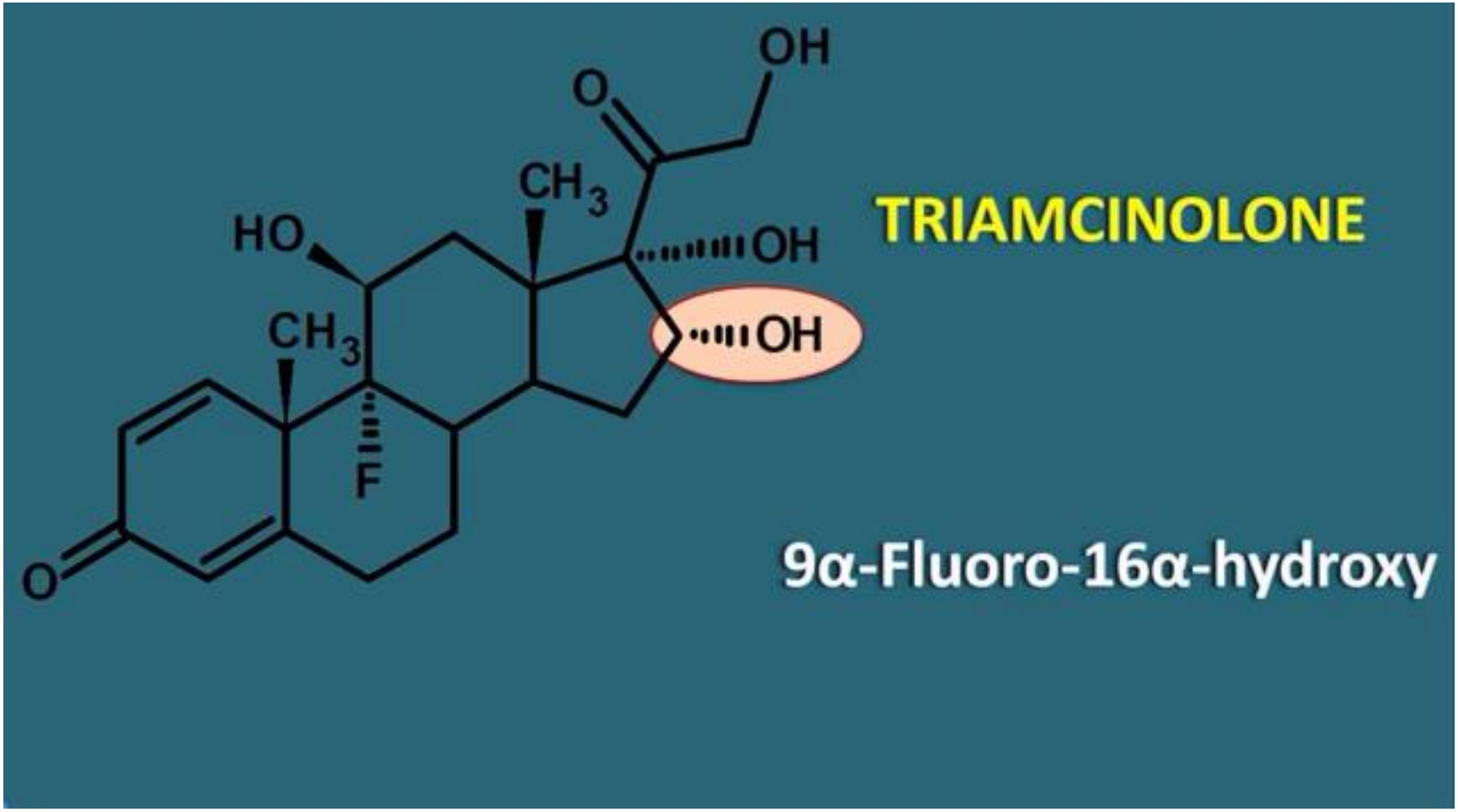
Cont'd...


OH
O
CH3
HO
OH
CH3
CH3
F
O
BETAMETHASONE
9α-Fluoro-16β-methyl
OH
O
CH3
HO
OH
CH3
CH3
F
O
DEXAMETHASONE
9α-Fluoro-16α-methyl

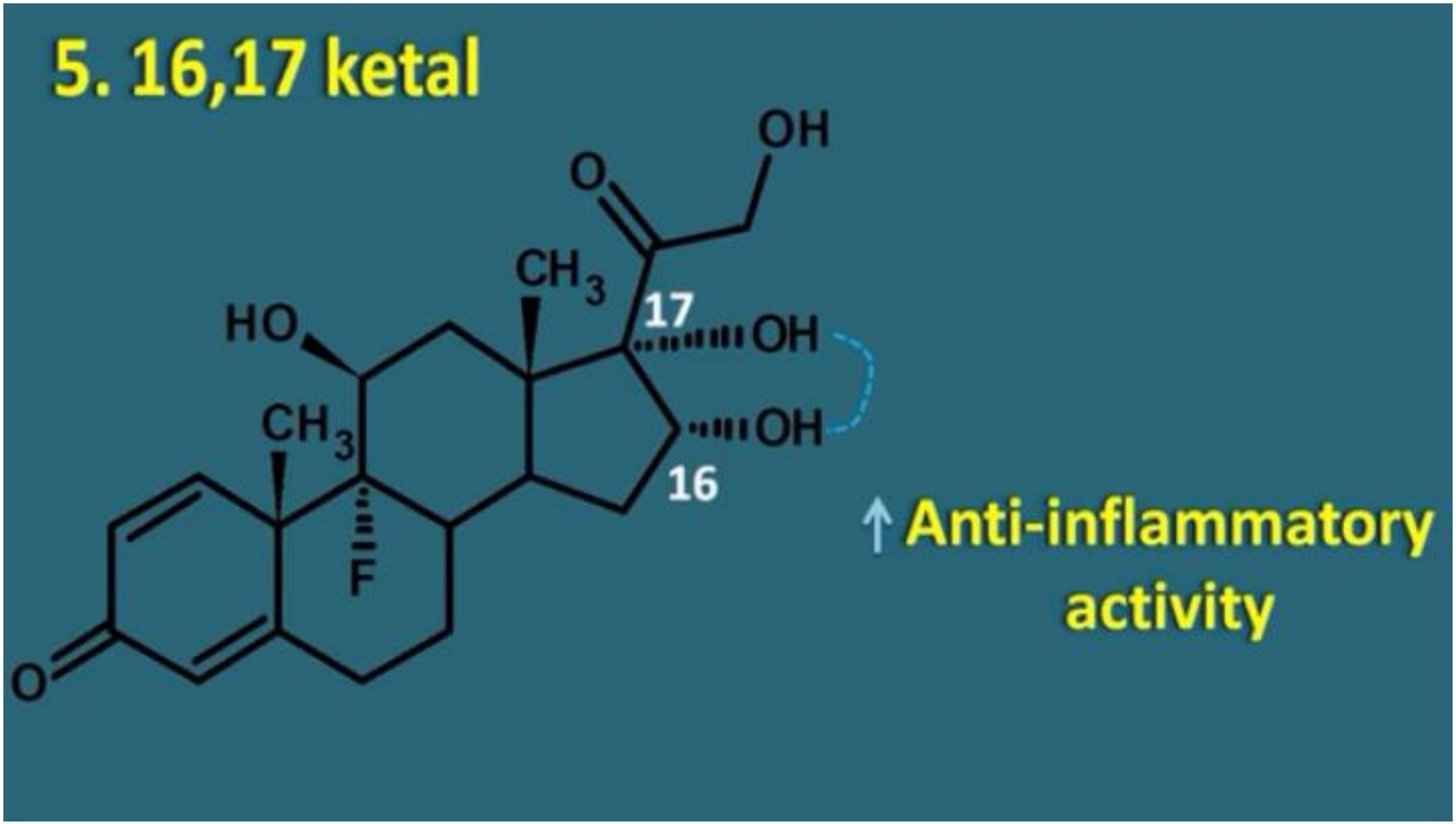
Cont'd...


OH
O
CH3
HO
OH
CH3
CH3
Cl
O
9α-Chloro-16β-methyl
BECLOMETHASONE

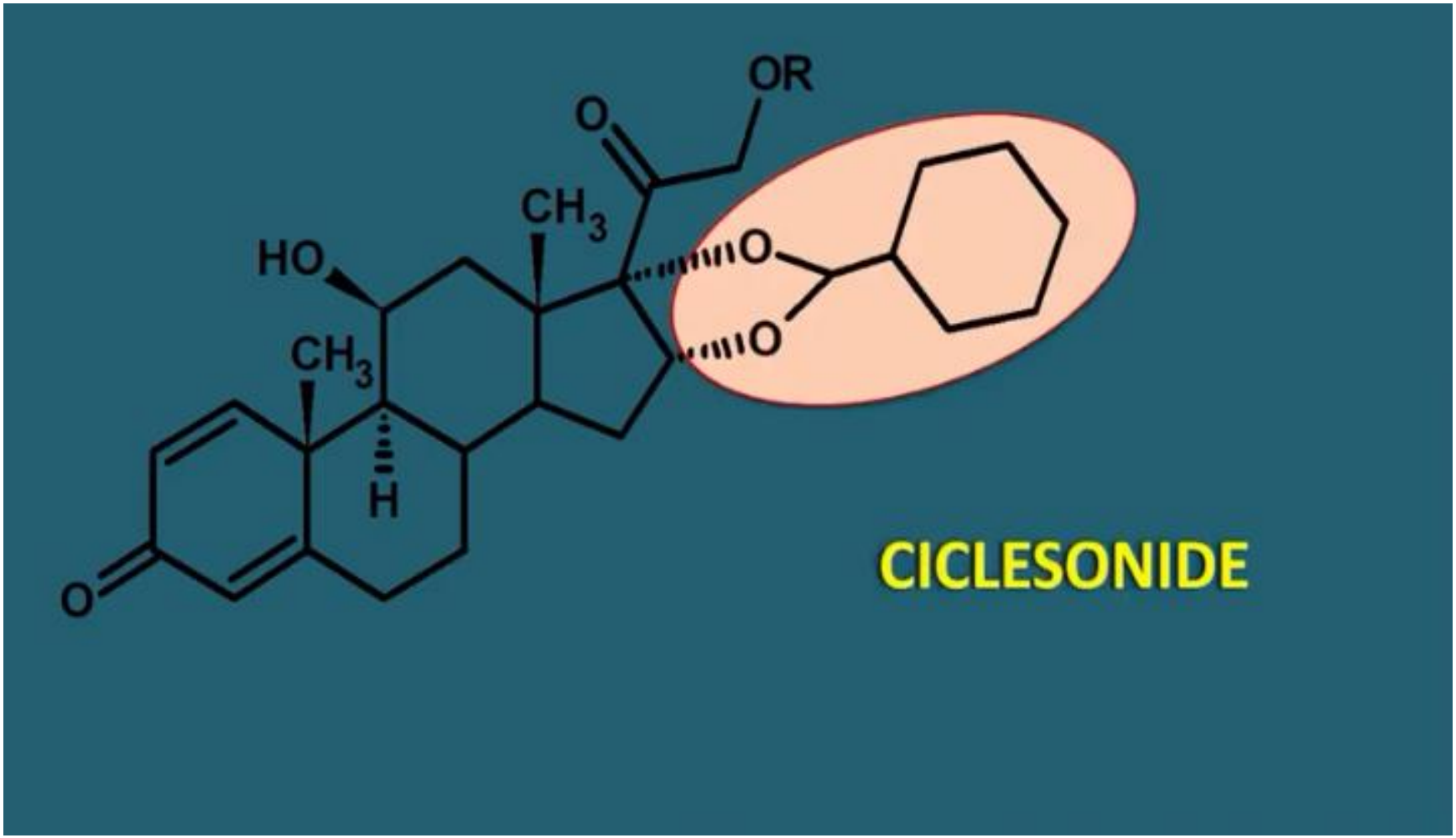
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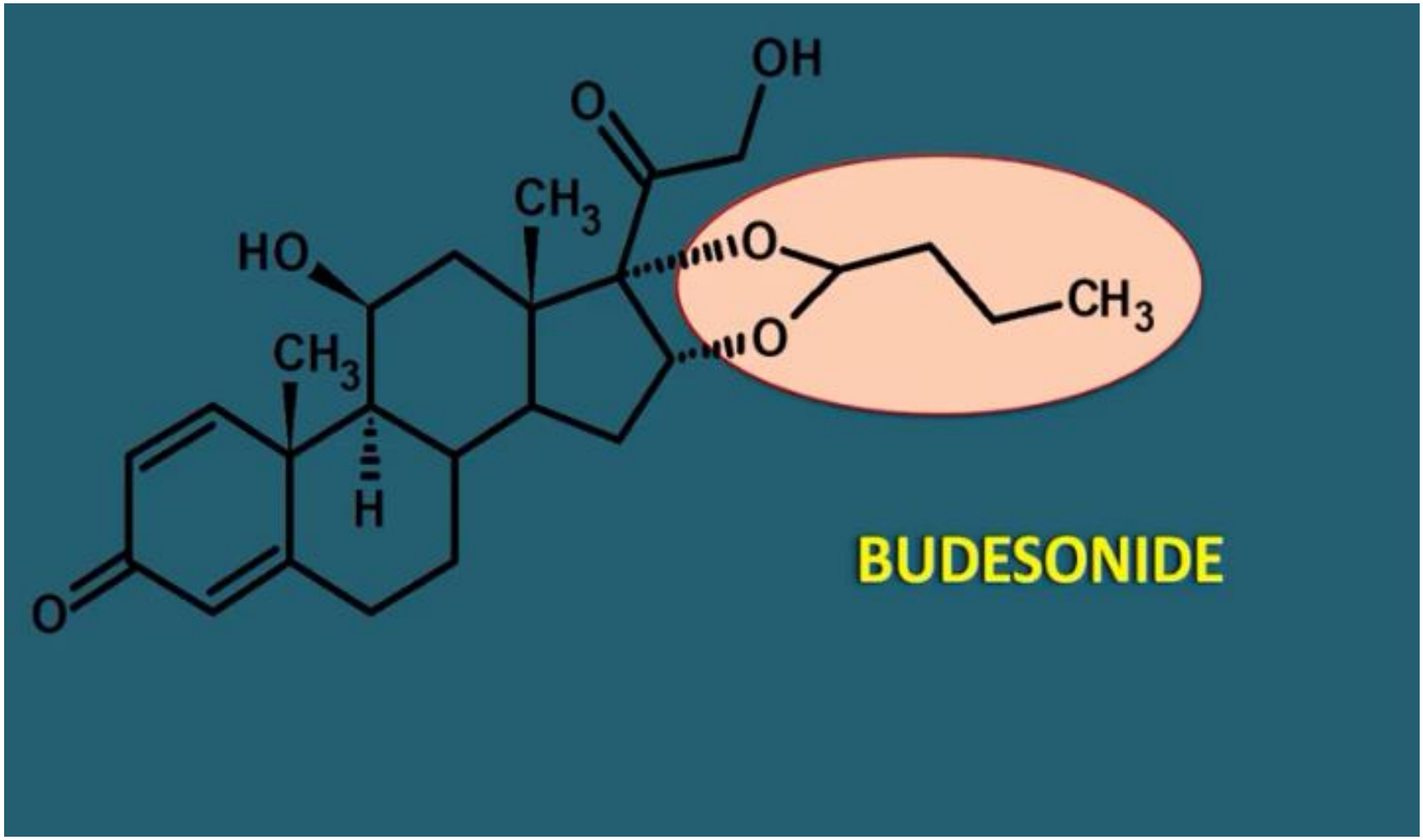
Cont'd...


5. 16,17 ketal
OH
O
CH3
17
HO
OH
CH3
OH
16
F
O
↑ Anti-inflammatory activity

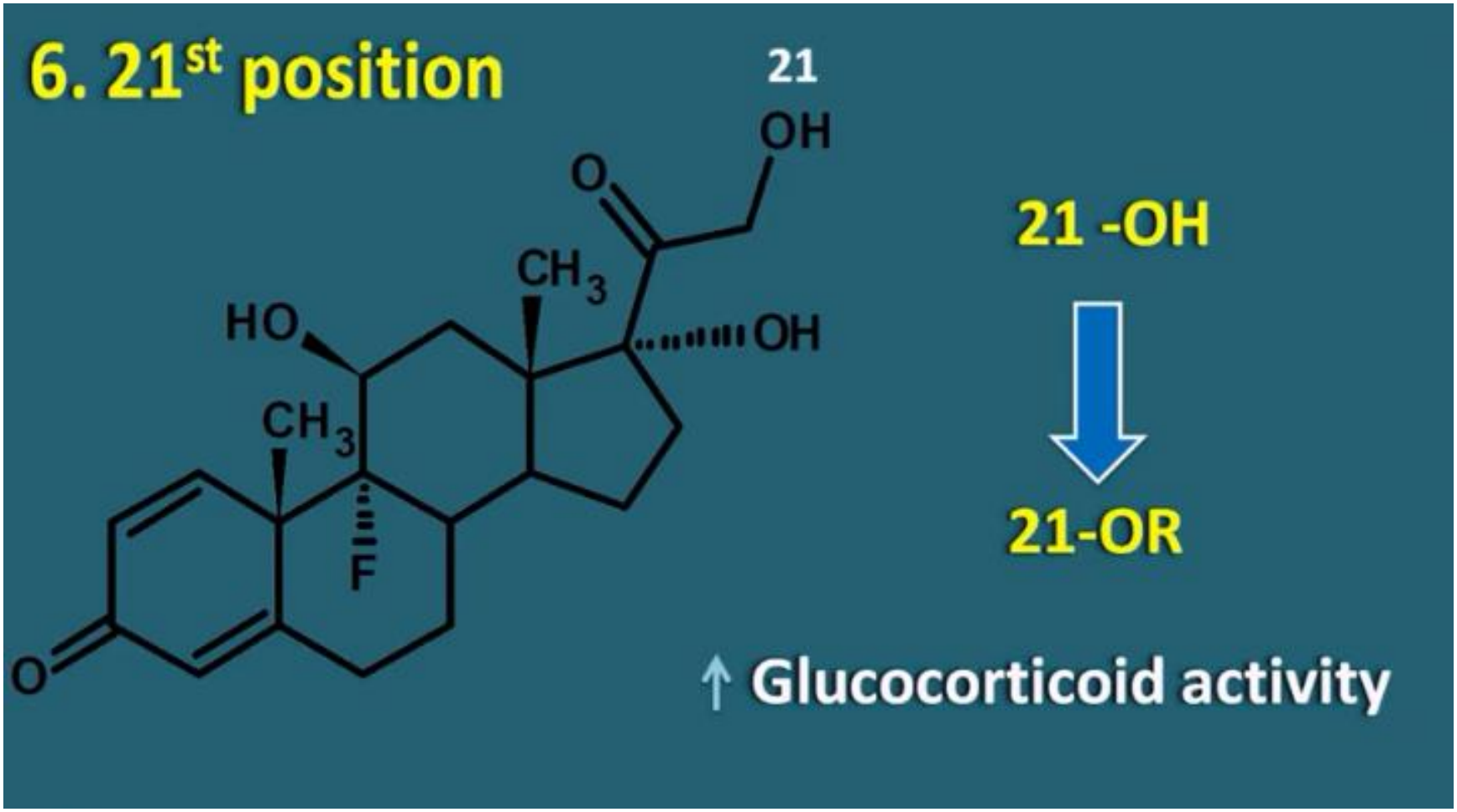
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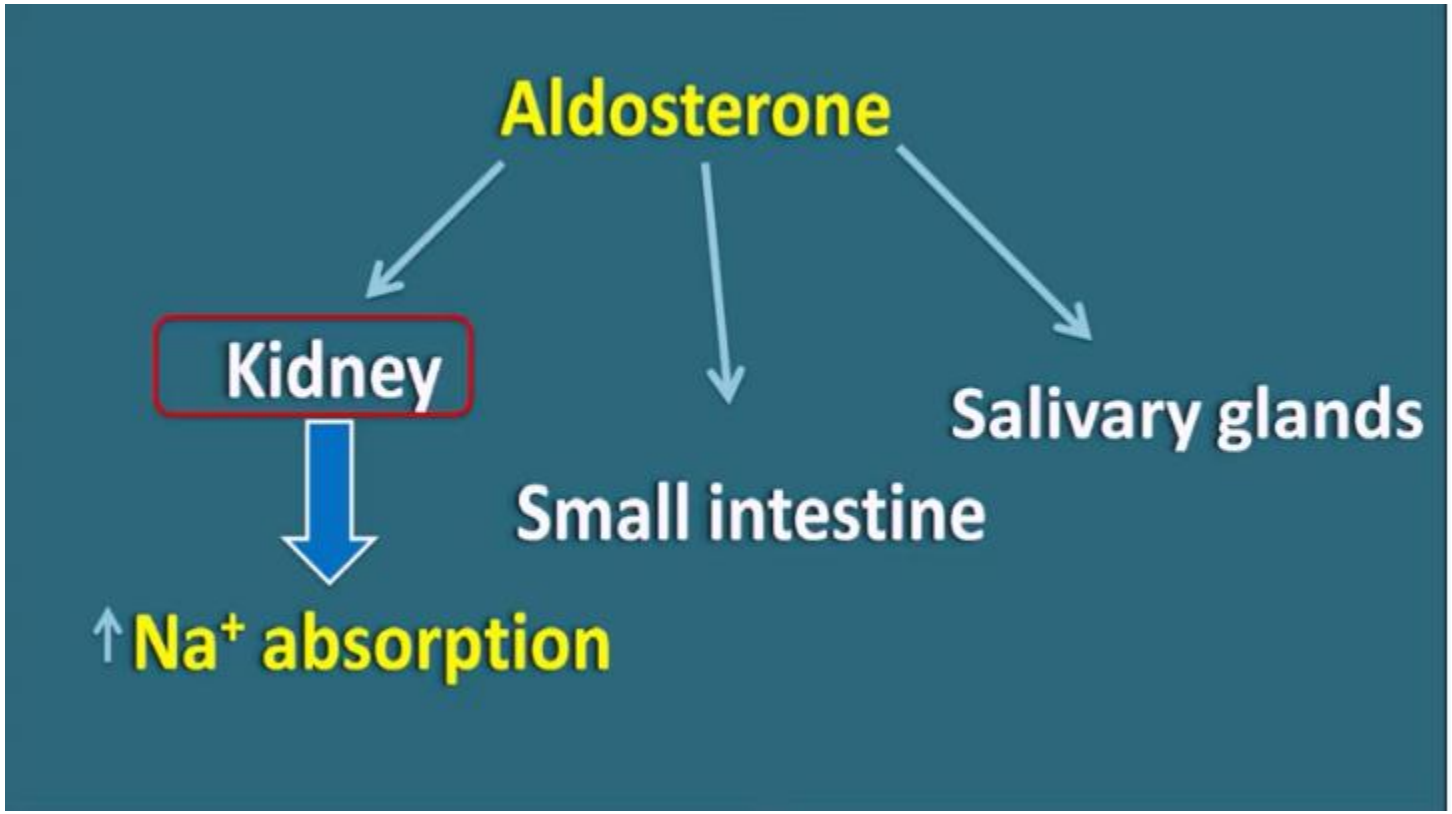
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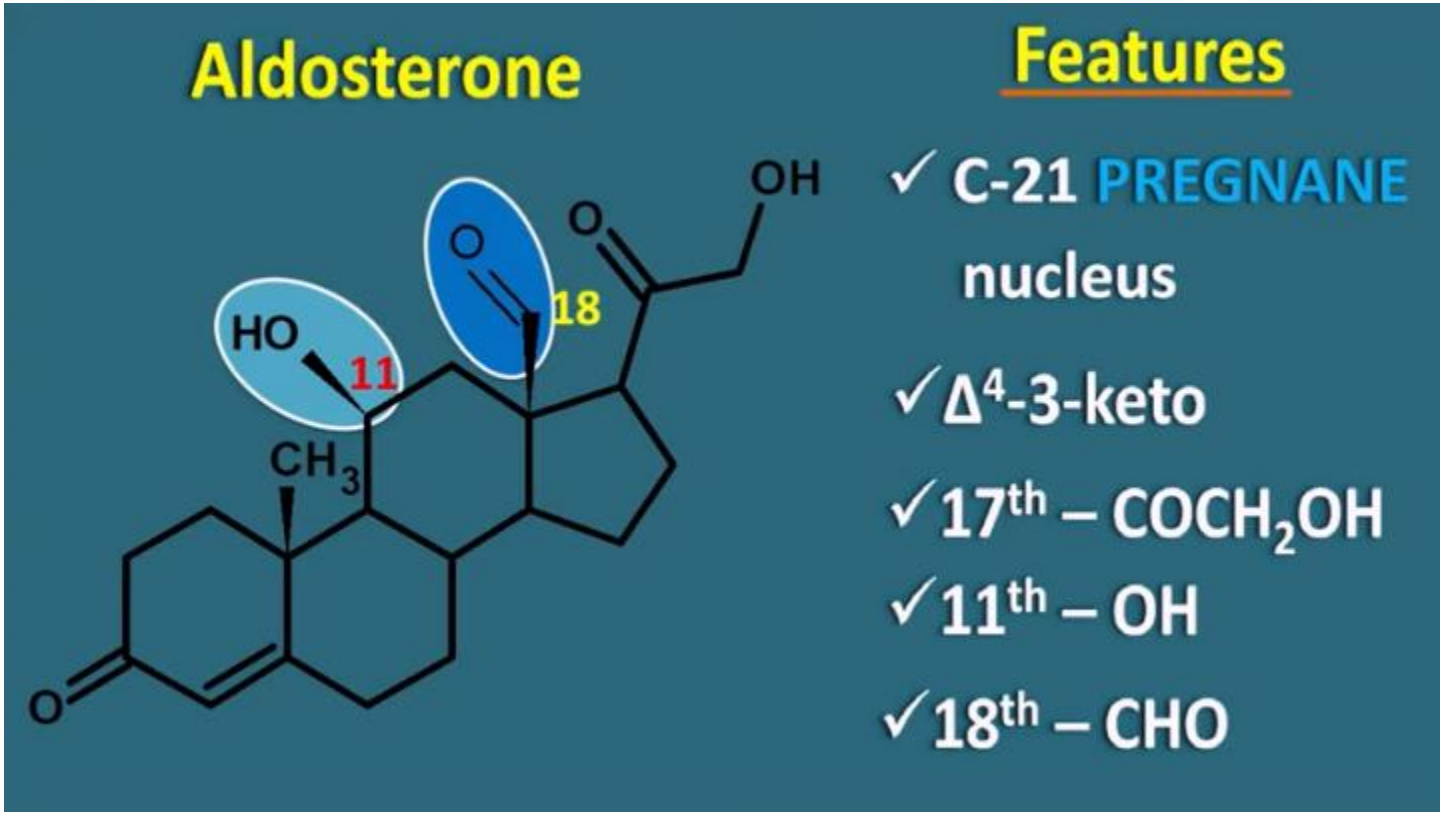
Cont'd...


6. 21st position
21
OH
O
CH3
HO
OH
CH3
F
O
21 -OH
21-OR
Glucocorticoid activity

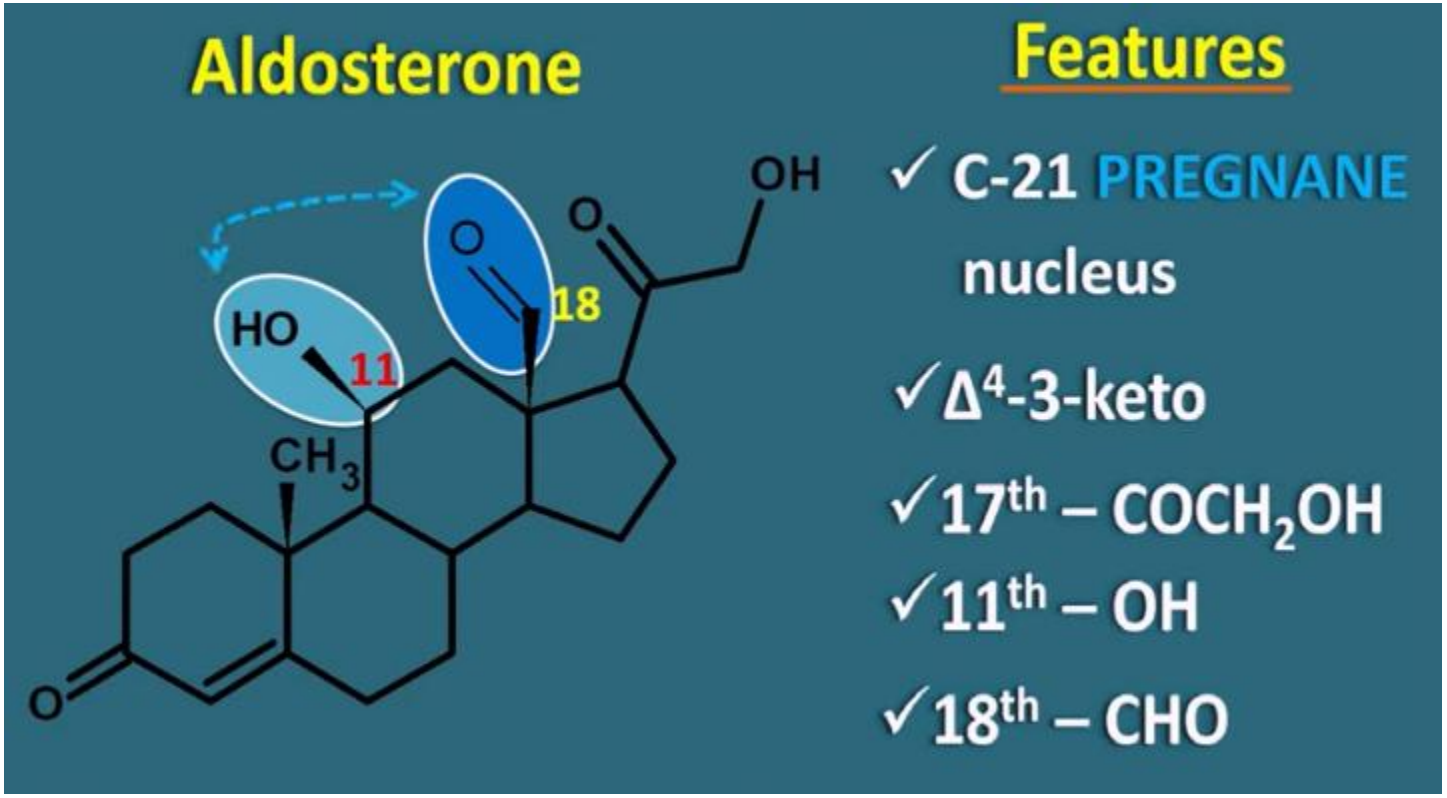
Mineralocorticoids



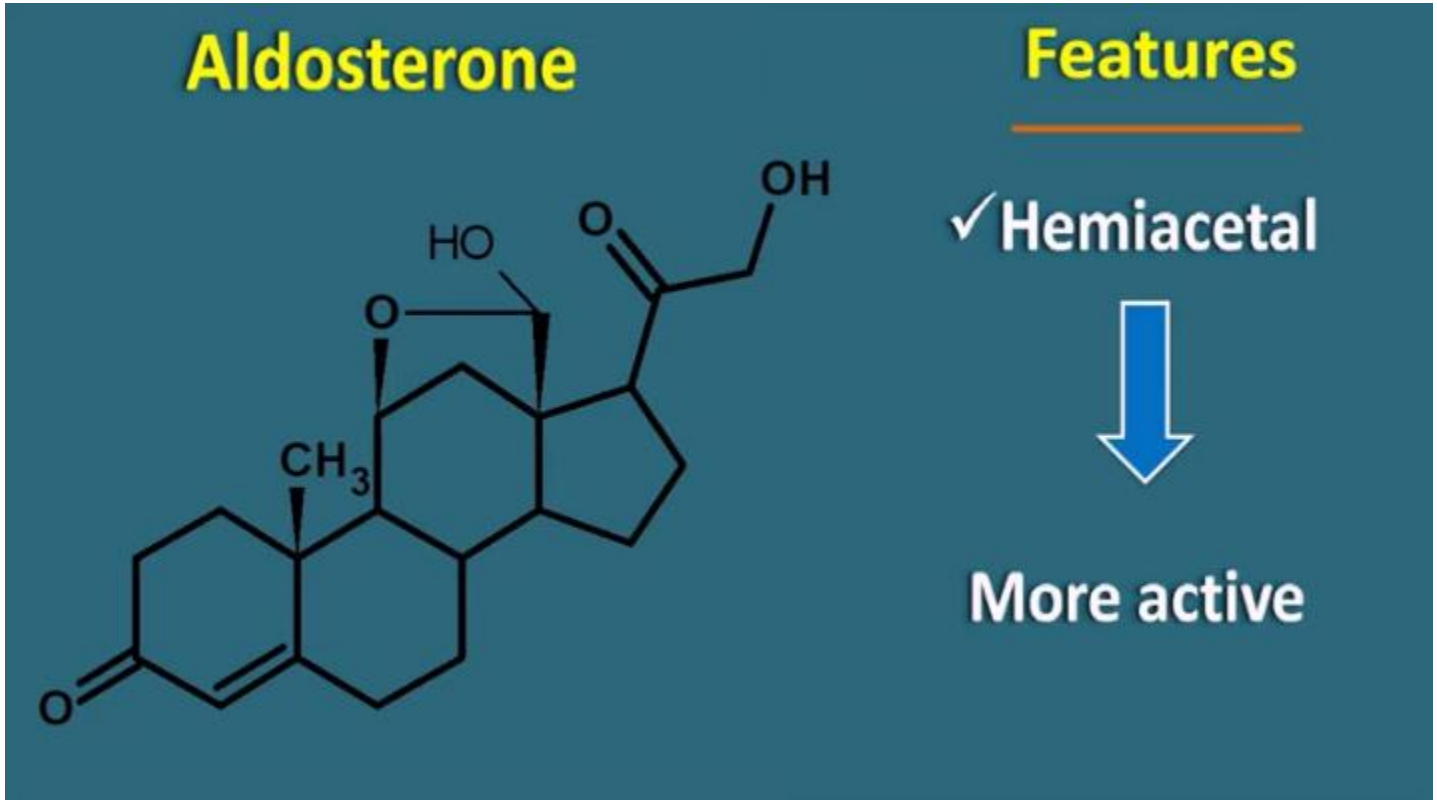
Cont'd...


Aldosterone
Features
OH
O
O
18
HO
11
CH3
O
✓ C-21 PREGNANE nucleus
✓ Δ4-3-keto
✓ 17th – COCH2OH
✓ 11th – OH
✓ 18th – CHO

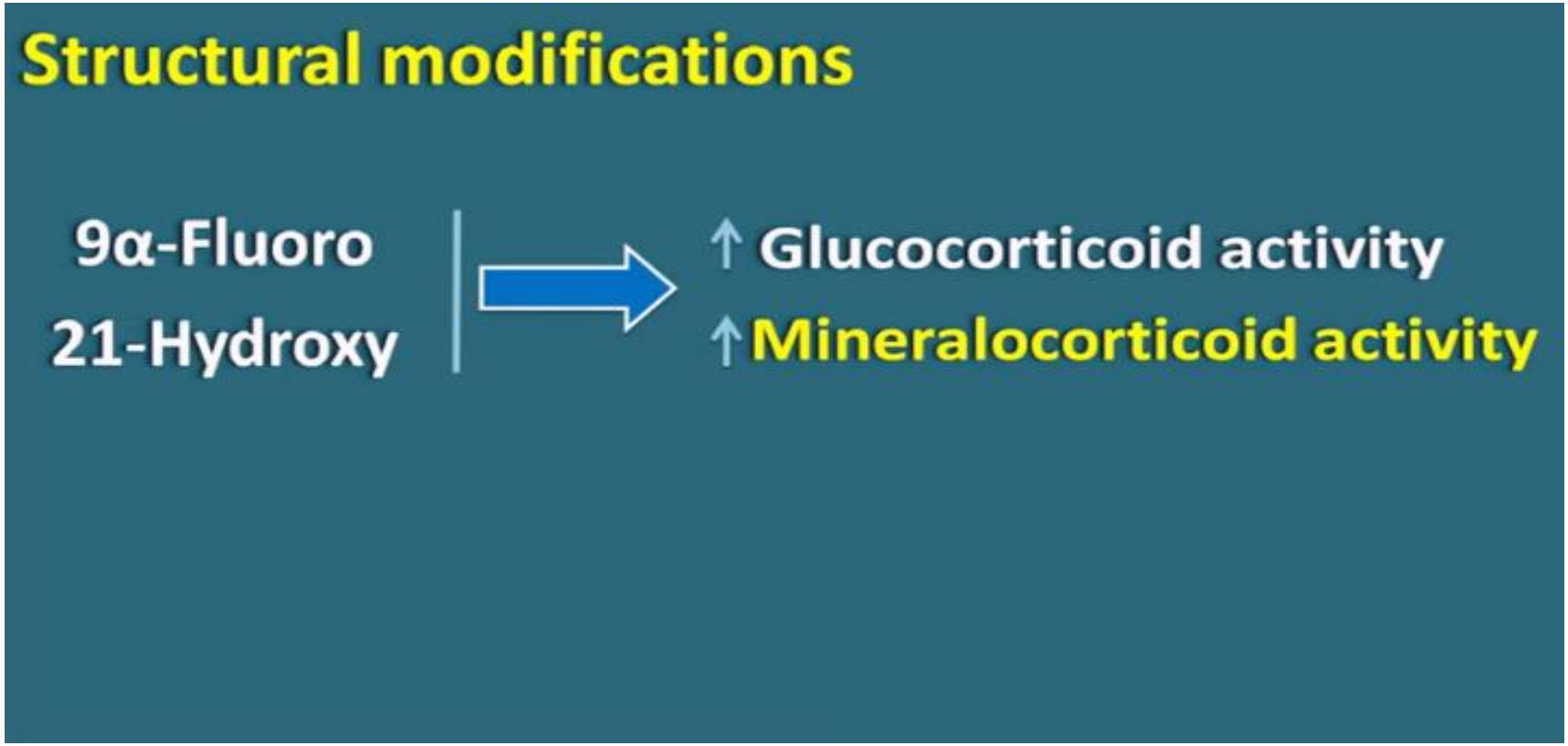
Cont'd...


Aldosterone
Features
OH
O
O
18
HO
11
CH3
O
✓ C-21 PREGNANE nucleus
✓ Δ4-3-keto
✓ 17th – COCH2OH
✓ 11th – OH
✓ 18th – CHO

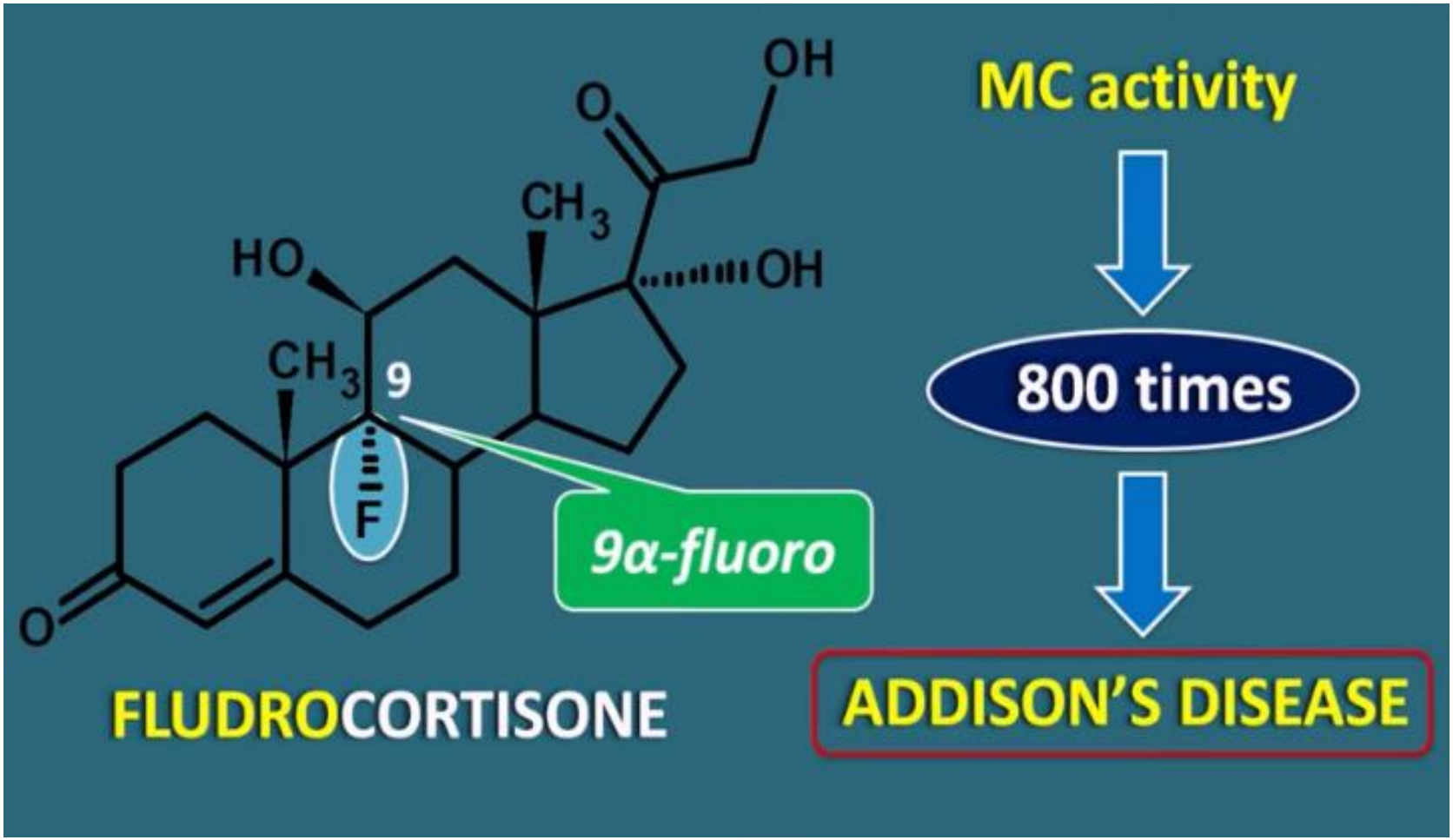
Cont'd...


Aldosterone
Features
✓Hemiacetal
More active
OH
HO
O
O
O
O
CH3

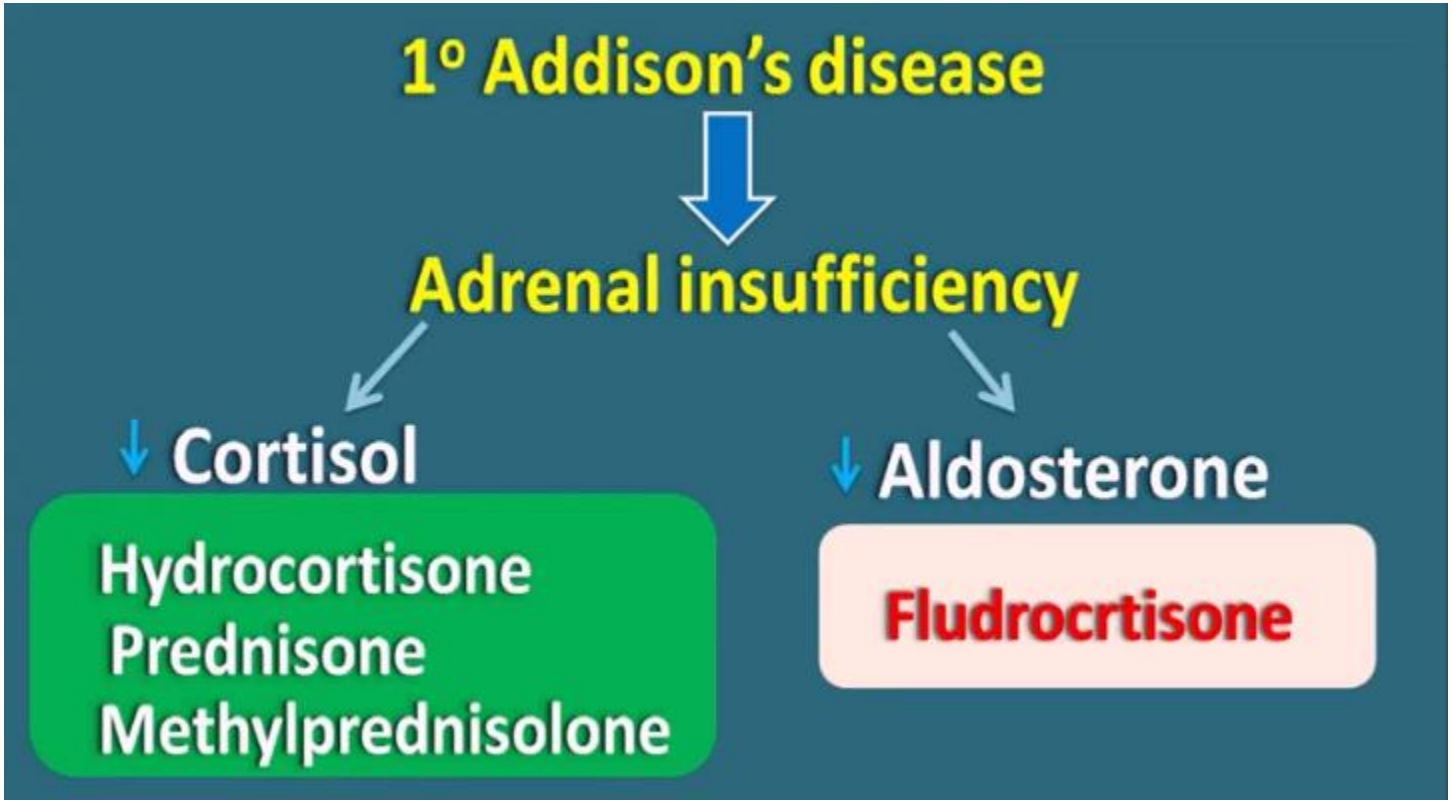
Cont'd...


Structural modifications
9α-Fluoro
21-Hydroxy
↑ Glucocorticoid activity
↑ Mineralocorticoid activity

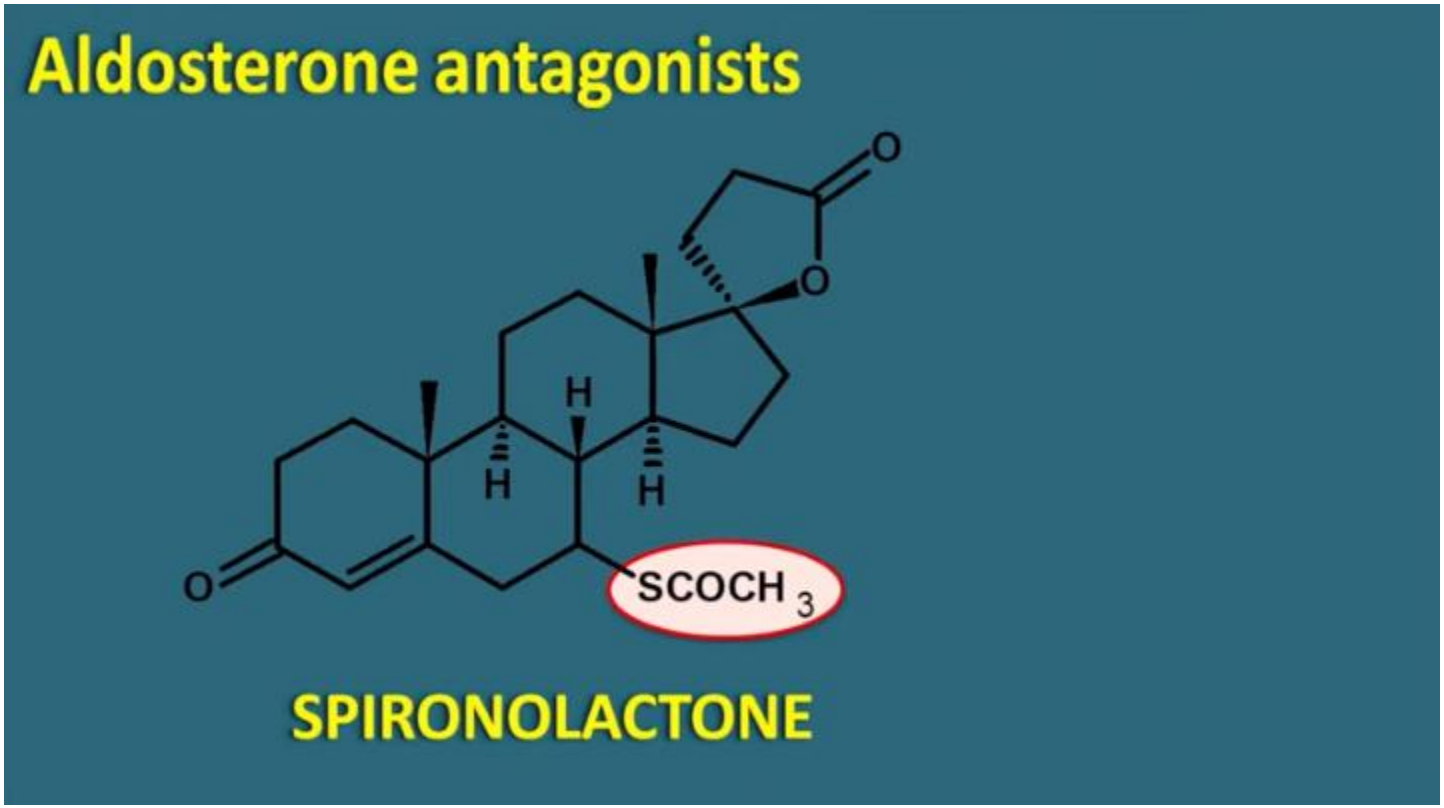
Cont'd...


OH
O
CH3
HO
OH
CH3
9
F
O
9α-fluoro
FLUDROCORTISONE
MC activity
800 times
ADDISON'S DISEASE

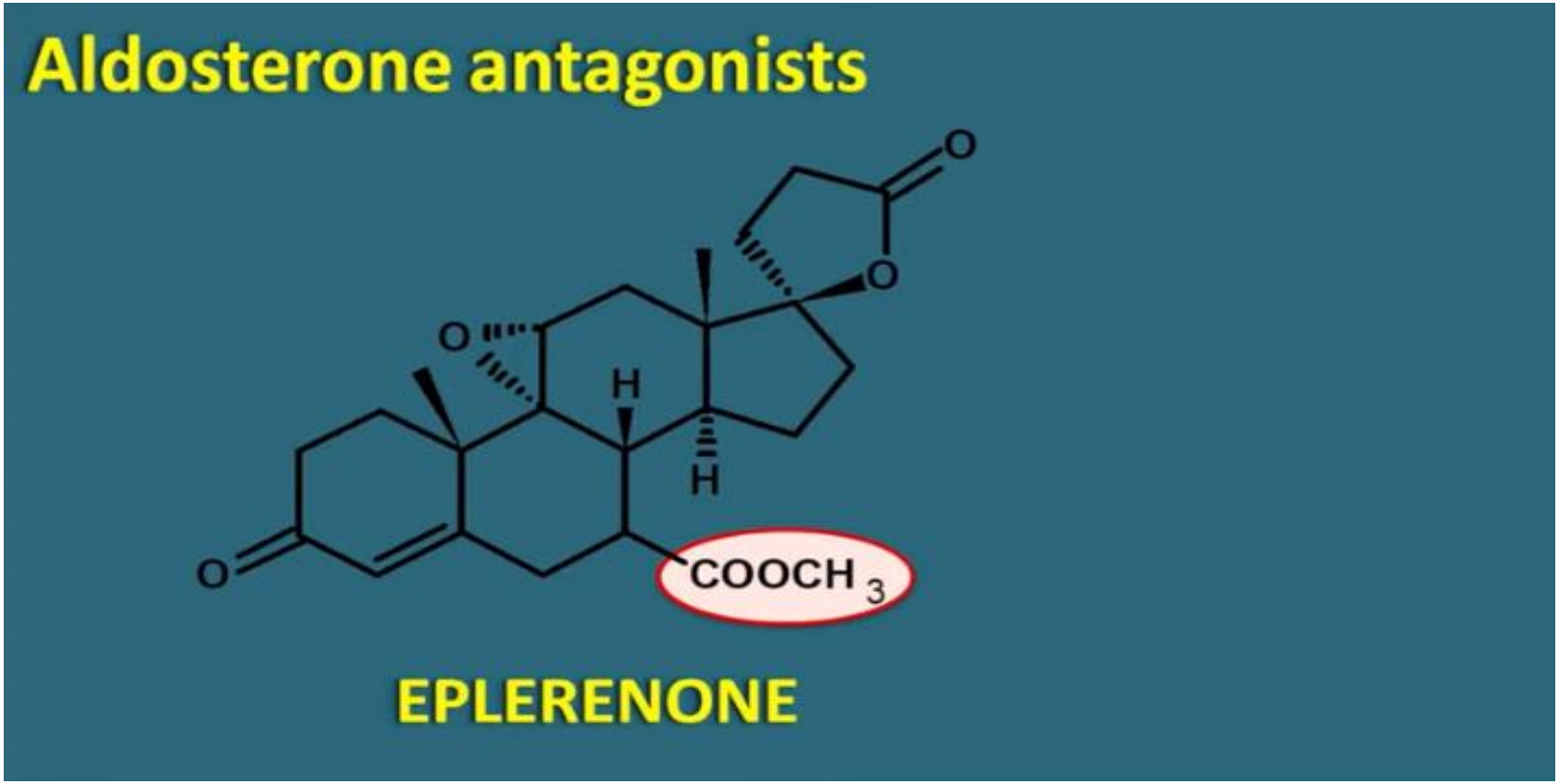
Cont'd...


1° Addison's disease
Adrenal insufficiency
↓ Cortisol
Hydrocortisone
Prednisone
Methylprednisolone
↓ Aldosterone
Fludrocrtisone

Aldosterone antagonist



Cont'd...



Reading assignments

Male and female sex hormones, derivatives and related drugs


THANK
YOU